Arithmetic constraints and covering curves for rational distances in a square

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Abstract

We study points in the strict interior of a square whose four vertex distances are rational. Primitive integral and Gaussian models yield exact denominator identities, prime-power bounds, and reciprocity restrictions on the squarefree classes assigned to opposite corners. We construct ordered elliptic sources and covering curves of genera five, nine and seventeen, with explicit maps retaining the four distances. A common squareclass descent gives infinite families of genus-nine curves without rational points, while a compensated source descent excludes an entire squarefree kernel. Explicit quaternion evaluations give a parity obstruction and an infinite family of complete genus-five failures of the Hasse principle. We describe nine elliptic factors of the genus-nine Jacobian and derive rank constraints. For the fixed opposite-distance difference $1/5$, we determine all twenty-eight bitangents of a genus-three quotient, its odd-theta algebra and the theta class defining the original double cover. Finally, we prove that rational boundary points and their real components produce locally soluble lifts at every fixed finite Abel–Jacobi level.

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1 Introduction

The rational-distance problem for a square asks for a point whose distances from all four vertices are rational. We consider the strict interior of the unit square and retain the four distances as part of the arithmetic data. Writing the vertices in the order $00, 10, 01, 11$, the equations are

$$\begin{aligned} r_0^2 &= x^2 + y^2, & r_1^2 &= (1-x)^2 + y^2, \\ r_2^2 &= x^2 + (1-y)^2, & r_3^2 &= (1-x)^2 + (1-y)^2, \end{aligned} \qquad 0 < x, y < 1,$$

with $r_i \in \mathbf{Q}_{>0}$. Differences of the squared distances recover x and y, so this is a rational-point problem with a specified real chamber.

The geometry of rational-distance varieties has been studied from several directions. Berry relates the square to a Kummer surface and a branched double cover, and studies associated curves of genus five [1]; his work on triangles develops the corresponding three-distance geometry [2]. Bremner and Ulas give constructions for rational-distance problems attached to several geometric objects [3]. Corvaja, Turchet and Zannier study distance varieties for general configurations of rational points, including their surfaces and product-quotient descriptions [6]. The present paper develops the arithmetic of explicit covering curves, the squareclasses carried by one ordered source, and congruences that hold at every denominator height.

The first part concerns the primitive integral equations. Let L be the least common denominator of x, y, and write $A_i = Lr_i$. We prove that $24 \mid L$, that the common radius content $G = \gcd(A_0, A_1, A_2, A_3)$ satisfies $G^2 \mid L$, and that the six reduced denominators of the pairwise radius sums and differences satisfy exact gcd and product identities (Theorem 2.6). Signed radius identities compare each prime-power divisor of L with the size of the square. They imply, in particular, that L has a prime divisor other than 2 or 3 (Corollary 2.11).

Gaussian factorization assigns four canonical, pairwise coprime squarefree integers $m_0, \ldots, m_3$ to the corners. Their product is the squarefree part of

$$(r_0 r_3 + r_1 r_2)^2 - 1.$$

All primes in these integers are 1 or 7 modulo 8. A reciprocity identity for each pair of corners retains the common denominator and the sign of the Gaussian norm. For the two opposite pairs it gives

$$m_0 m_3 \equiv 1 \text{ or } 15 \pmod{16}, \qquad m_1 m_2 \equiv 1 \text{ or } 15 \pmod{16}$$

(Theorem 3.5). We also classify the residue pattern arising when $v_p(L) = 2v_p(G)$. An elliptic residue curve identifies the exceptional primes 5 and 13; imposing the additional product-square condition gives the refined exceptional set $\{5, 13, 17\}$ (Theorems 4.3 and 4.7).

The second part gives explicit rational correspondences. An ordered pair $Q = (a, b)$, $R = (c, e)$ on

$$E_n : \quad Y^2 = X(X-n)(X+n)$$

encodes the original point through two ordered product squareclasses, the two ordinates, and the single further square

$$\frac{a+c}{2(c+n)}.$$

Theorem 5.2 gives the forward and inverse maps, including all signs and strict inequalities. Fixing a signed sum or difference of two distances instead gives a smooth curve of genus five. The resulting fixed-pair finiteness theorem follows from the explicit branch calculation and Faltings' theorem (Theorem 5.8).

The product parameter $t = r_0r_3 + r_1r_2$ leads to a genus-nine curve D_t with four ordered square roots over a common coordinate ℓ. Its valuation conditions constrain one global class $[\ell]$. A generally nonsymmetric Legendre-symbol matrix then reduces the possible classes to two, and a local identity at 37 excludes both under an explicit residue condition. This proves an infinite family of all-height no-point results (Theorem 6.7 and Corollary 6.8). The associated unramified double covers have genus seventeen. Nine explicit elliptic quotients describe $\mathrm{Jac}(D_t)$ and yield rank constraints on parameters admitting rational points (Theorems 7.1 and 7.6).

Further descent must compare local data arising from the same global source. We express this requirement through a character map and a secondary obstruction in its cokernel. Keeping the complete compensation coset gives the criterion of Theorem 8.2. A short arithmetic certificate applies it, together with a higher quotient, to exclude the entire kernel $n = 15569$ (Corollary 8.4). A separate construction attaches a proper quaternion class to a complete genus-five source. Its local evaluations give an explicit parity criterion through Hilbert reciprocity (Theorem 9.1). An explicit conic parametrization yields an infinite family of complete genus-five failures of the Hasse principle (Theorem 9.3).

The final section treats the fixed opposite-distance difference $1/5$ in detail. Its genus-five source has an unramified genus-three quotient which is a smooth plane quartic. We determine all twenty-eight bitangents, their contact divisors and Galois action, obtaining the odd-theta algebra

$$\mathbf{Q}^8 \times \mathbf{Q}(\sqrt{2})^4 \times \mathbf{Q}(\sqrt{51})^4 \times \mathbf{Q}(\sqrt{2}, \sqrt{51}).$$

The calculation supplies explicit theta fractions of the type used in generalized descent [4, 12]. A polynomial identity identifies the original double cover with a specified difference of theta characteristics. We then prove a local-solubility theorem for every finite Abel–Jacobi multiplication cover through a rational boundary point. Its application preserves the original double cover and a strict real point (Corollary 10.6).

For the models encoding the original four-distance configurations, we give explicit reconstruction formulas and the inequalities defining the strict real chamber. Finite residue calculations are given as explicit identities, congruences or complete small tables. We use the classical torsion theorem of Mazur, the finiteness theorem of Faltings, and standard facts about elliptic curves, covers and Brauer groups in the statements where they enter. The appendix describes the compact four-radius surface and the Brauer evaluations associated with its real components.

Companion material. The paper sources and a partial Lean formalization are archived at `https://doi.org/10.5281/zenodo.23638745`. The archive identifies the verified statements and the pinned build environment.

2 Integral and Gaussian forms of the four-distance equations

We label the vertices of the unit square by

$$P_0 = (0, 0), \qquad P_1 = (1, 0), \qquad P_2 = (0, 1), \qquad P_3 = (1, 1).$$

A *strict rational-distance point* is a point (x, y) with $0 < x, y < 1$ whose distances r_i to all four P_i belong to $\mathbf{Q}_{>0}$. Thus

$$\begin{aligned} r_0^2 &= x^2 + y^2, & r_1^2 &= (1-x)^2 + y^2, \\ r_2^2 &= x^2 + (1-y)^2, & r_3^2 &= (1-x)^2 + (1-y)^2. \end{aligned} \tag{1}$$

The labels in these equations will be retained throughout.

2.1 Recovering the point from its four radii

Subtracting adjacent equations gives

$$2x = 1 + r_0^2 - r_1^2, \qquad 2y = 1 + r_0^2 - r_2^2. \tag{2}$$

In particular the coordinates are rational. The following elimination keeps both the point and its four positive radii.

Proposition 2.1. *Positive rational numbers (r_0, r_1, r_2, r_3) arise from a unique strict rational-distance point if and only if*

$$r_0^2 + r_3^2 = r_1^2 + r_2^2, \tag{3}$$

$$(1 + r_0^2 - r_1^2)^2 + (1 + r_0^2 - r_2^2)^2 = 4r_0^2, \tag{4}$$

$$-1 < r_0^2 - r_1^2 < 1, \qquad -1 < r_0^2 - r_2^2 < 1. \tag{5}$$

The point is given by (2).

Proof. Equations (3) and (4) follow by adding and subtracting (1). Conversely, define x, y by (2). The inequalities give $0 < x, y < 1$, and (4) gives $x^2 + y^2 = r_0^2$. Subtracting $2x - 1$ and $2y - 1$ gives the equations for r_1, r_2; the remaining equation then follows from (3). Positivity fixes the four square roots, and (2) gives uniqueness. □

Let L be the least positive common denominator of x, y, and put

$$X = Lx, \qquad Y = Ly, \qquad A_i = Lr_i. \tag{6}$$

Then X, Y, L are integers with

$$0 < X, Y < L, \qquad \gcd(X, Y, L) = 1.$$

Each A_i is an integer: a rational number whose square is an integer is itself an integer. Equations (1) become

$$\begin{aligned} A_0^2 &= X^2 + Y^2, & A_1^2 &= (L - X)^2 + Y^2, \\ A_2^2 &= X^2 + (L - Y)^2, & A_3^2 &= (L - X)^2 + (L - Y)^2. \end{aligned} \tag{7}$$

Conversely, positive integers satisfying these equations and $0 < X, Y < L$ give a strict rational-distance point by division by L; the additional gcd condition makes L its least coordinate denominator.

2.2 Parity, symmetry lines, and the primitive denominator

We first record the elementary descent needed to deal with the symmetry lines of the square.

Lemma 2.2 (Square-area descent)**.** *A rational right triangle cannot have positive rational-square area.*

Proof. Clearing denominators would give an integer right triangle of square area. Dividing by the gcd of its sides preserves this property: the new area is an integer and a rational square, hence an integer square. Choose a primitive such triangle with least hypotenuse. Its legs and hypotenuse have the form

$$2rs, \quad r^2 - s^2, \quad r^2 + s^2, \qquad r > s > 0, \quad \gcd(r, s) = 1,$$

where r, s have opposite parity. To obtain this parametrization, write the odd leg as a and the hypotenuse as h. The coprime positive integers $(h+a)/2, (h-a)/2$ have square product and are therefore squares.

The area $rs(r-s)(r+s)$ is a square. Its four factors are pairwise coprime, so

$$r = u^2, \qquad s = v^2, \qquad r+s = w^2, \qquad r-s = z^2.$$

The numbers w, z are odd. Since $w^2 - z^2 = 2v^2$ and odd squares are 1 modulo 8, the integer v is even. The positive integers

$$a' = (w+z)/2, \qquad b' = (w-z)/2$$

satisfy

$$(a')^2 + (b')^2 = u^2, \qquad a'b'/2 = (v/2)^2.$$

They give a right triangle of square area with hypotenuse u strictly smaller than r^2+s^2. Primitive reduction, if required, decreases the hypotenuse further. This contradicts its minimal choice. □

Lemma 2.3. *A strict rational-distance point lies on neither a midline nor a diagonal of the square. Its four radii are pairwise distinct.*

Proof. On $y = 1/2$, put $u = 2x - 1$, $p = 2r_0$, and $q = 2r_1$. Then

$$p^2 = u^2 + 2u + 2, \qquad q^2 = u^2 - 2u + 2, \qquad (pq)^2 = u^4 + 4.$$

If $u \neq 0$, the rational numbers $|u|, 2/|u|, pq/|u|$ are the sides of a right triangle of area one, contrary to Lemma 2.2. If $u = 0$, the squared distance to a vertex is $1/2$, which is not a rational square. The other midline follows by exchanging coordinates. On $x = y$ the distance to P_0 is $\sqrt{2}\,x$; on $x + y = 1$ the distance to P_1 is $\sqrt{2}\,(1-x)$. These are irrational because the coordinates are rational and the indicated factors are positive. Finally, the perpendicular bisector of every pair of square vertices is one of these four lines. □

Proposition 2.4 (Primitive congruences)**.** *For every strict rational-distance point,*

$$24 \mid L.$$

After a single exchange of the coordinates, X is odd and $4 \mid Y$. All four A_i are odd and prime to 3. Every prime divisor of

$$(X-Y)(X+Y-L) \tag{8}$$

is congruent to 1 or 7 modulo 8.

Proof. If L were odd, one could choose an odd element from each of $\{X, X-L\}$ and $\{Y, Y-L\}$. The corresponding equation in (7) would have right side 2 modulo 4. Thus L is even. Primitivity excludes both X, Y even, and their squared sum excludes both odd. Exchange coordinates so that X is odd. All A_i are odd; reduction modulo 8 then gives $4 \mid Y$ and $4 \mid L - Y$, hence $4 \mid L$.

If $3 \nmid L$, each of $\{X, X-L\}$ and $\{Y, Y-L\}$ contains a unit modulo 3. The associated squared distance would be 2 modulo 3. Hence $3 \mid L$. Now exactly one of X, Y is divisible by 3: both units would again give a nonsquare sum, while both nonunits would contradict primitivity. Consequently every A_i is a 3-adic unit.

The integer in (8) is nonzero by Lemma 2.3, and both factors are odd. Let an odd prime p divide $X - Y$. If X, Y are units modulo p, then A_0^2 reduces to twice a nonzero square. Otherwise

both are zero modulo p; primitivity gives $p \nmid L$, and A_3^2 reduces to $2L^2$. In either case $\left(\frac{2}{p}\right) = 1$. If instead $p \mid X+Y-L$, then $A_1^2 \equiv 2Y^2$ and $A_2^2 \equiv 2X^2 \pmod{p}$; at least one of X, Y is a unit, again by primitivity. Thus $\left(\frac{2}{p}\right) = 1$ in this case as well. The supplementary law for 2 gives $p \equiv 1, 7 \pmod{8}$.

A nonzero integer supported on these primes, with either sign, is 1 or 7 modulo 8. Were $L \equiv 4 \pmod{8}$, our orientation would instead give

$$(X-Y)(X+Y-L) = X^2 - Y^2 - LX + LY \equiv 5 \pmod{8}.$$

This contradiction proves $8 \mid L$, and therefore $24 \mid L$. □

2.3 Common content of the integer radii

The common divisor of the four radii has a precise relation to the coordinate denominator.

Proposition 2.5. *Set*

$$G = \gcd(A_0, A_1, A_2, A_3), \qquad a_i = A_i/G.$$

Then $G^2 \mid L$. Every prime divisor of G is 1 modulo 4, and $\gcd(G, 6) = 1$. Writing

$$H = L/G^2, \tag{9}$$

one has

$$24 \mid H, \qquad r_i = \frac{a_i}{GH}, \qquad \gcd(a_0, a_1, a_2, a_3) = 1.$$

For every prime $p \mid H$, all four a_i are p-adic units.

Proof. The congruences already proved give $\gcd(G, 6) = 1$. Suppose an odd prime p divides G but not L. Subtracting the origin square from the two adjacent squares gives $X \equiv Y \equiv L/2 \pmod{p}$. Then $A_0^2 \equiv L^2/2 \neq 0 \pmod{p}$, a contradiction. Thus $p \mid L$. Primitivity and $X^2+Y^2 \equiv 0 \pmod{p}$ now force both X, Y to be units and $(Y/X)^2 \equiv -1 \pmod{p}$. In particular $p \equiv 1 \pmod{4}$.

Write $s = v_p(G)$ and $e = v_p(L)$. The identity

$$A_1^2 - A_0^2 = L(L - 2X)$$

has right side of exact valuation e, since $p \nmid X$. Its left side is divisible by p^{2s}; hence $e \geq 2s$. This proves $G^2 \mid L$ at every prime. Division of the adjacent squared differences by G^2 shows that all four a_i^2 have the same residue modulo each prime divisor of H. If one were divisible by that prime, all would be, contrary to their collective gcd. The remaining assertions follow from (9) and $24 \mid L$. □

There is also a uniform relation between the denominators of a sum and a difference of two radii. Here the denominator of a nonzero rational number always means its positive reduced denominator.

Theorem 2.6 (Complementary denominators)**.** *For any distinct i, j, let d_+ and d_- be the denominators of $r_i + r_j$ and $r_i - r_j$. Then*

$$\gcd(d_+, d_-) = G, \qquad d_+ d_- = L/2. \tag{10}$$

Proof. Both denominators divide L. Fix an odd $p \mid L$, and write $e = v_p(L)$, $s = v_p(G)$. If $s = 0$, the four A_i are units modulo p, because their squares have the same residue and at least one is a unit. Every difference $A_i^2 - A_j^2$ is divisible by p^e. Exactly one of $A_i + A_j, A_i - A_j$ is divisible by p, so their reduced denominator depths are 0 and e.

Suppose $s > 0$. The proof of Proposition 2.5 shows $Y/X \equiv \pm\sqrt{-1} \pmod{p}$. Thus $L - 2X$, $L - 2Y$, $X - Y$, and $X + Y - L$ are all units modulo p. The six squared differences are, up to sign,

$$L(L - 2X), \quad L(L - 2Y), \quad 2L(X + Y - L), \quad 2L(X - Y), \tag{11}$$

with repetitions in the parallel directions. Each has exact depth e. If $e > 2s$, the four a_i are units, and their sum and difference have depths 0 and $e - 2s$. The depths of the corresponding denominators are consequently $e - s$ and s. If $e = 2s$, the squared difference of the normalized pair is a unit, so both its sum and difference are units; both denominator depths are s. In all cases the sum of the two depths is e and their minimum is s.

At 2, write $e = v_2(L) \geq 3$ and keep X odd, $4 \mid Y$. All four A_i are odd. The horizontal and opposite squared differences in (11) have depth $e + 1$; the vertical ones have depth at least $e + 3$. Of a sum and a difference of two odd integers, precisely one has depth one. The other therefore has depth at least e. Their reduced denominator depths are $e - 1$ and 0. Combining the prime-by-prime statements proves (10). □

2.4 Gaussian roots and the four corner classes

Put $c_0 = 0$, $c_1 = 1$, $c_2 = i$, $c_3 = 1 + i$, and define the integer Gaussian displacements

$$U_j = X + iY - Lc_j.$$

For $j = (\epsilon, \eta)$ put

$$d_j = \gcd(|X - \epsilon L|, |Y - \eta L|), \qquad m_j = \mathrm{sf}(d_j),$$

where sf denotes the positive squarefree part of a positive integer. The coordinate orientation is the one in Proposition 2.4.

Theorem 2.7 (Canonical corner classes)**.** *The integers d_j are odd, pairwise coprime, and coprime to L. Every prime dividing any d_j is 1 or 7 modulo 8. There are nonzero $q_j \in \mathbf{Z}[i]$ such that*

$$U_j = m_j q_j^2, \qquad \mathrm{N}(q_j) = A_j/m_j > 0. \tag{12}$$

In particular m_0, m_1, m_2, m_3 are positive odd squarefree integers with disjoint supports, and, for $z = x + iy$,

$$[z - c_j] = [Lm_j] \quad \textit{in } \mathbf{Q}(i)^*/\mathbf{Q}(i)^{*2}. \tag{13}$$

Conversely, if a rational strict point has a primitive representation with X odd and admits (12) *with positive integers m_j and $q_j \in \mathbf{Z}[i]$, all four of its distances are rational.*

Proof. Every first coordinate $X - \epsilon L$ is odd, so all d_j are odd. A prime dividing d_j and L would divide X, Y, L. A prime dividing two different d_j would, by subtracting their coordinate pairs, divide L as well. This proves the asserted coprimalities. If $p \mid d_j$, the squared distance from the opposite vertex is $2L^2$ modulo p. Since $p \nmid L$, this implies $\left(\frac{2}{p}\right) = 1$.

Divide the two legs and hypotenuse of the jth Pythagorean triangle by d_j. This gives coprime integers a, b with a odd, b even, and $a^2 + b^2$ an integer square. The Gaussian integers $a + ib, a - ib$ are coprime: a common odd Gaussian prime would divide both a, b, while $1 + i$ does not divide

their odd norm. Unique factorization in $\mathbf{Z}[i]$ therefore makes $a+ib$ a unit times a square. Here unique factorization follows from the norm-Euclidean algorithm: rounding both coordinates of a complex quotient gives a remainder whose norm is at most half that of the divisor. The unit is not $\pm i$, because a Gaussian square of odd norm has odd real part and even imaginary part. The remaining units ± 1 are Gaussian squares. Writing $d_j = m_j t_j^2$ and absorbing t_j gives $U_j = m_j q_j^2$. Taking the positive square root of its norm proves $A_j = m_j \,\mathrm{N}(q_j)$. Finally U_j/L has squareclass m_j/L, which equals Lm_j. For the converse, taking norms in the four displayed root equations gives the four distances $m_j \,\mathrm{N}(q_j)/L \in \mathbf{Q}_{>0}$. □

The product

$$n = m_0 m_1 m_2 m_3 \tag{14}$$

is thus a positive odd squarefree integer, prime to L, supported on primes 1 and 7 modulo 8. Equation (13) records both its corner allocation and the single common factor L.

Proposition 2.8 (The common squarefree kernel)**.** *Put $t = r_0 r_3 + r_1 r_2$. Then $t > 1$ and*

$$[t^2 - 1] = [n] \quad \textit{in } \mathbf{Q}^*/\mathbf{Q}^{*2}.$$

Consequently the Gaussian corner kernel (14) *equals* $\mathrm{sf}(t^2-1)$*.*

Proof. Write $z = x + iy$, and set

$$A = z(z-1-i), \qquad B = (z-1)(z-i), \qquad a = r_0 r_3, \qquad b = r_1 r_2.$$

Then $B - A = i$, $A\overline{A} = a^2$ and $B\overline{B} = b^2$. The triangle inequality gives $a + b \geq 1$. Equality forces A and B to lie on the imaginary axis, and hence

$$\mathrm{Re}(A) = (x-y)(x+y-1) = 0.$$

Lemma 2.3 excludes equality, so $t > 1$. Expanding with the two norm identities gives

$$(bA + aB)^2 = AB\big(A\overline{B} + B\overline{A} + 2ab\big) = AB\big((a+b)^2 - |A-B|^2\big) = (t^2-1)AB.$$

Multiplication of the four Gaussian identities (12) gives $AB = n(q_0 q_1 q_2 q_3/L^2)^2$. Thus $(t^2-1)/n$ is a square in $\mathbf{Q}(i)$. A positive rational number which is a square in $\mathbf{Q}(i)$ is a square in $\mathbf{Q}$: if $(u+iv)^2$ is positive rational, then $uv = 0$ and positivity forces $v = 0$. This proves the claim. □

The common factor also occurs in the unit restrictions at ramified split primes. For $p \equiv 1 \pmod 8$, choose either square root ι of -1 in $\mathbf{F}_p$ and define

$$\epsilon(p) = \left(\frac{1+\iota}{p}\right).$$

This is independent of the choice: replacing ι by $-\iota$ multiplies $1+\iota$ by $-\iota$, a square in $\mathbf{F}_p^*$. Equivalently,

$$\epsilon(p) = 2^{(p-1)/4}(-1)^{(p-1)/8} \in \{1, -1\}, \tag{15}$$

where the power is evaluated in $\mathbf{F}_p$ and the result identified with its sign.

Proposition 2.9. *If $p \equiv 1 \pmod 8$ divides m_j, then*

$$\left(\frac{Lm_k}{p}\right) = 1$$

for either adjacent corner k, and

$$\left(\frac{Lm_{j^{\mathrm{op}}}}{p}\right) = \epsilon(p)$$

for the opposite corner j^{op}.

Proof. Use the reduction homomorphism $\mathbf{Z}[i] \longrightarrow \mathbf{F}_p$ given by $i \mapsto \iota$. From (12), $X + iY \equiv Lc_j \pmod p$, and therefore for $k \neq j$

$$m_k q_k^2 \equiv L(c_j - c_k) \pmod p.$$

All factors on the right are nonzero. Adjacent differences are ± 1 or $\pm i$, all squares modulo p. Opposite differences are these units times $1+i$. Taking quadratic characters proves the assertions. For (15), use $(1+\iota)^2 = 2\iota$ and raise to $(p-1)/4$. □

2.5 Signed radii and denominator growth

The integral equations also connect divisibility to the real size of the square. Set

$$d = 2 + \sqrt{2}.$$

For every strict point,

$$A_0 + A_1 + A_2 + A_3 < dL. \tag{16}$$

Indeed, the sum of the distances to the four vertices is a convex function on the square and equals d at each vertex. A strict point is a convex combination of all four vertices with positive weights. The triangle inequality for the distance to P_0 is strict in this combination, because the two adjacent nonzero vertex vectors are not parallel. This proves (16).

Choose arbitrary signs and write a, b, c, e for the signed versions of A_0, A_1, A_2, A_3. Put

$$S = a + b + c + e, \quad T = a - b - c + e, \quad P = ae - bc, \quad Q = ae + bc.$$

The four-distance equations give the identities

$$a^2 + e^2 = b^2 + c^2, \qquad PQ = -L^2(L - 2X)(L - 2Y), \qquad ST = 2P. \tag{17}$$

The second follows by expanding $A_0^2 A_3^2 - A_1^2 A_2^2$; the last follows from the first. Lemma 2.3 shows that the product on the right is nonzero. Thus S, T, P, Q are all nonzero, for every sign choice. Moreover

$$|T| \leq A_0 + A_1 + A_2 + A_3 < dL. \tag{18}$$

Theorem 2.10 (Prime-power comparisons)**.** *Let $K = 2^{v_2(L)}$. Then*

$$K^2 < \frac{2+\sqrt{2}}{4}\, L. \tag{19}$$

If p is an odd prime with $p \mid L$ and $p \mid XY$, and $P_p = p^{v_p(L)}$, then

$$P_p^2 < \frac{2+\sqrt{2}}{2p}\, L. \tag{20}$$

In particular

$$3^{2v_3(L)} < \frac{2+\sqrt{2}}{6}\, L. \tag{21}$$

Proof. For the dyadic assertion use X odd and $4 \mid Y$. Write $h = v_2(L) \geq 3$ and $j = v_2(L-2Y) \geq 3$; the latter is finite by Lemma 2.3. Choose the signs of a, b, c, e so that all four are 1 modulo 4. Every pair sum then has depth one. The horizontal and vertical squared differences give

$$v_2(a-b) = v_2(c-e) = h, \qquad v_2(a-c) = v_2(b-e) = h+j-1.$$

Consequently a, b, c, e are congruent modulo 2^h. It follows that

$$S \equiv 4a \pmod{2^h}, \qquad Q \equiv 2a^2 \pmod{2^h},$$

so $v_2(S) = 2$ and $v_2(Q) = 1$. Since $v_2(L-2X) = 1$, the identities (17) yield

$$v_2(P) = 2h+j, \qquad v_2(T) = 2h+j-1 \geq 2h+2.$$

The nonzero integer T therefore has $|T| \geq 4K^2$. Equation (18) proves (19).

For the odd-prime assertion, exchange the coordinates independently if needed so that $p \nmid X$ and $p \mid Y$. This is possible by primitivity and the hypothesis $p \mid XY$. The four A_i are p-adic units and their squares have the same residue. Choose their signs so that a, b, c, e have the same nonzero residue modulo p. Then $S \equiv 4a$ and $Q \equiv 2a^2 \pmod{p}$ are units. Set $h = v_p(L)$ and $j = v_p(L-2Y) \geq 1$, again finite. The same identities now give the exact depth

$$v_p(T) = 2h+j \geq 2h+1.$$

Every A_i is odd, so the signed sum T is even. Hence

$$2pP_p^2 \leq 2p^{2h+j} \leq |T| < dL,$$

which proves (20). At $p = 3$, the required divisibility $3 \mid XY$ was proved in Proposition 2.4, so (21) follows. □

Writing $L = Km$ with m odd, (19) takes the linear form

$$K < \frac{2+\sqrt{2}}{4}\, m.$$

For odd p as in the theorem, writing $L = P_p s$ similarly gives $P_p < (2+\sqrt{2})s/(2p)$. These comparisons apply to the complete prime powers in the denominator.

Corollary 2.11. *The least coordinate denominator of a strict rational-distance point has a prime divisor at least* 5*.*

Proof. If L were supported on 2 and 3, write $L = KP$ with $K = 2^{v_2(L)}$ and $P = 3^{v_3(L)}$. The preceding theorem gives

$$K^2 < dL/4, \qquad P^2 < dL/6.$$

Multiplication and $KP = L$ yield $1 < d^2/24$. But $d^2 = 6 + 4\sqrt{2} < 24$, a contradiction. □

The same signed identities give a precise statement in the other 5-adic direction, where all four radii have nontrivial content.

Proposition 2.12 (The unit-coordinate direction at 5)**.** *Suppose* $5 \mid L$ *and* $5 \nmid XY$*. Write* $L = 5^h m$ *with* $5 \nmid m$*. There is an integer* $s \geq 1$ *such that*

$$v_5(A_i) = s \quad (0 \leq i \leq 3), \qquad h > 2s.$$

Some signed sum T *of the four* A_i *has exact depth* $2h - 3s$*, and*

$$5^{h-3s} < \frac{2+\sqrt{2}}{2}\, m. \tag{22}$$

Proof. Nonzero squares modulo 5 are 1 and 4. Since X, Y are units and $X^2 + Y^2$ is a square, this sum must be zero modulo 5. Thus $Y/X \equiv \pm 2 \pmod 5$, and all four A_i are divisible by 5. Let $s = \min_i v_5(A_i)$. The adjacent squared differences have exact depth h, so $h \geq 2s$.

If $h = 2s$, division of the four squared-distance equations by 5^{2s} gives residues

$$c, \quad c+u, \quad c+v, \quad c+u+v, \qquad u, v \neq 0, \quad v/u = \pm 2$$

in $\mathbf{F}_5$. These four residues are distinct, whereas $\mathbf{F}_5$ has only three square residues. Therefore $h > 2s$. Every squared difference has depth at least h, and hence all four radii have depth exactly s.

Divide the four radii by 5^s and choose signs so that the resulting units a, b, c, e have the same residue modulo 5. Their sums $S = a + b + c + e$ and $Q = ae + bc$ are units. The determinant identity becomes

$$(ae - bc)(ae + bc) = -\frac{L^2(L - 2X)(L - 2Y)}{5^{4s}}.$$

Both $L - 2X, L - 2Y$ are units, so $ae - bc$ and $a - b - c + e$ have exact depth $2h - 4s$. Multiplying the latter by 5^s gives an actual signed sum of the integer radii with exact depth $2h - 3s$. It is nonzero and even. Its absolute value is at least $2 \cdot 5^{2h-3s}$ and less than dL, which proves (22). □

3 Reciprocity for the four corner colours

The squarefree parts of the four corner contents carry a finer reciprocity constraint than their product. We shall prove that the product belonging to either pair of opposite corners is congruent to 1 or 15 modulo 16. The proof also describes the prime divisors of all six sums of two integer distances. Its ingredients are the Gaussian square roots of the individual displacements and ordinary quadratic reciprocity.

3.1 Integer distances and Gaussian roots

Fix a strictly interior point with four positive rational distances, and use the corner order

$$c_0 = 0, \qquad c_1 = 1, \qquad c_2 = \mathrm{i}, \qquad c_3 = 1 + \mathrm{i}.$$

Write

$$x = X/L, \qquad y = Y/L, \qquad L > 0, \qquad \gcd(X, Y, L) = 1, \qquad A_j = Lr_j \in \mathbf{Z}_{>0}. \tag{23}$$

Interchanging x and y if necessary, take X odd. The integral normalization of Section 2 gives $4 \mid Y$ and $8 \mid L$. This interchange only exchanges the labels 1 and 2, so it preserves the two opposite pairs as unordered pairs.

Put

$$U_j = X + \mathrm{i}Y - Lc_j, \qquad g_j = \gcd(|\operatorname{Re} U_j|, |\operatorname{Im} U_j|), \qquad m_j = \mathrm{sf}(g_j). \tag{24}$$

The m_j are positive, odd, squarefree, pairwise coprime, and coprime to L. Every prime dividing them is 1 or 7 modulo 8. These are canonical positive integers attached to the primitive coordinate lattice.

We shall use Gaussian integers w_j satisfying

$$U_j = m_j w_j^2, \qquad N_{\mathbf{Q}(\mathrm{i})/\mathbf{Q}}(w_j) = \frac{A_j}{m_j} > 0. \tag{25}$$

For completeness, the elementary construction is as follows. The triple

$$\left(\frac{\operatorname{Re} U_j}{g_j}, \frac{\operatorname{Im} U_j}{g_j}, \frac{A_j}{g_j} \right)$$

is a primitive Pythagorean triple whose first leg is odd and whose second leg is even. If its legs and hypotenuse are denoted by a, b, k, the positive integers $(k+a)/2$ and $(k-a)/2$ are relatively prime, and their product is $(b/2)^2$. Thus they are squares, say s^2, t^2, and a choice of the sign of t gives $a + \mathrm{i}b = (s + \mathrm{i}t)^2$. Multiplying this root by the positive integer $\sqrt{g_j/m_j}$ gives w_j. Its norm is positive and its norm squared is A_j^2/m_j^2, which proves the second identity in (25).

In particular, A_j/m_j is an odd sum of two integer squares, whence

$$A_j \equiv m_j \pmod 4. \tag{26}$$

The signs in the norm identity will determine the characters at primes congruent to 7 modulo 8.

The diagonal pairs have squared separation 2, and adjacent pairs have squared separation 1. For $i \neq j$ define

$$s_{ij} = m_i m_j, \qquad d_{ij} = |c_i - c_j|^2 \in \{1, 2\}, \qquad V_{ij} = \frac{A_i + A_j}{2} \in \mathbf{Z}_{>0}. \tag{27}$$

Lemma 3.1 (The six pair identities)**.** *For every pair $i \neq j$,*

$$(A_i + A_j)^2 - 4 s_{ij} \operatorname{Re}(w_i \overline{w_j})^2 = d_{ij} L^2. \tag{28}$$

Consequently, with $\Sigma_{ij} = r_i + r_j$ and $T_{ij} = 2 \operatorname{Re}(w_i \overline{w_j})/L$, one has

$$\Sigma_{ij}^2 - s_{ij} T_{ij}^2 = d_{ij}.$$

Proof. Set $Z = w_i \overline{w_j}$ and $s = s_{ij}$. Equation (25) gives

$$N(Z) = \frac{A_i A_j}{s}, \qquad Z^2 = \frac{U_i \overline{U_j}}{s}.$$

The identity $2 \operatorname{Re}(Z)^2 = N(Z) + \operatorname{Re}(Z^2)$ therefore shows that the left side of (28) is

$$A_i^2 + A_j^2 - 2 \operatorname{Re}(U_i \overline{U_j}) = |U_i - U_j|^2 = d_{ij} L^2.$$

□

3.2 Prime divisors of all six half-sums

Theorem 3.2 (Six-pair support)**.** *Let p be an odd prime dividing V_{ij}. Then*

$$p \nmid s_{ij}, \qquad \left(\frac{-s_{ij}}{p}\right) = 1. \tag{29}$$

Equivalently, every odd prime divisor of V_{ij} is unramified and split in $\mathbf{Q}(\sqrt{-s_{ij}})$.

Proof. Write $s = s_{ij}$, $d = d_{ij}$, and $V = V_{ij}$. If $p \mid m_i$, the point reduces to corner i modulo p: $U_i \equiv 0$ and $A_i \equiv 0$. Since $p \nmid L$,

$$A_j^2 \equiv dL^2 \not\equiv 0 \pmod{p}.$$

Thus $p \nmid V$. The same argument applies to a prime dividing m_j, and proves the first assertion.

Suppose next that $p \mid L$. If $p \equiv 1 \pmod 4$, choose a square root of -1 modulo p for which $X + \mathrm{i}Y$ is nonzero. Such a choice exists: vanishing for both choices would force $p \mid X, Y, L$. In this one embedding all the U_k have the same nonzero residue, and hence all w_k have nonzero residues. Reducing

$$\left(\frac{w_i}{w_j}\right)^2 = \frac{m_j}{m_i}\frac{U_i}{U_j}$$

gives $\left(\frac{s}{p}\right) = 1$. Since $\left(\frac{-1}{p}\right) = 1$, the required conclusion follows.

If $p \equiv 3 \pmod 4$, the form $X^2 + Y^2$ is anisotropic modulo p. Primitivity therefore makes its value nonzero, and every A_k and w_k is a unit at p. Work in $\mathbf{F}_{p^2} = \mathbf{F}_p(\mathrm{i})$, where conjugation is Frobenius. For $h = w_i/w_j$ and $a = m_j/m_i \in \mathbf{F}_p^*$, we have $h^2 = a$, and consequently

$$N_{\mathbf{F}_{p^2}/\mathbf{F}_p}(h) = h^{p+1} = a^{(p+1)/2} = a\left(\frac{a}{p}\right).$$

The actual norm identity in (25) gives at the same time

$$N(h) = \frac{m_j}{m_i}\frac{A_i}{A_j}.$$

It follows that $A_i/A_j = \left(\frac{s}{p}\right)$ modulo p. The assumption $p \mid V$ makes this ratio -1, so $\left(\frac{s}{p}\right) = -1$ and $\left(\frac{-s}{p}\right) = 1$.

It remains to treat $p \nmid L$. Reducing (28), and using $p \mid V$ and $p \nmid s$, gives

$$-4s\,\mathrm{Re}(w_i\overline{w_j})^2 = dL^2.$$

All factors on the right are nonzero modulo p. For an adjacent pair, $d = 1$, this proves $\left(\frac{-s}{p}\right) = 1$. For an opposite pair the same argument proves the assertion whenever $\left(\frac{2}{p}\right) = 1$.

Finally, an opposite sum cannot vanish modulo a prime with $\left(\frac{2}{p}\right) = -1$. For the pair $(0, 3)$, the equality $A_0 = -A_3$ modulo p implies $x + y = 1$ in $\mathbf{F}_p$. The other two distance squares then become $2x^2$ and $2y^2$. At least one is nonzero, which is impossible when 2 is a nonsquare. For the pair $(1, 2)$, the same reasoning gives $x = y$ and the other two squares $2x^2$ and $2(x-1)^2$, again with at least one nonzero. This exhausts the cases. □

3.3 The characters at a corner prime

Define a character on the subgroup $\{1, 7, 9, 15\}$ of $(\mathbf{Z}/16\mathbf{Z})^*$ by

$$\sigma(1) = \sigma(15) = 1, \qquad \sigma(7) = \sigma(9) = -1. \tag{30}$$

We use the same notation for positive integers supported on primes congruent to 1 or 7 modulo 8. Multiplication of residues shows that this is multiplicative. For a prime in the support,

$$\sigma(p) = \begin{cases} (-1)^{(p-1)/8}, & p \equiv 1 \pmod 8, \\ (-1)^{(p+1)/8}, & p \equiv 7 \pmod 8. \end{cases} \tag{31}$$

Lemma 3.3 (Ramified character)**.** *For $p \mid m_i$ and $j \neq i$,*

$$\left(\frac{V_{ij}}{p}\right) = \left(\frac{m_j}{p}\right) \sigma(p)^{d_{ij}-1}. \tag{32}$$

Proof. At the corner prime p we have $A_i \equiv 0$ and

$$U_j \equiv L(c_i - c_j) \pmod p.$$

Since $\left(\frac{2}{p}\right) = 1$, the characters of V_{ij} and A_j agree. The difference $c_i - c_j$ is a Gaussian unit for adjacent corners, and a Gaussian unit times $1 + \mathrm{i}$ for opposite corners.

First suppose $p \equiv 7 \pmod 8$. In $\mathbf{F}_{p^2}$, put $u = c_i - c_j$. The two identities for w_j give

$$\frac{A_j}{m_j} = N(w_j) = \left(\frac{Lu}{m_j}\right)^{(p+1)/2},$$

and hence

$$A_j = L\left(\frac{Lm_j}{p}\right) u^{(p+1)/2}. \tag{33}$$

The exponent $(p+1)/2$ is divisible by 4. It therefore kills every Gaussian unit. In the adjacent case its final factor is 1. In the opposite case that factor is

$$z_p = (1+\mathrm{i})^{(p+1)/2} = (-1)^{(p+1)/8}\, 2^{(p+1)/4} \in \mathbf{F}_p, \qquad z_p^2 = 2.$$

Because $\left(\frac{-1}{p}\right) = -1$ and $\left(\frac{2}{p}\right) = 1$, its character is $\left(\frac{z_p}{p}\right) = \sigma(p)$. Taking characters in (33), a sign ± 1 has its own value as character, so

$$\left(\frac{A_j}{p}\right) = \left(\frac{L}{p}\right)\left(\frac{Lm_j}{p}\right) \sigma(p)^{d_{ij}-1} = \left(\frac{m_j}{p}\right) \sigma(p)^{d_{ij}-1}.$$

Now suppose $p \equiv 1 \pmod 8$ and work in a Gaussian embedding into $\mathbf{F}_p$. All four Gaussian units are squares. For an adjacent pair, $w_j^2 = Lu/m_j$ therefore gives $\left(\frac{Lm_j}{p}\right) = 1$, while $A_j^2 = L^2$ gives $\left(\frac{A_j}{p}\right) = \left(\frac{L}{p}\right) = \left(\frac{m_j}{p}\right)$.

For an opposite pair write $\lambda_p = \left(\frac{1+\mathrm{i}}{p}\right)$. This value is unchanged on replacing i by $-\mathrm{i}$, because their ratio is a Gaussian unit. The root equation gives $\left(\frac{Lm_j}{p}\right) = \lambda_p$. Choose $J \in \mathbf{F}_p$ with $J^2 = \mathrm{i}$. It has order 8, so

$$\left(\frac{J}{p}\right) = (-1)^{(p-1)/8} = \sigma(p).$$

Since $((1+\mathrm{i})/J)^2 = 2$, either square root of 2 has character $\lambda_p\sigma(p)$; changing its sign does not change that character. The equation $A_j^2 = 2L^2$ now yields

$$\left(\frac{A_j}{p}\right) = \left(\frac{L}{p}\right)\lambda_p\sigma(p) = \left(\frac{m_j}{p}\right)\sigma(p).$$

This proves the formula for both congruence classes and both separations. □

3.4 A pair reciprocity identity

For a positive integer a, let

$$\mathcal{G}(a) = \prod_{p\equiv 3\ (4)} (-1)^{v_p(a)}. \tag{34}$$

Thus $\mathcal{G}(a)$ is 1 or -1 according as the odd part of a is 1 or 3 modulo 4. Let $e_j \in \{0,1\}$ be defined by

$$m_j \equiv (-1)^{e_j} \pmod 4. \tag{35}$$

Equivalently, e_j is the parity of the number of prime factors of m_j which are 7 modulo 8.

Theorem 3.4 (Pair reciprocity)**.** *For every pair $i \neq j$, let $e \in \{0,1\}$ satisfy $e \equiv e_i + e_j \pmod 2$. Then*

$$\mathcal{G}(V_{ij})^{1+e} = (-1)^{e_ie_j}\sigma(m_im_j)^{d_{ij}-1}. \tag{36}$$

Proof. Put $s = m_im_j$ and $V = V_{ij}$. All Jacobi symbols below have coprime positive arguments, by Theorem 3.2. Multiplication of (32) over the primes of m_i and m_j gives

$$\left(\frac{V}{s}\right) = \left(\frac{m_j}{m_i}\right)\left(\frac{m_i}{m_j}\right)\sigma(s)^{d_{ij}-1} = (-1)^{e_ie_j}\sigma(s)^{d_{ij}-1}.$$

Here the second equality is quadratic reciprocity; a Jacobi symbol with denominator 1 is interpreted as 1.

Let V_{odd} be the odd part of V. The six-pair support theorem gives, for every odd $p \mid V$,

$$\left(\frac{s}{p}\right) = \left(\frac{-1}{p}\right).$$

Multiplication with valuations therefore gives

$$\left(\frac{s}{V_{\mathrm{odd}}}\right) = \mathcal{G}(V).$$

Also $\left(\frac{2}{s}\right) = 1$, since every prime factor of s is 1 or 7 modulo 8. A second application of quadratic reciprocity yields

$$\left(\frac{V}{s}\right) = \left(\frac{V_{\mathrm{odd}}}{s}\right) = \left(\frac{s}{V_{\mathrm{odd}}}\right)\mathcal{G}(V)^e = \mathcal{G}(V)^{1+e}.$$

Equating the two expressions proves the result. □

3.5 The two opposite products

Theorem 3.5 (Opposite-colour congruences)**.** *For every strictly interior point with four positive rational distances, the canonical corner colours in* (24) *satisfy*

$$\boxed{m_0m_3 \equiv 1 \text{ or } 15 \pmod{16}, \qquad m_1m_2 \equiv 1 \text{ or } 15 \pmod{16}.} \tag{37}$$

More precisely, an opposite product with an even number of prime factors congruent to 7 *modulo* 8 *is* 1 *modulo* 16*; one with an odd number is* 15 *modulo* 16*.*

Proof. Fix an opposite pair i, j and put $s = m_im_j$ and $V = V_{ij}$. Here $d_{ij} = 2$. If $e_i \neq e_j$, the left side of (36) is $\mathcal{G}(V)^2 = 1$, and $e_ie_j = 0$. Hence $\sigma(s) = 1$. Since $s \equiv 7 \pmod 8$ in this case, the conclusion is $s \equiv 15 \pmod{16}$.

Suppose instead that $e_i = e_j$. By (26), the integers A_i, A_j have the same residue $(-1)^{e_i}$ modulo 4. Thus V is odd and their half-difference $D = (A_i - A_j)/2$ is even. The squared-distance identities give the exact formulas

$$A_0^2 - A_3^2 = 2L(X + Y - L), \qquad A_2^2 - A_1^2 = 2L(X - Y).$$

The factors $X + Y - L$ and $X - Y$ are odd. Consequently, in either opposite pair,

$$v_2(VD) = v_2(L) - 1 \geq 2.$$

Since V is odd, $4 \mid D$. It follows that

$$V \equiv A_i \equiv (-1)^{e_i} \pmod 4, \qquad \mathcal{G}(V) = (-1)^{e_i}.$$

Now $e = 0$ and $e_ie_j = e_i$, so the master identity reads

$$(-1)^{e_i} = (-1)^{e_i}\sigma(s).$$

Therefore $\sigma(s) = 1$. As $s \equiv 1 \pmod 8$, this gives $s \equiv 1 \pmod{16}$. □

Write $b = m_0m_3$, $c = m_1m_2$ and $n = bc$. The allowable residue patterns take the particularly simple form

$n \pmod{16}$	$(b, c) \pmod{16}$
1	$(1, 1),\ (15, 15)$
15	$(1, 15),\ (15, 1)$.

(38)

Corollary 3.6. *If every prime dividing* $m_0m_1m_2m_3$ *is* 1 *modulo* 8*, then*

$$m_0m_3 \equiv m_1m_2 \equiv 1 \pmod{16}.$$

Proof. Every e_j is zero, so both opposite products fall in the even case of Theorem 3.5. □

For adjacent pairs the exponent $d_{ij} - 1$ is zero, and the same reciprocity theorem gives

$$\mathcal{G}(V_{ij})^{1+e} = (-1)^{e_ie_j}.$$

In particular, if two adjacent colours have the same residue modulo 4, the odd part of their half-sum has that same residue modulo 4. Together with Theorem 3.2, this supplies simultaneous arithmetic information for every pair of corners.

The support conclusion also has a primitive-radius formulation. Let $G = \gcd(A_0, A_1, A_2, A_3)$ and write $M_j = A_j/G$. Since all A_j are odd, G and the M_j are odd, and

$$\frac{A_i + A_j}{2} = G\frac{M_i + M_j}{2}.$$

Every odd prime dividing $(M_i + M_j)/2$ consequently satisfies (29). This formulation uses the same canonical colours (24) and removes the common content of the four integer radii.

4 Critical radius content and an elliptic residue curve

The common content of the four integral radii produces a finite-field curve independent of the original denominator. Its elliptic structure gives a classification at every content prime. A second elliptic curve records the square class of the opposite-product parameter.

4.1 Critical reduction

Write the primitive integral equations as

$$\begin{aligned} A_0^2 &= X^2 + Y^2, & A_1^2 &= (X-L)^2 + Y^2, \\ A_2^2 &= X^2 + (Y-L)^2, & A_3^2 &= (X-L)^2 + (Y-L)^2, \end{aligned} \tag{39}$$

with corner order $00, 10, 01, 11$, positive radii, $0 < X, Y < L$, and $\gcd(X, Y, L) = 1$. Thus $r_j = A_j/L$. Put

$$G = \gcd_j A_j, \qquad H = L/G^2, \qquad a_j = A_j/G.$$

Recall from Proposition 2.5 that $G^2 \mid L$ and $p \mid G \Rightarrow p \equiv 1 \pmod 4$. The local argument also fixes the valuations needed here. If an odd $p \mid G$ did not divide L, subtraction would give $X = Y = L/2 \pmod p$, contrary to $X^2 + Y^2 = 0$. Hence $p \mid L$, and primitivity makes X, Y units. Consequently $(Y/X)^2 = -1 \pmod p$ and

$$v_p(A_1^2 - A_0^2) = v_p(L(L-2X)) = v_p(L) \geq 2v_p(G).$$

The prime 2 cannot divide G, since the equations modulo four would then force X, Y, L all even.

Definition 4.1. A prime $p \mid G$ is *critical* if $v_p(L) = 2v_p(G)$.

At a critical prime set $s = v_p(G) > 0$ and, in $\mathbf{F}_p$,

$$\iota = Y/X, \qquad \iota^2 = -1, \qquad b = -2HX \neq 0, \qquad c = a_0^2.$$

The four normalized radii satisfy

$$(a_0^2, a_1^2, a_2^2, a_3^2) = \big(c, c+b, c+\iota b, c+(1+\iota)b\big). \tag{40}$$

The shifts are distinct, so at most one normalized radius vanishes. At a noncritical content prime, $p \mid H$; all a_j^2 are then equal modulo p, and $\gcd_j a_j = 1$ makes every a_j a unit.

4.2 The fixed elliptic curve

Over a field k of characteristic different from two containing $\iota^2 = -1$, define

$$\mathcal{C}_\iota : \quad \begin{cases} V_2^2 - V_0^2 = \iota(V_1^2 - V_0^2), \\ V_3^2 - V_0^2 = (1+\iota)(V_1^2 - V_0^2) \end{cases} \subset \mathbf{P}^3_k. \tag{41}$$

Write $b = V_1^2 - V_0^2$, and let U_ι be the open subset $bV_0V_1V_2V_3 \neq 0$.

Proposition 4.2. *The curve $\mathcal{C}_\iota$ is smooth, geometrically connected, and k-isomorphic to the smooth projective curve $E : y^2 = x^3 - x$. Its eight points with $b = 0$ are $[1 : \epsilon_1 : \epsilon_2 : \epsilon_3]$, with $\epsilon_j = \pm 1$.*

Proof. The parameter $T = V_0^2/b$ expresses its function field as

$$k(T)\left(\sqrt{\frac{T+1}{T}}, \sqrt{\frac{T+\iota}{T}}, \sqrt{\frac{T+1+\iota}{T}}\right).$$

The three square classes are geometrically independent, by their respective odd valuations at $-1, -\iota, -1-\iota$. The degree-eight cover has four branch values $0, -1, -\iota, -1-\iota$, each with four points of ramification index two, and is unramified at infinity. Riemann–Hurwitz gives $2g-2 = -16+16 = 0$. The diagonal pencil of the two quadrics has coefficients

$$(\iota-1)\lambda + \iota\mu, \quad -\iota\lambda - (1+\iota)\mu, \quad \lambda, \quad \mu.$$

Their zero positions are distinct. A singular point would therefore have at most one nonzero coordinate, and no such point satisfies the two equations. Thus the displayed intersection is smooth.

Even coordinate-sign changes, modulo simultaneous sign change, give a degree-four quotient

$$W^2 = T(T+1)(T+\iota)(T+1+\iota), \qquad W = V_0V_1V_2V_3/b^2. \tag{42}$$

A nonidentity element has a projective fixed point only if two coordinates vanish, which is impossible. After choosing a rational origin, each such involution is translation by nonzero two-torsion. Indeed an automorphism $P \mapsto \alpha(P) + Q$ with $\alpha \neq 1$ has a fixed point because $1-\alpha$ is surjective. The quotient group is therefore the entire two-torsion translation group. Factoring multiplication by two shows that the quotient is k-isomorphic to $\mathcal{C}_\iota$.

In (42), set $R = (1+\iota)T + \iota$, $S = 2W$. Then $S^2 = 1 - R^4$, and

$$X_E = \frac{2(1+S)}{R^2}, \quad Y_E = \frac{4(1+S)}{R^3}, \qquad R = \frac{2X_E}{Y_E}, \quad S = \frac{X_E^2 - 4}{X_E^2 + 4}$$

identify its function field with $Y_E^2 = X_E^3 + 4X_E$. Scaling by $x = X_E/(1+\iota)^2$, $y = Y_E/(1+\iota)^3$, and using $(1+\iota)^4 = -4$, gives $y^2 = x^3 - x$. The identifications extend to the smooth projective curves. □

For $p \equiv 1 \pmod 4$, write χ_p for the quadratic character, with $\chi_p(0) = 0$, and put

$$\delta_p = \begin{cases} 1, & \chi_p(\iota) = \chi_p(1+\iota) = 1, \\ 0, & \text{otherwise.} \end{cases}$$

The zero-coordinate fibres are disjoint; each has four points when $\delta_p = 1$, and none otherwise. For example $V_0 = 0, V_1 = 1$ gives $V_2^2 = \iota, V_3^2 = 1+\iota$. Dividing by a nonzero coordinate in each other fibre gives the same criterion. Therefore

$$\#U_\iota(\mathbf{F}_p) = \#E(\mathbf{F}_p) - 8 - 16\delta_p. \tag{43}$$

Each admissible $T = c/b$ gives eight projective points. Equivalently, $(\#E(\mathbf{F}_p) - 8 - 16\delta_p)/8$ counts the T for which $T, T+1, T+\iota, T+1+\iota$ are nonzero and have a common quadratic character. The square class of b supplies either common character.

4.3 The complete prime classification

Theorem 4.3. *For a prime $p \equiv 1 \pmod 4$, the following are equivalent:*

1. *$p \notin \{5, 13\}$.*
2. *A pattern (40) exists with $b \neq 0$ and all four entries nonzero squares in $\mathbf{F}_p$.*
3. *For every $s \geq 1$, there are infinitely many primitive triples (X, Y, L), with $0 < X, Y < L$ and $v_p(L) = 2s$, such that all four radicands in (39) have roots in $\mathbf{Q}_p$ of valuation exactly s.*

At 5 and 13 no critical pattern exists even with zero entries. In particular, for a rational four-distance point,

$$p \in \{5, 13\}, \quad p \mid G \quad \Longrightarrow \quad p \mid H.$$

Proof. Hasse's bound $\#E(\mathbf{F}_p) \geq p + 1 - 2\sqrt{p}$ [13] exceeds 24 for $p \geq 37$. Equation (43) then gives a point in U_ι. The remaining existence cases are

p	ι	(V_0, V_1, V_2, V_3)	b	c
17	4	$(1, 2, 8, 4)$	3	1
29	12	$(2, 8, 12, 1)$	2	4.

Interchanging V_1, V_2 replaces ι by $\iota^{-1} = -\iota$, covering the other root of minus one.

For the exclusions, division by a nonzero square reduces b to one of its two square classes. The square sets including zero are

$$S_5 = \{0, 1, 4\}, \qquad S_{13} = \{0, 1, 3, 4, 9, 10, 12\}.$$

All possibilities after the first two square tests are listed below:

p	ι	b	$c\ ;\ (c + \iota b, c + (1 + \iota)b)$
5	2	1	$0; (2, 3) \quad 4; (1, 2)$
5	2	2	$4; (3, 0)$
13	5	1	$0; (5, 6) \quad 3; (8, 9) \quad 9; (1, 2) \quad 12; (4, 5)$
13	5	2	$1; (11, 0) \quad 10; (7, 9) \quad 12; (9, 11).$

Every pair contains a nonsquare.

To realize an allowed signed unit pattern (V_0, V_1, V_2, V_3), put $c = V_0^2, b = V_1^2 - V_0^2$, and fix $s \geq 1$. Choose positive integers X, Y with

$$X \equiv -b/2 \pmod p, \qquad Y \equiv \tilde{\iota} X + p^{2s} z \pmod{p^{2s+1}}, \qquad z \equiv \frac{c}{2\iota X} \pmod p, \tag{44}$$

where Hensel's lemma supplies $\tilde{\iota}^2 \equiv -1 \pmod{p^{2s+1}}$. For sufficiently large positive integers k, put

$$M = 1 + pXYk, \qquad L = p^{2s} M. \tag{45}$$

Then $L > X, Y$, $v_p(L) = 2s$, and $\gcd(X, Y, L) = 1$, since $p \nmid XY$ and $M \equiv 1 \pmod{XY}$. Expanding (44) gives $(X^2 + Y^2)/p^{2s} \equiv c \pmod p$. Subtracting $2LX$ or $2LY$, and adding L^2, gives the other three reductions in (40). Hensel's lemma now supplies roots $A_j = p^s v_j$, with $v_j \in \mathbf{Z}_p^\times$ and $v_j \equiv V_j \pmod p$, of the four original radicands. Varying k gives infinitely many coordinate pairs. Conversely, roots with these valuations reduce to a critical pattern, to which the exclusions at 5 and 13 apply. □

Corollary 4.4. *Let $\mathcal{P}$ be a finite set of primes congruent to one modulo four, excluding 5 and 13, and prescribe $s_p \geq 1$. There are infinitely many primitive triples (X, Y, L), with $0 < X, Y < L$, for which $v_p(L) = 2s_p$ and all four original radicands have roots of valuation s_p in $\mathbf{Q}_p$, simultaneously for every $p \in \mathcal{P}$.*

Proof. Choose X, Y by the Chinese remainder theorem using (44) at the selected primes. If $D = \gcd(X, Y)$, no selected prime divides D. The congruences

$$L \equiv p^{2s_p} \pmod{p^{2s_p+1}} \quad (p \in \mathcal{P}), \qquad L \equiv 1 \pmod{D}$$

are compatible. Arbitrarily large solutions give the strict inequalities and primitivity. The prescribed unit-square reductions persist, so Hensel's lemma supplies the roots. □

4.4 The opposite product and the prime 17

Define

$$t = r_0 r_3 + r_1 r_2 = \frac{Q}{G^2 H^2}, \qquad Q = a_0 a_3 + a_1 a_2, \quad Q' = a_0 a_3 - a_1 a_2. \tag{46}$$

At a critical prime, multiplication in $\mathbf{F}_p$ gives

$$QQ' = c(c + (1+\iota)b) - (c+b)(c+\iota b) = -\iota b^2 \neq 0. \tag{47}$$

Thus both products are units, even with a zero radius reduction. For $s = v_p(G)$, the denominator unit in (46) is a square, and hence

$$v_p(t) = -2s, \qquad [p^{2s}t] = [Q] \text{ in } \mathbf{Q}_p^\times/\mathbf{Q}_p^{\times 2}, \qquad \frac{t+1}{t} \in \mathbf{Q}_p^{\times 2}. \tag{48}$$

The last assertion follows from $(t+1)/t \in 1 + p^{2s}\mathbf{Z}_p$.

Theorem 4.5. *If 17 is a critical content prime of a rational four-distance point, then none of $t, 2t, t+1, 2(t+1)$ is a square in $\mathbf{Q}_{17}$. Therefore, if any one is a rational square, $17 \mid G$ implies $17 \mid H$. At a critical 17, with $s = v_{17}(G)$, one also has*

$$a_0^2 + a_3^2 \equiv a_1^2 + a_2^2 \equiv 0 \pmod{17}, \qquad 17^{2s+1} \mid (2X - L)^2 + (2Y - L)^2.$$

Proof. Take $\iota = 4$. The square set is $S_{17} = \{0, 1, 2, 4, 8, 9, 13, 15, 16\}$, and the scaling classes of b are represented by 1 and 3. Successive square tests give

b	$c, c+b \in S_{17}$	also $c+4b \in S_{17}$	also $c+5b \in S_{17}$
1	$\{0, 1, 8, 15, 16\}$	$\{0, 15\}$	$\varnothing$
3	$\{1, 13, 15, 16\}$	$\{1, 13\}$	$\{1\}$.

Thus all coordinates are nonzero, and division by a_0 gives $[1 : \pm 2 : \pm 8 : \pm 4]$. Hence $Q/a_0^2 \in \{3, 5, 12, 14\}$, all nonsquares modulo 17. Interchanging the middle coordinates handles $\iota = -4$. Since 2 and $(t+1)/t$ are local squares, (48) gives the four exclusions.

The same table yields $c/b = -(1+\iota)/2$, so $c = H(X+Y)$ and $2c + (1+\iota)b = 0$. The identity

$$\frac{(2X - L)^2 + (2Y - L)^2}{G^2} = 4a_0^2 - 4H(X+Y) + 2G^2H^2 \tag{49}$$

proves the divisibility. At a noncritical 17, its right side is a unit, since $17 \mid H$ and a_0 is a unit, so the corresponding valuation is exactly $2s$. □

4.5 The normalized product-square cover

On $\mathcal{C}_\iota$ put $f = V_0V_3 + V_1V_2$, $f' = V_0V_3 - V_1V_2$, so that $ff' = -\iota b^2$. Let $\widetilde{\mathcal{C}}_\iota$ be the smooth projective curve obtained by adjoining

$$S^2 = 2f/V_0^2. \tag{50}$$

On $V_j \neq 0$, the same extension is $S_j^2 = 2f/V_j^2$.

Proposition 4.6. *The map $\widetilde{\mathcal{C}}_\iota \to \mathcal{C}_\iota$ is a geometrically connected unramified double cover of genus-one curves with rational points. For $p \equiv 1 \pmod 4$, its number of rational points above $b = 0$ is*

$$B_p = \begin{cases} 16, & p \equiv 1 \pmod 8, \\ 8, & p \equiv 5 \pmod 8. \end{cases}$$

Moreover,

$$\#\{V \in \mathcal{C}_\iota(\mathbf{F}_p) : b(V) \neq 0,\ \chi_p(2f(V)) = 1\} = \frac{\#E(\mathbf{F}_p) - B_p}{2}. \tag{51}$$

The character is unchanged by rescaling projective coordinates.

Proof. At $b = 0$, the four zeros of f are $[1 : \epsilon_1 : \epsilon_2 : -\epsilon_1\epsilon_2]$. There $f' \neq 0$, and b/V_0^2 is a uniformizer, because the map $T = V_0^2/b$ is unramified at infinity. Thus $ff' = -\iota b^2$ makes every zero of f/V_0^2 have order two. On $V_0 = 0$, one has $f = V_1V_2 \neq 0$, so its poles also have even order, proving that the cover is geometrically unramified.

If the extension were geometrically trivial, its square root times V_0 would be a regular global section of $\mathcal{O}_{\mathcal{C}_\iota}(1)$ vanishing precisely at the four displayed points. This degree-four line bundle has four sections by Riemann–Roch; the four independent coordinate sections span them. But the coordinate matrix of these points has determinant ± 16, so no linear form vanishes at all four. The cover is connected. Riemann–Hurwitz gives genus one, and $S = \pm 2$ above $[1 : 1 : 1 : 1]$ gives rational points. With compatible origins the map is a degree-two isogeny.

At the four points with $b = 0, f \neq 0$, the values $2f/V_0^2 = \pm 4$ give eight rational preimages. At each remaining point, the normalized local equation has residue

$$\left(\frac{S}{b/V_0^2}\right)^2 = \frac{-2\iota V_0^2}{f'} = \pm\iota.$$

It gives two rational points precisely when ι is square, equivalently $p \equiv 1 \pmod 8$. This proves the boundary count. Isogenous elliptic curves over a finite field have equal point counts. Apply this to the double cover and Proposition 4.2. For $b \neq 0$, one has $f \neq 0$, and each square value has exactly two preimages, proving (51). □

Theorem 4.7. *For a prime $p \equiv 1 \pmod 4$, the following are equivalent:*

1. *$p \notin \{5, 13, 17\}$.*
2. *There is a signed critical pattern with every $V_j \neq 0$ and $2(V_0V_3 + V_1V_2) \in \mathbf{F}_p^{\times 2}$.*
3. *For every $s \geq 1$, there are infinitely many primitive strictly interior rational coordinate pairs $(X/L, Y/L)$, with $v_p(L) = 2s$, and roots $A_j \in \mathbf{Q}_p$ of (39), each of valuation s, such that $t = (A_0A_3 + A_1A_2)/L^2$ satisfies $2t, 2(t+1) \in \mathbf{Q}_p^{\times 2}$.*

Proof. If $p \equiv 5 \pmod 8$, zero-coordinate base points are impossible because ι is nonsquare. For every such prime $p \geq 29$, Hasse's lower bound on the product cover exceeds 16 and therefore its boundary count, giving the required unit point.

If $p \equiv 1 \pmod 8$, the base has at most sixteen geometric zero-coordinate points, giving at most 32 points on the double cover. Together with its sixteen points above $b = 0$, at most 48 points are excluded. For $p \geq 73$, Hasse's lower bound exceeds 48. Below 73 the primes in this class are 17 and 41. At 41 take $\iota = 9$ and $(V_0, V_1, V_2, V_3) = (6, 18, 2, 13)$. The squared coordinates are $(36, 37, 4, 5)$, so $b = 1$, $Q = 32$, and $2Q = 23 = 8^2 \pmod{41}$. Swapping V_1, V_2 preserves f and handles the other choice of ι. The exclusions follow from Theorems 4.3 and 4.5, including every root sign.

For local realization repeat (44)–(45) with the prescribed signed pattern. Hensel's lemma gives $A_j = p^s v_j$, $v_j \equiv V_j \pmod p$. The product is

$$t = \frac{v_0 v_3 + v_1 v_2}{p^{2s} M^2}.$$

Twice the numerator is a unit square, so $2t$ is square. The ratio $(t+1)/t \in 1 + p^{2s}\mathbf{Z}_p$ supplies the second root. Conversely, reduction gives an admissible signed pattern. □

4.6 Excess radius valuations

Corollary 4.8. *Suppose at least one of $t, 2t, t+1, 2(t+1)$ is a rational square. If $p \mid G$ and $v_p(A_j) > v_p(G)$ for some corner, then*

$$v_p(L) = 2v_p(G), \qquad p \equiv 1 \pmod{16}, \qquad 2^{(p-1)/4} \equiv 1 \pmod p.$$

Proof. At a noncritical prime all normalized radii are units. Hence p is critical. A square symmetry moves the unique zero to corner zero and preserves the opposite-product sum. Dividing by a_1 gives

$$a_0 = 0, \quad a_1 = 1, \quad a_2^2 = \iota, \quad a_3^2 = 1 + \iota, \quad Q = a_2.$$

Thus $p \equiv 1 \pmod 8$, and 2 is square in $\mathbf{Q}_p$. By (48), any of the four square hypotheses forces $\chi_p(Q) = 1$. But

$$\chi_p(a_2) = \iota^{(p-1)/4} = (-1)^{(p-1)/8},$$

so $p \equiv 1 \pmod{16}$. Finally $(1+\iota)^2 = 2\iota$ gives

$$1 = \chi_p(1+\iota) = 2^{(p-1)/4}(-1)^{(p-1)/8} = 2^{(p-1)/4}.$$

□

5 Ordered elliptic sources and fixed-distance fibres

We label the vertices $00, 10, 01, 11$ by $0, 1, 2, 3$, respectively. The distance surface is given by

$$r_0^2 = x^2 + y^2, \quad r_1^2 = (x-1)^2 + y^2, \quad r_2^2 = x^2 + (y-1)^2, \quad r_3^2 = (x-1)^2 + (y-1)^2. \tag{52}$$

Throughout, a strict rational point has $0 < x, y < 1$ and $r_0, r_1, r_2, r_3 \in \mathbf{Q}_{>0}$. This section gives two descriptions of the complete source of such points: an ordered pair on a congruent-number elliptic curve, with one final square condition, and a genus-five curve obtained by fixing a signed pair of distances.

5.1 Diagonal sums and a simultaneous square model

Set

$$s = r_0 + r_3, \qquad d = r_0 - r_3, \qquad \tau = r_1 + r_2, \qquad e = r_2 - r_1.$$

Subtracting the equations in (52) gives

$$sd = 2(x + y - 1), \qquad \tau e = 2(x - y).$$

The four equations are equivalently encoded by

$$s^2 + d^2 = \tau^2 + e^2, \qquad (s^2 - 2)(2 - d^2) = \tau^2 e^2, \tag{53}$$

together with the return formulas

$$\begin{aligned} (r_0, r_1, r_2, r_3) &= \tfrac{1}{2}(s + d, \tau - e, \tau + e, s - d), \\ x &= \tfrac{1}{2} + \tfrac{1}{4}(sd + \tau e), \qquad y = \tfrac{1}{2} + \tfrac{1}{4}(sd - \tau e). \end{aligned} \tag{54}$$

For example, eight times the difference between the first two sides of $x^2 + y^2 = r_0^2$ is $-(s^2 - 2)(2 - d^2) + \tau^2 e^2$; the other three equations follow by subtraction and the first equation of (53).

Lemma 5.1. *Neither diagonal of the square contains a strict rational four-distance point. The square symmetries let us choose the chamber*

$$x + y > 1, \qquad x > y. \tag{55}$$

Proof. On $x = y$ one has $r_0 = \sqrt{2}x$, and on $x + y = 1$ one has $r_1 = \sqrt{2}(1 - x)$. The square symmetries act on $(x + y - 1, x - y)$ by signed permutations. □

In (55) we have

$$s > d > 0, \qquad \tau > e > 0, \qquad s, \tau > \sqrt{2}.$$

The strict triangle inequalities give $d^2, e^2 < 2$. In fact

$$d^2 + e^2 \le 2\big((x + y - 1)^2 + (x - y)^2\big) = 4\big((x - \tfrac{1}{2})^2 + (y - \tfrac{1}{2})^2\big) < 2. \tag{56}$$

Define

$$p = \frac{2 - e^2}{d^2}, \quad \beta = \frac{s^2}{2 - e^2}, \quad U = \frac{s}{d}, \quad V = \frac{2e}{d}, \quad W = \frac{2\tau}{d}, \quad T = \frac{2}{d}. \tag{57}$$

Here $p, \beta > 1$, and (53) becomes

$$\begin{aligned} U^2 &= p\beta, \qquad V^2 = 2(p - 1)(\beta - 1), \\ W^2 &= 2(p + 1)(\beta + 1), \qquad T^2 = (p - 1)\beta + p + 1. \end{aligned} \tag{58}$$

Conversely these four equations give

$$\begin{aligned} (r_0, r_1, r_2, r_3) &= \left(\frac{U + 1}{T}, \frac{W - V}{2T}, \frac{W + V}{2T}, \frac{U - 1}{T}\right), \\ x &= \frac{1}{2} + \frac{U + WV/4}{T^2}, \qquad y = \frac{1}{2} + \frac{U - WV/4}{T^2}. \end{aligned} \tag{59}$$

Thus all four square roots remain attached to the same p, β.

5.2 The complete ordered elliptic source

For a positive squarefree integer n, let

$$E_n: \quad Y^2 = X(X-n)(X+n).$$

An ordered source consists of two points $Q=(a,b)$ and $R=(c,\ell)$ on E_n, and three rational roots u,v,h satisfying

$$u^2 = -\frac{ac}{n}, \qquad v^2 = -\frac{(a-n)(c-n)}{n}, \qquad h^2 = \frac{a+c}{2(c+n)}. \tag{60}$$

The third product root is fixed by the ordinates:

$$w = \frac{b\ell}{nuv}, \qquad w^2 = (a+n)(c+n). \tag{61}$$

In particular the first two equations specify the ordered square classes

$$[ac] = [-n], \qquad [(a-n)(c-n)] = [-n] \quad \text{in } \mathbf{Q}^*/\mathbf{Q}^{*2}.$$

The condition on h is a further square condition on this same pair.

Theorem 5.2 (Exact ordered-source correspondence)**.** *Strict rational points in the chamber* (55) *are in bijection with the union, over positive squarefree n, of the rational solutions of* (60) *for which*

$$-n < a < 0 < c-n, \qquad b,\ell,u,v,h > 0, \tag{62}$$

and

$$\frac{2u}{a+c}\left(-\frac{b}{a}+\frac{\ell}{c}\right) < 2. \tag{63}$$

For this correspondence the final square is

$$f(Q,R) := \frac{a+c}{2(c+n)} = \frac{1}{(r_0+r_3)^2}. \tag{64}$$

Proof. We first give rational maps that preserve the actual roots and ordinates. From an ordered source set

$$\begin{aligned} &p = -\frac{n}{a}, && k = \frac{b}{a^2}, && \beta = \frac{c+n}{c-n}, \\ &U = \frac{(n-a)w}{bv}, && V = -\frac{2b}{av}, && W = \frac{2(a-n)u}{av}, \quad T = 2Uh. \end{aligned} \tag{65}$$

The curve equations and (60) imply (58) and

$$nk^2 = p(p^2-1). \tag{66}$$

For instance, $k^2 = b^2/a^4 = (a^2-n^2)/a^3$ gives (66); the other identities follow by cancelling the factors $a, a\pm n, c, c\pm n$. Conversely, from (58) and (66) put

$$\begin{aligned} &a = -\frac{n}{p}, && b = ka^2, && c = n\frac{\beta+1}{\beta-1}, \quad \ell = \frac{n}{k}\frac{UVW}{(\beta-1)^2}, \\ &u = \frac{(p-1)W}{kV}, && v = \frac{2(p^2-1)}{kV}, && h = \frac{T}{2U}. \end{aligned} \tag{67}$$

These formulas give, separately,

$$b^2 = \frac{n^3(p^2-1)}{p^3} = a(a-n)(a+n),$$
$$\ell^2 = \frac{4n^3\beta(\beta+1)}{(\beta-1)^3} = c(c-n)(c+n),$$
$$u^2 = \frac{n(\beta+1)}{p(\beta-1)} = -\frac{ac}{n},$$
$$v^2 = \frac{2n(p+1)}{p(\beta-1)} = -\frac{(a-n)(c-n)}{n},$$
$$h^2 = \frac{(p-1)\beta+p+1}{4p\beta} = \frac{a+c}{2(c+n)}.$$

Inserting (67) into (65) returns U, V, W, T with their signs. Conversely it returns b and ℓ as well as a, c, u, v, h.

To obtain n and k from a strict original point, set

$$t = r_0 r_3 + r_1 r_2 = \frac{s^2 - e^2}{2}, \quad J = d^2 + e^2 - 2, \quad K = 2 - d^2,$$

and choose the unique positive squarefree integer n and positive $q \in \mathbf{Q}$ with $t^2 - 1 = nq^2$. This is possible because (53) gives

$$t^2 - 1 = \left(\frac{de}{J}\right)^2 K(2-e^2) > 0. \tag{68}$$

For completeness, writing $D = d^2$ and $E = e^2$, the two equations of (53) give

$$s^2 = -\frac{(2-E)(2+E-D)}{D+E-2}, \qquad \tau^2 = -\frac{(2-D)(2+D-E)}{D+E-2}.$$

These expressions prove (68) and the identity

$$\frac{p(p^2-1)}{t^2-1} = \left(\frac{\tau J^2}{d^4 eK}\right)^2.$$

Consequently

$$k = \frac{q\tau J^2}{d^4 eK} > 0$$

satisfies (66). Equations (57) and (67) now give the required source. All their denominators are nonzero by (56), $d, e > 0$, and $p, \beta > 1$.

On the source, direct use of (65) gives

$$q = \frac{2u}{a+c}, \qquad t = \frac{c-a}{c+a}, \qquad \frac{WV}{T^2} = -\frac{qb}{a}, \qquad \frac{4U}{T^2} = \frac{q\ell}{c}. \tag{69}$$

Thus (63) is precisely $(4U + WV)/T^2 < 2$. The signs in (62) give $U > 1$, $W > V > 0$ and $T > 0$. All four radii in (59) are positive. That formula shows that the metric inequality is equivalent to $x < 1$ in the chamber; it also gives $x > 1/2$ and $|y - 1/2| < 1/2$. Finally $r_0 + r_3 = 2U/T = 1/h$ proves (64). The displayed inverse formulas and the uniqueness of $n, q > 0$ prove bijectivity. $\square$

The formulas above also describe the signed algebraic covers. Let S denote the surface (52), and let X_n be the normalization of $E_n \times E_n$ in the field obtained by adjoining u, v, h from (60).

Proposition 5.3. *There is an equality of function fields*

$$\mathbf{Q}(X_n) = \mathbf{Q}(S)(q), \qquad q^2 = \frac{t^2-1}{n}, \qquad t = r_0 r_3 + r_1 r_2.$$

The extension has degree two.

Proof. The four radicands in (52) have independent square classes over $\overline{\mathbf{Q}}(x, y)$: each factors into two lines, and each of these eight lines occurs for only one corner. Hence S is geometrically integral. The preceding maps are rational identities before signs are chosen. Their inverse recovers q from the simultaneous-square coordinates by

$$q = \frac{kVW}{(p^2-1)T^2}.$$

The two root classes defining u, v are geometrically independent: their half-divisor classes on $E_n \times E_n$ restrict on either factor to the independent classes $[(0,0) - O]$ and $[(n,0) - O]$ in $E_n[2]$. Their degree-four cover is therefore connected and unramified. The zero divisor $a + c = 0$ has simple generic multiplicity, which remains odd after this unramified pullback. Adjoining h is therefore ramified there, so this last quadratic extension is also connected. These statements hold over $\overline{\mathbf{Q}}$. The resulting involution

$$(a, b, c, \ell, u, v, h) \longmapsto (a, -b, c, -\ell, -u, -v, h)$$

fixes U, V, W, T, hence fixes S, and sends q to $-q$. It is nontrivial, proving the degree assertion. The normalization of the graph gives the corresponding birational identification of proper models. □

5.3 Fixing an opposite pair of distances

Let $c_0 = \mu/\nu$ be a fixed positive rational number, with coprime positive integers μ, ν. Suppose

$$c_0 = r_3 + \varepsilon r_0, \qquad d_0 = r_3 - \varepsilon r_0, \qquad \varepsilon \in \{1, -1\};$$

for a difference we orient the pair so that $r_3 > r_0$. The strict real bounds give $0 < c_0 < 2$. With

$$A = 2\nu^2 - \mu^2, \quad B = 4\nu^2 - \mu^2, \quad P = \nu d_0, \quad Z = \nu^2\big((1-2x) - (1-2y)\big),$$

and $Q = 2\nu r_1$, $R = 2\nu r_2$, the full equations are

$$Z^2 = A(P^2 - 2\nu^2), \qquad Q^2 = P^2 + \mu^2 + 2Z, \qquad R^2 = P^2 + \mu^2 - 2Z. \tag{70}$$

Indeed, if $u_0 = 1 - 2x$ and $v_0 = 1 - 2y$, then

$$c_0 d_0 = u_0 + v_0, \qquad c_0^2 + d_0^2 = 2 + u_0^2 + v_0^2,$$

so $(u_0 - v_0)^2 = (2 - c_0^2)(d_0^2 - 2)$. The inverse is

$$\begin{aligned} x &= \frac{1}{2} - \frac{\mu P + Z}{4\nu^2}, \quad y = \frac{1}{2} - \frac{\mu P - Z}{4\nu^2}, \\ r_3 &= \frac{\mu + P}{2\nu}, \qquad r_0 = \frac{\varepsilon(\mu - P)}{2\nu}, \qquad r_1 = \frac{|Q|}{2\nu}, \quad r_2 = \frac{|R|}{2\nu}. \end{aligned} \tag{71}$$

Its strict locus is specified by positivity of these radii and

$$|\mu P + Z| < 2\nu^2, \qquad |\mu P - Z| < 2\nu^2. \tag{72}$$

Substitution into (52) proves the inverse.

Proposition 5.4. *For every rational $0 < c_0 < 2$, the smooth projective curve determined by* (70) *is geometrically connected of genus five.*

Proof. Neither A nor B is zero. Over the Z-line the three radicands, after multiplication by the same nonzero constant A, are

$$f_0 = Z^2 + 2A\nu^2, \quad f_+ = (Z + A)^2 + 2A\mu^2, \quad f_- = (Z - A)^2 + 2A\mu^2.$$

All three have two simple roots. A common root of f_0, f_+ would be $Z = -\mu^2/2$, where $f_0 = B^2/4 \neq 0$; the pair f_0, f_- is identical after $Z \mapsto -Z$. A common root of f_+, f_- would be $Z = 0$, where $f_+ = A(2\nu^2 + \mu^2) \neq 0$. Thus there are six distinct branch values. The three square classes are independent over $\overline{\mathbf{Q}}(Z)$, since each has simple zeros belonging to no other radicand. The cover has degree eight, with four points of ramification index two above each branch value; infinity is unramified. Riemann–Hurwitz gives $2g - 2 = -16 + 6 \cdot 4 = 8$. □

Multiplication of the last two equations in (70) produces an elliptic quotient:

$$F_{\mu,\nu}: \quad \eta^2 = \xi^3 + \alpha\xi^2 + B^2\xi, \qquad \alpha = 2(3\mu^2 - 4\nu^2), \qquad (\xi, \eta) = (P^2, PQR). \tag{73}$$

Its cubic discriminant is nonzero because $\alpha^2 - 4B^2 = -32\mu^2 A \neq 0$. Every strict image satisfies

$$0 < \xi < B. \tag{74}$$

For the upper bound use $c_0^2 + d_0^2 < 4$. For the lower bound, $P = 0$ for a sum would place the original point on $x + y = 1$, while for a difference $P = \nu(r_3 + r_0) > 0$. The quotient always has the four torsion points

$$O, \quad (0, 0), \quad (B, 2\mu B), \quad (B, -2\mu B). \tag{75}$$

The duplication formula

$$\xi(2S) = \frac{(\xi(S)^2 - B^2)^2}{4\eta(S)^2} \tag{76}$$

shows that the last two double to $(0, 0)$.

Theorem 5.5 (Uniform torsion for odd pair parameters)**.** *If μ, ν are odd, the rational torsion subgroup of $F_{\mu,\nu}$ is cyclic of order four, with points* (75)*. Any strict four-distance point with opposite-pair value μ/ν gives a point of infinite order on $F_{\mu,\nu}$.*

Proof. Put $S = (0, 0)$ and $T = (B, 2\mu B)$. The quadratic factor $\xi^2 + \alpha\xi + B^2$ has discriminant $32\mu^2(\mu^2 - 2\nu^2)$, of exact 2-adic valuation five. Consequently S is the unique nonzero rational point killed by two. The duplication formula gives $2T = S$, so T has exact order four. We have

$$B \equiv 3 \pmod 8, \qquad \alpha \equiv 14 \pmod{16}, \qquad B^2 \equiv 9 \pmod{16}.$$

We first exclude odd torsion. For the integral Weierstrass equation over $\mathbf{Z}_2$, let G_1 be the subgroup of $F_{\mu,\nu}(\mathbf{Q}_2)$ reducing to infinity. Its parameter $z = -\xi/\eta$ lies in $2\mathbf{Z}_2$. The formal group law has integral coefficients and linear part $z_1 + z_2$. Indeed, in coordinates $z = -\xi/\eta$ and $w = -1/\eta$ the equation has a unit derivative in w at $(0, 0)$; successive coefficient comparison solves for w as an integral power series in z. Substitution into the addition law gives

$$[m]z = mz + z^2 g_m(z), \qquad g_m(z) \in \mathbf{Z}_2[[z]].$$

If m is odd and $z \neq 0$, the two terms on the right have different valuations, with the first strictly smaller. Thus G_1 has no nonzero odd torsion. This argument uses the smooth point at infinity and applies to the present integral equation at 2.

For an arbitrary rational point P, put $Q = 2P$. If Q is finite, its abscissa ξ is a rational square by (76). If $\xi = 0$, then $Q = S$ and $4P = O$. Suppose $\xi \neq 0$, and write $v = \operatorname{ord}_2(\xi)$. The case $v = 0$ is impossible. The curve equation makes the ordinate integral, and a unit square is 1 or 9 modulo 16, so

$$\xi(\xi^2 + \alpha\xi + B^2) \equiv 8 \pmod{16},$$

whereas a square in $\mathbf{Z}_2$ cannot have this residue.

If $v > 0$, the term $B^2\xi$ has uniquely least valuation in the curve equation. Hence $2\operatorname{ord}_2(\eta) = v$, and (76) gives $\operatorname{ord}_2(\xi(2Q)) = -v - 2 \leq -4$. If $v < 0$, the cubic term has uniquely least valuation, so $2\operatorname{ord}_2(\eta) = 3v$ and $\operatorname{ord}_2(\xi(2Q)) = v - 2 \leq -4$. The numerator and denominator of the duplication formula are nonzero in both cases. A point with a pole in its abscissa belongs to G_1. We have therefore proved that $4P \in G_1$ for every rational point P, including the cases $4P = O$. If P has odd order, then $4P = O$ by the preceding formal-group argument, and coprimality with four gives $P = O$.

Every rational torsion point has 2-power order. If $4P = O$, then $2P$ is O or S. In the first case P is O or S; in the second, $2(P - T) = O$, so P is T or $-T$. There is no point of order eight: a half of T or $-T$ would make B a rational square by (76), contrary to $B \equiv 3 \pmod 8$. A point of larger 2-power order would have a multiple of order eight. Thus the torsion subgroup is precisely $\{O, S, T, -T\}$. Finally, every strict image satisfies (74), whereas the finite torsion abscissas are 0 and B. This proves the infinite-order assertion. □

Corollary 5.6. *If μ, ν are odd and $\operatorname{rank} F_{\mu,\nu}(\mathbf{Q}) = 0$, then μ/ν is neither an opposite-pair sum nor a nonzero absolute difference at a strict rational four-distance point.*

Proof. Such a point would give a rational point of $F_{\mu,\nu}$ in the interval (74). Rank zero makes every rational point torsion, whereas Theorem 5.5 places all torsion points outside that interval. □

5.4 Adjacent pairs and fixed-pair finiteness

For an adjacent pair choose signed radii R_0, R_1 with $R_0 + R_1 = c_0 > 0$, where $R_i^2 = r_i^2$. Write $\lambda = c_0^2$. A strict adjacent sum is greater than one and an absolute difference is less than one, so a nonzero permitted value has $\lambda \neq 0, 1$. With $U_0 = 2x - 1$ and $V_0 = 2y$, the first two distance equations give the conic

$$(\lambda - 1)U_0^2 + \lambda V_0^2 - \lambda(\lambda - 1)H_0^2 = 0. \tag{77}$$

It is smooth and contains $(c_0, 0, 1)$. In the chart $H_0 = 1$,

$$R_0 = \frac{c_0 + U_0/c_0}{2}, \qquad R_1 = \frac{c_0 - U_0/c_0}{2}.$$

The remaining roots are given by

$$f_2 = \frac{(U_0 + H_0)^2 + (V_0 - 2H_0)^2}{H_0^2}, \qquad f_3 = \frac{(U_0 - H_0)^2 + (V_0 - 2H_0)^2}{H_0^2}.$$

Adjoin both square roots to the conic function field. On its affine chart the return is

$$x = \frac{U_0 + 1}{2}, \quad y = \frac{V_0}{2}, \quad (r_0, r_1, r_2, r_3) = \left(|R_0|, |R_1|, \frac{|\sqrt{f_2}|}{2}, \frac{|\sqrt{f_3}|}{2}\right).$$

Proposition 5.7. *For every rational $c_0 > 0$ with $c_0 \neq 1$, this normalized cover is geometrically connected of genus five.*

Proof. Over $\overline{\mathbf{Q}}$, the numerator of f_2 factors into the two lines $U_0 + H_0 \pm i(V_0 - 2H_0) = 0$. For the line with sign $\delta = \pm 1$, evaluation of the dual conic is

$$\frac{1}{\lambda - 1} - \frac{1}{\lambda} - \frac{(1 - 2\delta i)^2}{\lambda(\lambda - 1)} = \frac{4 + 4\delta i}{\lambda(\lambda - 1)} \neq 0.$$

Both intersections are therefore transverse. The two lines meet at $(-1, 2, 1)$, which would belong to the conic only if $\lambda^2 - 6\lambda + 1 = 0$; this is impossible for rational λ. Thus f_2 has four distinct simple zeros. The same calculation with $U_0 \mapsto -U_0$ proves this for f_3.

The two zero sets are disjoint. Their numerator difference is $4U_0H_0$. At $H_0 = 0$ a common zero would satisfy $U_0^2 + V_0^2 = 0$, which together with the conic forces $U_0 = V_0 = 0$. In the chart $H_0 = 1$, a common zero has $U_0 = 0$ and $V_0 = 2 \pm i$, whence the conic requires $\lambda = 4 \pm 4i$, again impossible. The two points of the conic at infinity are distinct and neither numerator vanishes there; both poles of each f_i have order two. The radicands are consequently independent square classes, and the degree-four cover has eight branch values, each contributing two to the ramification divisor. Hence $2g - 2 = 4(-2) + 8 \cdot 2 = 8$. □

Theorem 5.8 (Fixed-pair finiteness)**.** *Fix two distinct vertices, signs $\varepsilon_i, \varepsilon_j \in \{1, -1\}$, and a nonzero rational number c_0. There are only finitely many strict rational four-distance points satisfying $\varepsilon_i r_i + \varepsilon_j r_j = c_0$.*

Proof. Changing both signs and applying a square symmetry reduces the pair to one of the two constructions above. Their return formulas show that every strict point is the image of a rational point on the corresponding smooth projective genus-five curve. Faltings' theorem [8] makes this rational-point set finite. □

5.5 The fibre with opposite difference one fifth

For $c_0 = 1/5$, take $\mu = 1$, $\nu = 5$, $\varepsilon = -1$ and $Z = 7z$ in (70). The curve becomes

$$C: \quad P^2 = z^2 + 50, \qquad Q^2 = z^2 + 14z + 51, \qquad R^2 = z^2 - 14z + 51. \tag{78}$$

Its strict positive branch has $P, Q, R > 0$ and

$$\begin{aligned} x &= \tfrac{1}{2} - \frac{P + 7z}{100}, & y &= \tfrac{1}{2} - \frac{P - 7z}{100}, \\ (r_0, r_1, r_2, r_3) &= \tfrac{1}{10}(P - 1, Q, R, P + 1), & |P + 7z| &< 50, & |P - 7z| &< 50. \end{aligned} \tag{79}$$

The projective model in coordinates $[P : Q : R : z : T]$ is obtained by homogenizing the three equations. Above $T = 0$ it has the eight rational points $[\pm 1 : \pm 1 : \pm 1 : 1 : 0]$. These points are smooth: in the chart $z = 1$ the derivatives in P, Q, R form an invertible diagonal matrix, and T is a local parameter.

Proposition 5.9 (Elliptic decomposition of the fixed fibre)**.** *There is a $\mathbf{Q}$-isogeny*

$$\mathrm{Jac}(C) \sim_{\mathbf{Q}} E_1 \times F \times E_3 \times E_4^2,$$

where

$$\begin{aligned} E_1 &: \; y^2 = x^3 - 4606x^2 + 6245001x, \\ F &: \; y^2 = x^3 - 194x^2 + 9801x, \\ E_3 &: \; y^2 = x^3 - 9902x^2 + 25492401x, \\ E_4 &: \; y^2 = x^3 - 202x^2 + 9801x. \end{aligned}$$

Proof. Put $f_0 = z^2 + 50$, $f_\pm = z^2 \pm 14z + 51$. The seven nontrivial characters of the sign-change group $(\mathbf{Z}/2\mathbf{Z})^3$ give quotients $w^2 = \prod_{j\in I} f_j$. Their genera are $0, 0, 0, 1, 1, 1, 2$, according as $|I|$ is one, two, or three. The pullbacks of their differential spaces are the respective character spaces in $H^0(C, \Omega^1_C)$, and their dimensions sum to five. The product of the associated Jacobian pullback maps is therefore an isogeny over $\mathbf{Q}$.

The quotient $w^2 = f_+ f_- = z^4 - 94z^2 + 2601$ maps with degree two to $y^2 = x^3 - 94x^2 + 2601x$ by $(x, y) = (z^2, zw)$. The scaling $(x, y) \mapsto (49x, 343y)$ gives E_1. The two quotients $w^2 = f_0 f_\pm$ are isomorphic by $z \mapsto -z$. For the plus sign the substitutions

$$W_0 = w + z^2 + 7z + 26, \qquad x = 2W_0 + 49, \qquad y = (4z + 14)W_0 + 336$$

give E_4. Their rational inverse on $W_0 \neq 0$ is

$$W_0 = (x - 49)/2, \qquad z = \frac{y - 14W_0 - 336}{4W_0}, \qquad w = W_0 - z^2 - 7z - 26.$$

The remaining genus-two curve is

$$D: \quad w^2 = (z^2 + 50)(z^4 - 94z^2 + 2601).$$

Its involutions $(z, w) \mapsto (-z, w)$ and $(z, w) \mapsto (-z, -w)$ have elliptic quotients. The first is $(x, y) = (z^2 + 50, w)$ on F. For the second put $\zeta = z^2$ and $v = zw$, so that $v^2 = \zeta(\zeta + 50)(\zeta^2 - 94\zeta + 2601)$. More generally, for $v^2 = \zeta(\zeta + c_1)(\zeta^2 + a_1\zeta + b_1)$, the substitutions

$$X = \frac{c_1 b_1}{\zeta} + b_1, \qquad Y = \frac{c_1 b_1 v}{\zeta^2}$$

give

$$Y^2 = X^3 + (c_1 a_1 - 2b_1)X^2 + b_1(b_1 - c_1 a_1 + c_1^2)X.$$

Taking $(c_1, a_1, b_1) = (50, -94, 2601)$ yields E_3. The two one-dimensional differential eigenspaces on D prove $\mathrm{Jac}(D) \sim_{\mathbf{Q}} F \times E_3$, completing the decomposition. □

Example 5.10. The elliptic quotient F contains

$$(\xi, \eta) = \left(\frac{1476225}{18496}, -\frac{589188735}{2515456} \right).$$

This follows by direct substitution in its cubic equation. Its first coordinate is $(1215/136)^2$ and satisfies $50 < \xi < 99$. A lift to (78) would require

$$Z^2 = 49(\xi - 50) = \left(\frac{35}{136} \right)^2 22057.$$

Since $148^2 < 22057 < 149^2$, the required rational root does not exist. Thus the interval (74) and a square elliptic coordinate do not characterize the image of $C(\mathbf{Q})$; the three simultaneous equations in (78) give the complete lift.

6 A common squareclass descent

The sum of the products of opposite distances gives a useful one-parameter family of covering curves. The four square roots in this family determine one point of the square. In particular, the squareclass of their common coordinate can be compared at different places.

6.1 The four-root curve and its return map

Let $t \in \mathbf{Q} \setminus \{0, 1, -1\}$, and put

$$s = t - 1, \qquad T = t + 1, \qquad d = sT = t^2 - 1.$$

Let $\mathcal{U}_t$ be the affine curve

$$P^2 = \ell + T, \qquad C^2 = \ell - s, \qquad M^2 = \frac{s(T-\ell)}{\ell}, \qquad B^2 = \frac{T(\ell+s)}{\ell}, \qquad \ell PCMB \neq 0. \quad (80)$$

We denote its smooth projective completion by D_t. The order of the four roots will be fixed throughout this section.

For a point whose distances from $(0,0), (1,0), (0,1), (1,1)$ are r_0, r_1, r_2, r_3, respectively, the coordinates are

$$\begin{aligned} t &= r_0 r_3 + r_1 r_2, \qquad P = r_0 + r_3, \qquad M = r_3 - r_0, \\ B &= r_1 + r_2, \qquad C = r_2 - r_1, \qquad \ell = P^2 - t - 1. \end{aligned} \quad (81)$$

To verify the forward construction and its range, put $u = 1 - 2x$, $v = 1 - 2y$, $A = r_0 r_3$, and $A' = r_1 r_2$. Expansion of the four squared-distance identities gives

$$A^2 - (A')^2 = -uv, \qquad (A + A')^2 + (A - A')^2 = 1 + (u^2 + v^2)^2/4.$$

Thus $t^2 = (A + A')^2$ is the larger root of

$$F_t(Z) = Z^2 - (1 + (u^2 + v^2)^2/4)Z + u^2 v^2.$$

The diagonal exclusion in Lemma 5.1 gives $u^2 \neq v^2$, and hence $F_t(1) = -(u^2 - v^2)^2/4 < 0$. For $a = u^2, b = v^2 \in [0, 1)$,

$$16F_t(5/4) = 5 - 5a^2 + 6ab - 5b^2 > 0.$$

Indeed this polynomial is strictly concave on $[0,1]^2$, with vertex values $5, 0, 0, 1$, and its two zero vertices are excluded. The axis of F_t is below 1, so $1 < t < \sqrt{5}/2$. Furthermore

$$PM = u + v, \quad BC = v - u, \quad P^2 - C^2 = B^2 - M^2 = 2t, \quad P^2 + M^2 = u^2 + v^2 + 2.$$

These identities give $(P^2 - T)(M^2 + s) = d$ by expansion. The first two equations of (80) follow from the definition of ℓ; the last two then follow from this product identity and $B^2 - M^2 = 2t$. The diagonal exclusion also gives $C, M \neq 0$. Since $s > 0$ and $d > 0$, the product identity first gives $\ell > 0$, and the equations for C^2, M^2 then give $s < \ell < T$.

Conversely, a tuple in (80) determines

$$\begin{aligned} u &= \frac{PM - BC}{2}, \qquad v = \frac{PM + BC}{2}, \qquad x = \frac{1-u}{2}, \qquad y = \frac{1-v}{2}, \\ r_0 &= \frac{P - M}{2}, \quad r_3 = \frac{P + M}{2}, \quad r_1 = \frac{B - C}{2}, \quad r_2 = \frac{B + C}{2}. \end{aligned} \quad (82)$$

These formulas give the four squared-distance identities over every field of characteristic zero. Indeed, the root equations imply

$$B^2 - M^2 = P^2 - C^2 = 2t, \qquad P^2M^2 + B^2C^2 = 2(P^2 + M^2 - 2).$$

Consequently $u^2 + v^2 = P^2 + M^2 - 2 = B^2 + C^2 - 2$. Since $u + v = PM$ and $u - v = -BC$, substitution into $\big((1-u)^2 + (1-v)^2\big)/4$ and its three translates gives $r_0^2, r_1^2, r_2^2, r_3^2$. Moreover,

$$r_0r_3 + r_1r_2 = \frac{P^2 - M^2 + B^2 - C^2}{4} = t.$$

For $t > 1$, the strict real return chamber is

$$s < \ell < T, \qquad P, B > 0, \qquad |PM| + |BC| < 2. \tag{83}$$

In this chamber $P > |M|$ and $B > |C|$, so all four returned distances are positive. The last inequality is equivalent to $|u|, |v| < 1$, and hence to $0 < x, y < 1$. These identities also show directly that the transformation

$$(\ell, P, C, M, B) \longmapsto (d/\ell, B, M, C, P) \tag{84}$$

is the reflection $x \mapsto 1 - x$ on the square.

Proposition 6.1. *The curve D_t is geometrically connected of genus nine. The map $\ell : D_t \to \mathbf{P}^1$ has degree sixteen and is branched precisely at*

$$0, \quad \infty, \quad s, \quad T, \quad -s, \quad -T.$$

If $t > 1$, none of the points above these six values is rational.

Proof. Over $\overline{\mathbf{Q}}(\ell)$, the four radicands in (80) have independent squareclasses: their valuations at $-T, s, T, -s$, respectively, each isolate one of the four classes. The extension therefore has degree sixteen. At each of the six displayed branch values the inertia group has order two. For example, the valuation vectors at 0 and ∞, in the order (P, C, M, B), are $(0, 0, 1, 1)$ and $(1, 1, 0, 0)$ modulo two. Thus Riemann–Hurwitz gives

$$2g(D_t) - 2 = -32 + 6 \cdot 8 = 16.$$

The following table records a regular function on the normalization and its value at each omitted fibre:

ℓ	s	T	$-s$	$-T$	0	∞
regular square	M^2	C^2	P^2	B^2	C^2	M^2
value	2	2	2	2	$-s$	$-s$.

The first four require $\sqrt{2} \in \mathbf{Q}$, and the last two require a square root of the negative number $-s$. This excludes every rational point on the six fibres. □

6.2 Prime support and the ramified unit conditions

Write $[a]_K$ for the class of $a \in K^\times$ modulo squares; omit the subscript when $K = \mathbf{Q}$. For a nonzero rational number d, set

$$S(d) = \{p : \ p \text{ is odd and } v_p(d) > 0 \text{ is odd}\}.$$

Lemma 6.2 (Support of the common coordinate)**.** *If* $(\ell, P, C, M, B) \in \mathcal{U}_t(\mathbf{Q})$*, then*

$$[\ell] \in \langle [-1], [2], [p] : p \in S(d) \rangle.$$

For $t > 1$*, the sign is positive and the generator* $[-1]$ *can be omitted.*

Proof. Fix an odd prime p and put $m = v_p(t)$ and $n = v_p(\ell)$. Suppose first that $m < 0$, so $v_p(s) = v_p(T) = m$. If $n > m$, the first and third equations of (80) have valuations m and $2m - n$; both must be even, so n is even. If $n < m$, the fourth and first have valuations m and n, giving the same conclusion. If $n = m$, the reductions of P^2/t and C^2/t differ by 2. At least one is a unit; its square has valuation $m = n$, which is even.

Now let $m \geq 0$. If both s and T are units, then $n < 0$ is even by the first equation and $n > 0$ is even by the third, whose valuation is $-n$. If $e = v_p(s) > 0$, then T is a unit. Negative n is even by the first equation, and $0 < n < e$ is even by the second. For $n > e$, the second and third equations have valuations e and $e - n$, so both e and n are even. Hence odd n can occur only at $n = e$ with e odd. In that case $e = v_p(d) > 0$. If instead $e = v_p(T) > 0$, the first equation handles $n < 0$ and $0 < n < e$, while the first and fourth have valuations $e, e - n$ for $n > e$. Again odd n forces $n = e = v_p(d) > 0$ odd.

This accounts for every odd prime, including equal-valuation cases. Squareclasses in $\mathbf{Q}$ are determined by sign and valuation parity. Finally $C^2 = \ell - s$ implies $\ell > s > 0$ when $t > 1$. □

Lemma 6.3 (Ramified classes)**.** *Let* p *be odd and suppose* $v_p(d) > 0$ *is odd. If* $\mathcal{U}_t(\mathbf{Q}_p)$ *is nonempty, then* $\left(\frac{2}{p}\right) = 1$ *and every point satisfies*

$$[\ell]_{\mathbf{Q}_p} \in \{1, [d]_{\mathbf{Q}_p}\}. \tag{85}$$

If $\left(\frac{2}{p}\right) = 1$*, both classes in* (85) *occur.*

Proof. First suppose $e = v_p(s) > 0$ is odd. If $v_p(\ell) > 0$, then $P^2 \equiv 2 \pmod{p}$. Otherwise $B^2 = T(1 + s/\ell) \equiv 2 \pmod{p}$. Thus 2 is a square modulo p. A negative $n = v_p(\ell)$ makes $M^2 = -s(1 - T/\ell)$ have odd valuation e. For $0 < n < e$, the valuations n of C^2 and $e - n$ of M^2 cannot both be even. For $n > e$, C^2 has odd valuation e. Only $n = 0$ and $n = e$ remain.

If $n = 0$, the third equation forces $v_p(T - \ell) > 0$, so $\ell \equiv T \equiv 2 \pmod{p}$ and $[\ell]_{\mathbf{Q}_p} = 1$. If $n = e$, then

$$M^2 = (d/\ell)(1 - \ell/T),$$

and the second factor is a principal unit, hence a square. Thus $[\ell]_{\mathbf{Q}_p} = [d]_{\mathbf{Q}_p}$. The substitution $(t, P, C, M, B) \mapsto (-t, C, P, B, M)$ leaves ℓ, d unchanged and proves the assertion when $v_p(T) > 0$.

For existence near $t = 1$, choose $\ell = s + p^{2h}$ with $2h > e$. Then $C = p^h$, and the other three radicands have square unit residues $2, 2, 4$. Hensel's lemma gives all three roots. This realizes $[d]$; the reflection (84) realizes 1. The sign substitution treats the other case. □

Lemma 6.4 (The dyadic exponent)**.** *If* $\mathcal{U}_t(\mathbf{Q}_2)$ *is nonempty, then either* $v_2(t) \geq 2$ *or* $v_2(t)$ *is negative and odd. If additionally* d *is a square in* $\mathbf{Q}_2$*, then*

$$v_2(\ell) = v_2(t) < 0 \quad \textit{is odd.}$$

Proof. Apply (82), without real sign restrictions, to obtain coordinates x, y and four distances r_j in $\mathbf{Q}_2$. Choose $N = 2^h$ with $h \geq 0$ minimal such that $X = Nx, Y = Ny \in \mathbf{Z}_2$. Then $A_j = Nr_j \in \mathbf{Z}_2$, since their squares are integral. If $h = 0$, one of the four distances has square 2

(mod 4), a contradiction. Hence $h \geq 1$, and X, Y are not both even. They cannot both be odd, since $A_0^2 = X^2 + Y^2$. After exchanging the coordinate axes, take X odd and Y even. Every A_j is odd. The first and third squared distances modulo eight now give $4 \mid Y$ and $4 \mid Y - N$, so $h \geq 2$.

Put $U = N - 2X$, $V = N - 2Y$, and $S_\pm = A_0 A_3 \pm A_1 A_2$. Direct expansion gives

$$S_+ S_- = -N^2 UV, \qquad t = S_+/N^2, \qquad v_2(U) = 1.$$

For $V \neq 0$, we have $v_2(V) \geq 2$. Exactly one of the sum and difference of two odd units has valuation one. The product identity therefore gives either $v_2(t) = 1 - 2h$ or $v_2(t) = v_2(V) \geq 2$. If $V = 0$, then $S_+ \neq 0$ because $t \neq 0$, so $S_- = 0$ and $S_+ = 2A_0 A_3$; again $v_2(t) = 1 - 2h$.

If d is a dyadic square, the alternative $v_2(t) \geq 2$ is impossible: it would give $d \equiv -1 \pmod 8$. Write $m = v_2(t) < 0$, which is odd, and $n = v_2(\ell)$. For $n > m$, $P^2 = \ell + T$ has odd valuation m. For $n < m$, $B^2 = T(1 + s/\ell)$ has odd valuation m. Thus $n = m$. □

6.3 The residue matrix

Let $\Delta = p_1 \cdots p_r > 1$ be positive, odd and squarefree. Define $A_\Delta = (a_{ij}) \in M_r(\mathbf{F}_2)$ by

$$(-1)^{a_{ij}} = \left(\frac{p_j}{p_i}\right) \quad (i \neq j), \qquad a_{ii} = \sum_{j \neq i} a_{ij}. \tag{86}$$

Thus every row sums to zero. Quadratic reciprocity does not in general make this matrix symmetric; the denominator p_i in (86) specifies the place at which the row is imposed.

Proposition 6.5. *Suppose*

$$\Delta \equiv 1 \pmod 8, \qquad p_i \equiv 1 \text{ or } 7 \pmod 8, \qquad t > 1, \qquad d \in \Delta \mathbf{Q}^{\times 2}. \tag{87}$$

Every rational point of $\mathcal{U}_t$ has

$$[\ell] = \left[2 \prod_{i=1}^{r} p_i^{e_i}\right], \qquad (e_1, \ldots, e_r)^{\mathsf{T}} \in \ker A_\Delta. \tag{88}$$

If $\operatorname{rank} A_\Delta = r - 1$*, then*

$$[\ell] \in \{[2], [2\Delta]\}. \tag{89}$$

Proof. Write $d = \Delta q^2$. An odd valuation of d cannot be negative, since $v_p(t) < 0$ implies $v_p(t^2 - 1) = 2v_p(t)$. Thus $S(d) = \{p_1, \ldots, p_r\}$. The support lemma leaves $[\ell] = [2^e \prod p_i^{e_i}]$. As $\Delta \equiv 1 \pmod 8$, d is a dyadic square; the dyadic lemma forces $e = 1$.

At p_i, if $e_i = 0$, the ramified-class lemma requires ℓ to be a local square; if $e_i = 1$, it requires ℓ/d to be a local square. All valuation parities have already been accounted for. Moreover $\left(\frac{2}{p_i}\right) = 1$. The remaining unit conditions are exactly

$$\sum_{j \neq i} a_{ij}(e_j + e_i) = 0,$$

which is row i of (88). The zero and all-one vectors belong to the kernel. If its dimension is one, they are its only vectors, proving (89). □

Lemma 6.6 (A local squareclass restriction at 37)**.** *Suppose $t \in \mathbf{Z}_{37}$ and $\bar{t} \in \{14, -14, 16, -16\}$. Every point of $\mathcal{U}_t(\mathbf{Q}_{37})$ has $[\ell]_{\mathbf{Q}_{37}} = 1$.*

Proof. The substitution $t \mapsto -t$ used above preserves ℓ, so it is enough to treat $\bar{t} = 14, 16$. In either case T is a nonsquare unit. A positive valuation of ℓ would give $P^2 \equiv T \pmod{37}$; a negative valuation would give $B^2 \equiv T \pmod{37}$. Thus ℓ is a unit. We prove that its reduction λ is square.

Work in $\mathbf{F}_{37}$. The elements $13, 15, 17$ are twice squares:

$$13 = 2 \cdot 5^2, \qquad 15 = 2 \cdot 10^2, \qquad 17 = 2 \cdot 8^2.$$

The element 2 is nonsquare, while -1 is square. If λ were nonsquare, the last two radicands, together with the first two, would imply that all four elements

$$\lambda + \bar{t} + 1, \quad \lambda + \bar{t} - 1, \quad \lambda - \bar{t} + 1, \quad \lambda - \bar{t} - 1$$

are squares, allowing zero.

Here is a polynomial test for two squares differing by 2. If $a^2 = z + 1$ and $b^2 = z - 1$, put $w = (a + b)^2/2$. Since $(a + b)(a - b) = 2$, we have $w \neq 0$ and

$$z = (w + w^{-1})/2, \qquad w^{18} = -1.$$

Let T_j be the Chebyshev polynomials, characterized by $T_j((w + w^{-1})/2) = (w^j + w^{-j})/2$. Then $T_9(z) = 0$. Reduction of $-T_9/3$ modulo 37 gives

$$F(Z) = Z^9 + 7Z^7 + 4Z^5 + 3Z^3 - 3Z.$$

Apply this to $z = \lambda + \bar{t}$ and $z = \lambda - \bar{t}$. Both F-values vanish. Expanding and factoring their sum gives the identities in $\mathbf{F}_{37}[X]$

$$F(X + 14) + F(X - 14) = 2X^3(X^2 - 14)(X^2 + 4)(X^2 + 6),$$
$$F(X + 16) + F(X - 16) = 2X^3(X^2 - 14)(X^2 + 6)(X^2 + 18).$$

The three elements $14, 31 = -6, 19 = -18$ are nonsquares: each has 18th power -1. Since $\lambda \neq 0$, the second identity is impossible. The first leaves only $\lambda^2 = -4$, hence $\lambda = 12$ or 25, both squares ($12 = 7^2$ and $25 = 5^2$). This too contradicts the assumed nonsquareness. Thus λ is square; the square criterion for odd local fields then gives $[\ell]_{\mathbf{Q}_{37}} = 1$. □

Theorem 6.7 (Residue-matrix exclusion)**.** *Let $\Delta = p_1 \cdots p_r > 1$ be positive, odd and squarefree, and suppose*

$$\Delta \equiv 1 \pmod{8}, \qquad p_i \equiv 1 \text{ or } 7 \pmod{8}, \qquad \operatorname{rank}_{\mathbf{F}_2} A_\Delta = r - 1.$$

For every rational $t > 1$ satisfying

$$t^2 - 1 \in \Delta \mathbf{Q}^{\times 2}, \qquad t \in \mathbf{Z}_{37}, \qquad \bar{t} \in \{14, -14, 16, -16\}, \tag{90}$$

one has $D_t(\mathbf{Q}) = \varnothing$.

Proof. A rational point in the finite chart would satisfy $[\ell] \in \{[2], [2\Delta]\}$ by Proposition 6.5. The residues of d at the two positive values of $\bar{t}$ are $10 = 11^2$ and $33 = 12^2$. The prime 37 does not divide Δ, since its residue modulo eight is 5. From $d = \Delta q^2$ it follows that q is a 37-adic unit and Δ is a square unit there. Thus both 2 and 2Δ are nonsquares in $\mathbf{Q}_{37}$. This contradicts Lemma 6.6. Proposition 6.1 excludes every rational point outside the finite chart. □

The theorem constrains one global squareclass. A different choice of ℓ at each completion cannot replace the compatibility in (88). In particular, the theorem applies without a bound on the height of ℓ, on the height of t, or on the number of prime factors of Δ.

6.4 An explicit infinite family

Consider

$$\Delta_0 = 1926281441 = 7 \cdot 17 \cdot 23 \cdot 31 \cdot 73 \cdot 311.$$

In the indicated prime order, the matrix (86) is

$$A_{\Delta_0} = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 \end{pmatrix}. \tag{91}$$

Its entries follow from quadratic reciprocity and reduction of the six primes modulo one another. Its kernel has a short explicit certificate. Row three gives $e_1 = e_2$; row five then gives $e_4 = e_5$; row four gives $e_2 = e_3$; row six gives $e_4 = e_6$; and row one gives $e_2 = e_4$. Hence all six coordinates agree. The rank is therefore five.

Corollary 6.8. *Put $C_0 = 269297$ and $K = 2^{20} \cdot 7 \cdot 37$. For every integer $j \geq 1$, define*

$$q_j = \frac{1}{C_0} + \frac{K}{1 + K^2 j}, \qquad t_j = \frac{1 + \Delta_0 q_j^2}{1 - \Delta_0 q_j^2}. \tag{92}$$

The parameters t_j are distinct and satisfy

$$1 < t_j < \frac{17}{16} < \frac{\sqrt{5}}{2}, \qquad D_{t_j}(\mathbf{Q}) = \varnothing.$$

Every D_{t_j} has a real point in the strict return chamber (83)*. In addition, $t_j + 1$ is neither a rational square nor twice a rational square, and $[t_j^2 - 1] = [\Delta_0] \notin \{1, [2]\}$.*

Proof. For $q_0 = 1/C_0$, direct cancellation gives

$$t(q_0) = \frac{138225}{131072} =: t_0, \qquad 1 - \Delta_0 q_0^2 = \frac{2^{18}}{C_0}, \qquad t_0 \equiv 14 \pmod{37}.$$

The perturbation $\delta_j = K/(1 + K^2 j)$ has 37-adic valuation one. Since the denominator of $t(q_0)$ is a 37-adic unit, $t_j \equiv 14 \pmod{37}$ for all j. Also

$$t_j^2 - 1 = \Delta_0 \left(\frac{2q_j}{1 - \Delta_0 q_j^2} \right)^2.$$

At the real place $0 < \delta_j < 1/K < 10^{-7}$. The integer inequality

$$33\Delta_0 \left(\frac{1}{269297} + \frac{1}{10^7} \right)^2 = \frac{24893364230896160241}{26929700000000000000} < 1$$

gives $0 < \Delta_0 q_j^2 < 1/33$, and hence $1 < t_j < 17/16$. The sequence q_j is strictly decreasing, and the function $(1 + \Delta_0 q^2)/(1 - \Delta_0 q^2)$ is strictly increasing on $0 < q < 1/\sqrt{\Delta_0}$. Thus the t_j are distinct. Theorem 6.7 now applies to each one.

For the real assertion choose $\ell = \sqrt{d}$ and positive roots in (80). Since $s < \sqrt{sT} < T$, all four radicands are positive. Furthermore

$$(PM)^2 = (BC)^2 = 2\sqrt{d}.$$

The inequality $t < \sqrt{5}/2$ gives $d < 1/4$, so $|PM| + |BC| = 2\sqrt{2\sqrt{d}} < 2$. Finally $v_7(q_0) = -1$ and $v_7(\delta_j) = 1$, whence $v_7(q_j) = -1$ and $v_7(1 - \Delta_0 q_j^2) = -1$. As $t_j + 1 = 2/(1 - \Delta_0 q_j^2)$, we get $v_7(t_j + 1) = 1$, which proves both square exclusions. □

The limit parameter itself also satisfies the exclusion theorem:

$$t_0 = \frac{138225}{131072}, \qquad t_0^2 - 1 = \frac{\Delta_0}{2^{34}}, \qquad D_{t_0}(\mathbf{Q}) = \varnothing. \tag{93}$$

For a one-prime squareclass $\Delta = p \equiv 1 \pmod 8$, the matrix is (0), so its rank hypothesis is automatic. Thus Theorem 6.7 also applies to every such p and every t satisfying (90).

6.5 The unramified genus-seventeen covers

The support lemma also gives a uniform finite descent for each fixed parameter. For $t > 1$ define

$$\mathcal{K}_t = \left\{ 2^e \prod_{p \in S(d)} p^{e_p} : e, e_p \in \{0, 1\} \right\}.$$

For $\kappa \in \mathcal{K}_t$, let $\widetilde{D}_{t,\kappa}$ be the smooth projective curve obtained by adjoining to (80) the further equation

$$Z^2 = \ell/\kappa. \tag{94}$$

Proposition 6.9. *The morphism $\widetilde{D}_{t,\kappa} \to D_t$ is a geometrically connected unramified double cover, and $g(\widetilde{D}_{t,\kappa}) = 17$. Every rational point of D_t lifts to one of these covers; on the affine chart the lift retains the same coordinates ℓ, P, C, M, B.*

Proof. On D_t, the function ℓ has eight zeros and eight poles, all of order two, since the map to the ℓ-line has ramification index two over 0 and ∞. Thus adjoining its square root is unramified. To prove geometric connectedness, suppose that ℓ became a square in the four-radical function field over $\overline{\mathbf{Q}}$. In a multiquadratic extension the base-field squareclasses that become squares are exactly the span of its defining classes. The valuations at $-T, s, T, -s$ force each coefficient of such an expression for $[\ell]$ to vanish. The remaining class $[\ell]$ is nonzero on $\overline{\mathbf{Q}}(\ell)$, a contradiction. Riemann–Hurwitz now gives

$$2g(\widetilde{D}_{t,\kappa}) - 2 = 2(2 \cdot 9 - 2) = 32.$$

All rational points of D_t are in the affine chart by Proposition 6.1. For such a point, Lemma 6.2 supplies $\kappa \in \mathcal{K}_t$ with $[\ell] = [\kappa]$, giving a rational Z in (94). □

Proposition 6.10 (Finite local compatibility)**.** *For fixed $t > 1$, define*

$$\mathfrak{S}_\ell(t) = \{\kappa \in \mathcal{K}_t : \widetilde{D}_{t,\kappa}(\mathbf{Q}_v) \neq \varnothing \text{ for every place } v\}.$$

It suffices to impose these conditions at the real place, at the primes $p \leq 1153$, and at the primes at which one of $t, t-1, t+1$ has nonzero valuation. If $\mathfrak{S}_\ell(t)$ is empty, then $D_t(\mathbf{Q})$ is empty.

Proof. Let p be an odd prime outside the stated set. The parameters t, s, T, κ are units. The six branch sections on the ℓ-line remain distinct modulo p: their differences are products of units among $2, t, s, T$. The normalization of the five-radical cover is smooth over $\mathbf{Z}_p$. More explicitly, away from the branch sections it is finite *étale*; near a branch section, after an étale extension extracting square roots of unit factors, its local models are $w^2 = z$, with z a parameter on the base. These models are smooth, and smoothness descends. The private-valuation argument in the preceding proposition persists on the geometric special fibre, so the latter is connected of genus seventeen.

The Hasse–Weil bound gives

$$\#\widetilde{D}_{t,\kappa}(\mathbf{F}_p) \geq p + 1 - 34\sqrt{p} > 0 \qquad (p \geq 1163).$$

The next prime after 1153 is 1163. Smooth Hensel lifting therefore supplies a $\mathbf{Q}_p$-point. A smooth curve over a local field has infinitely many points in a sufficiently small neighbourhood of any rational point, so the point can also be chosen off the finite boundary. The final assertion follows by applying Proposition 6.9 to a rational point of D_t. □

Thus the common-coordinate descent produces finitely many geometrically connected covers while retaining all four ordered roots. The residue-matrix theorem supplies domains on which the associated compatibility set is empty, and Corollary 6.8 gives an explicit infinite sequence of such domains inside the strict real return range.

7 Nine elliptic factors and rank constraints

The four roots in (80) give nine elliptic quotients. Their rational twists can be identified explicitly, and the images of a single strict point impose rank conditions on every factor. Throughout this section put

$$s = t - 1, \qquad T = t + 1, \qquad d = sT, \qquad z = \ell + T.$$

Thus the coordinate denoted by ℓ satisfies

$$P^2 = z, \qquad C^2 = z - 2t, \qquad M^2 = -\frac{s(z - 2T)}{z - T}, \qquad B^2 = \frac{T(z - 2)}{z - T}. \tag{95}$$

The smooth projective curve with this function field is D_t. For a strict rational-distance point, its four roots satisfy

$$t > 1, \qquad s < \ell < T, \qquad P, B > 0, \qquad CM \neq 0.$$

Here $CM \neq 0$ also follows directly from the square: $C = 0$ would give $x = y$, and $M = 0$ would give $x + y = 1$. Either diagonal forces a positive rational distance to be a nonzero rational multiple of $\sqrt{2}$.

7.1 Character quotients and their rational models

Define the elliptic curves

$$\begin{aligned} E_d &: Y^2 = X(X-d)(X+d), \\ G_t &: Y^2 = X(X-d^2)(X+d), \\ F_+ &: Y^2 = X(X-2d)(X+2td), \\ F_- &: Y^2 = X(X-2d)(X-2td), \\ H_+ &: Y^2 = X(X-2T)(X-T^2), \\ H_- &: Y^2 = X(X-2s)(X-d). \end{aligned} \tag{96}$$

Each curve carries the point at infinity as its origin. All six cubics are nonsingular for $t \notin \{0, 1, -1\}$.

Theorem 7.1. *For every $t \in \mathbf{Q} \setminus \{0, 1, -1\}$ there is an isogeny over $\mathbf{Q}$*

$$\mathrm{Jac}(D_t) \sim_{\mathbf{Q}} E_d^2 \times G_t \times F_+^2 \times F_-^2 \times H_+ \times H_-. \tag{97}$$

The rank of the group of rational Néron–Severi classes of $\mathrm{Jac}(D_t)$ *is at least twelve.*

Proof. The sign changes of the four roots give a group $V = (\mathbf{Z}/2\mathbf{Z})^4$, in the order (P, C, M, B). On the z-line its six inertia generators, over $0, 2t, \infty, 2T, 2, T$, are

$$e_1, \quad e_2, \quad e_1 + e_2, \quad e_3, \quad e_4, \quad e_3 + e_4.$$

A nonzero character supported on just one of the blocks $\langle e_1, e_2\rangle$ and $\langle e_3, e_4\rangle$ is ramified at two points. Its double-cover quotient is rational. There are six such characters. Each of the nine characters nonzero on both blocks is ramified at four points and has a genus-one quotient.

The latter characters are the products of one member of (P, C, PC) and one member of (M, B, MB). Their elliptic models are

	M	B	MB
P	E_d	H_+	F_+
C	H_-	E_d	F_-
PC	F_-	F_+	G_t.

(98)

Here are birational maps establishing all the models and their rational twists. The character in the first column specifies the genus-one function field $\mathbf{Q}(z, \text{character})$:

character	target	X	Y
PM	E_d	$-s\ell$	$sPM\ell$
BC	E_d	$T\ell$	$TBC\ell$
PB	H_+	Tz	$TPB\ell$
CM	H_-	$s(2T-z)$	$sCM\ell$
PCB	F_+	$4tT\ell/z$	$4tT^2\ell PCB/z^2$
CMB	F_-	$4td/(T-\ell)$	$-4td\ell CMB/(T-\ell)^2$
$PCMB$	G_t	$dT(2t/z-1)$	$dtT\ell PCMB/z^2$.

(99)

For example, the PB and $PCMB$ equations before transformation are

$$(PB\ell)^2 = Tz(z-2)(z-T), \qquad (PCMB\ell)^2 = -dz(z-2t)(z-2)(z-2T).$$

Substitution of the corresponding rows gives the stated cubics. The other rows follow by multiplying their radicands in (95). Every displayed abscissa is an invertible Möbius function of z, and its ordinate is a nonzero rational function of z times the indicated character. It therefore recovers that character and defines a birational map of genus-one curves. All maps extend uniquely between the smooth projective models. The inverse image of the point at infinity is a rational point on the smooth quotient; this point supplies the origin used for its Jacobian. More explicitly, the origins in the seven rows lie above $z = \infty, \infty, \infty, \infty, 0, 2T, 0$, respectively. Each is the unique ramified point of its genus-one double cover above the indicated value.

The involution

$$(\ell, P, C, M, B) \longmapsto (d/\ell, B, M, C, P) \tag{100}$$

exchanges the PCB and PMB characters, and exchanges CMB and PCM. Composing the two corresponding rows of (99) with this involution gives

$$\begin{aligned} PMB \longrightarrow F_+ : \quad (X,Y) &= \left(\frac{4td}{\ell + s}, \frac{4td\ell PMB}{(\ell+s)^2} \right), \\ PCM \longrightarrow F_- : \quad (X,Y) &= \left(\frac{4ts\ell}{\ell - s}, -\frac{4ts^2\ell PCM}{(\ell-s)^2} \right). \end{aligned}$$

Their origins lie above $z = 2$ and $z = 2t$, respectively. This gives the two remaining entries of (98).

The pullback of the differential of each elliptic quotient lies in its one-dimensional character eigenspace in $H^0(D_t, \Omega^1)$. These nine distinct eigenspaces span that space, since $g(D_t) = 9$. The product norm map from $\mathrm{Jac}(D_t)$ to the nine quotient Jacobians has the direct sum of these differential pullbacks as its cotangent map. It therefore has invertible differential and is an isogeny over $\mathbf{Q}$. Its dual gives the product pullback isogeny and proves (97).

For an elliptic square E^2, the two factor divisors and the diagonal have intersection matrix

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \qquad \det = 2.$$

They give three independent rational divisor classes. Apply this to E_d^2, F_+^2, F_-^2, and take the polarization class on each of G_t, H_+, H_-. The twelve classes are independent on the product. Pullback and the norm along a rational isogeny identify Néron–Severi spaces after tensoring with $\mathbf{Q}$, proving the assertion. □

The genus-three quotient and its three Pryms also follow from the same character decomposition. Let $K \subset V$ be generated by the sign changes of (P, M) and of (C, B). None of the six inertia subgroups meets K nontrivially. Thus $D_t \to D_t/K$ is an unramified cover of degree four and its quotient has genus three. Its invariant field is generated by

$$u = \frac{PM - BC}{2}, \qquad v = \frac{PM + BC}{2}, \qquad z = \frac{u^2 + v^2 + 2}{2} + t - \frac{uv}{t}.$$

Consequently it is the smooth model of

$$C_t : \quad t^2(u^2+v^2)^2 - 4u^2v^2 - 4t^2 d = 0. \tag{101}$$

The characters trivial on K give

$$\mathrm{Jac}(C_t) \sim_{\mathbf{Q}} E_d^2 \times G_t.$$

The three intermediate double covers of C_t have genus five. Writing $\kappa_1 = e_1 + e_3$ and $\kappa_2 = e_2 + e_4$, their Pryms are

$$\begin{aligned} \operatorname{Prym}(D_t/\langle\kappa_1\rangle \to C_t) &\sim_{\mathbf{Q}} F_+ \times F_-, \\ \operatorname{Prym}(D_t/\langle\kappa_2\rangle \to C_t) &\sim_{\mathbf{Q}} F_+ \times F_-, \\ \operatorname{Prym}(D_t/\langle\kappa_1+\kappa_2\rangle \to C_t) &\sim_{\mathbf{Q}} H_+ \times H_-. \end{aligned}$$

The additional characters in these three quotients are respectively (PMB, PCM), (PCB, CMB), and (PB, CM). Removing the three characters of C_t from each quotient proves the displayed isogenies.

7.2 Two-descent and four auxiliary point sets

We recall the elementary forms of two-descent used below; see also [5, 13]. On a split cubic

$$E: \quad y^2 = (x-e_1)(x-e_2)(x-e_3)$$

with distinct rational roots, the Kummer homomorphism has coordinates $[x - e_i]$ in $\mathbf{Q}^\times/\mathbf{Q}^{\times 2}$. At $(e_i, 0)$ its ith coordinate is $[(e_i - e_j)(e_i - e_k)]$; at another root $(e_j, 0)$ it is $[e_j - e_i]$, and at infinity it is 1. The product of the three coordinates is 1. The map has kernel $2E(\mathbf{Q})$. Indeed, the tangent-and-chord law proves multiplicativity, and when all three coordinates are squares one may choose rational roots $r_i^2 = x - e_i$ with $r_1 r_2 r_3 = -y$; the usual halving formulas give a point with abscissa

$$x + r_1 r_2 + r_1 r_3 + r_2 r_3.$$

The same calculation at a root shows that $(e_i, 0)$ has a rational half if and only if both $e_i - e_j$ and $e_i - e_k$ are squares.

For the integral model $y^2 = x^3 + ax^2 + bx$, the one-coordinate isogeny descent sends an ordinary point to $[x]$, sends $(0,0)$ to $[b]$, and sends infinity to 1. Its kernel is the image of the dual isogeny from

$$E': \quad y^2 = x^3 - 2ax^2 + (a^2 - 4b)x.$$

Its image consists of signed squarefree divisors q of b for which

$$W^2 = qU^4 + aU^2V^2 + (b/q)V^4, \qquad \gcd(U, V) = 1, \tag{102}$$

is soluble in integers, together with the stated values at zero and infinity. To obtain this equation write $x = q(U/V)^2$ and substitute in the cubic. A prime not dividing b cannot occur to an odd exponent in x: negative valuations follow from the leading cubic term, and positive valuations from the constant term of $x^2 + ax + b$. These observations give the support assertion and the homogeneous quartic.

If E has full rational two-torsion, write $\alpha(E)$ and $\alpha'(E')$ for these two images. Then

$$2^{\operatorname{rank} E(\mathbf{Q})} = \frac{|\alpha(E)|\,|\alpha'(E')|}{4}. \tag{103}$$

For completeness, the dual compositions are multiplication by two. The other rational two-torsion points of E map onto the kernel of the dual isogeny. The index calculation therefore gives $[E(\mathbf{Q}) : 2E(\mathbf{Q})] = |\alpha(E)|\,|\alpha'(E')|$. Mordell–Weil and $E(\mathbf{Q})[2] \simeq (\mathbf{Z}/2\mathbf{Z})^2$ identify the left side with $2^{\operatorname{rank} E(\mathbf{Q})+2}$, giving (103).

Lemma 7.2. *The following are complete rational point sets:*

$$E_1(\mathbf{Q}) = \{O, (0,0), (1,0), (-1,0)\},$$
$$E_2(\mathbf{Q}) = \{O, (0,0), (2,0), (-2,0)\},$$
$$A(\mathbf{Q}) = \{O, (0,0), (6,0), (8,0)\}, \qquad A:\ y^2 = x(x-6)(x-8),$$
$$E_0(\mathbf{Q}) = \{O, (0,0), (-1,0), (-4,0), (2,\pm 6), (-2,\pm 2)\}, \qquad E_0:\ y^2 = x(x+1)(x+4).$$

Here E_1 and E_2 are the curves $y^2 = x^3 - x$ and $y^2 = x^3 - 4x$.

Proof. For E_1, the first descent image is $\{1, -1\}$. Its dual $y^2 = x^3 + 4x$ has nonnegative rational abscissae, so its image is contained in $\{1, 2\}$. Both classes occur, the latter at $(2, 4)$. Equation (103) gives rank zero. For E_2, the first image is $\{1, -1, 2, -2\}$, all represented by two-torsion. Its dual $y^2 = x^3 + 16x$ has only the candidates $1, 2$. The class 2 would require

$$W^2 = 2U^4 + 8V^4.$$

If U is odd the right side has valuation one at 2; otherwise V is odd and it has valuation three. Thus the second image is $\{1\}$ and the rank is zero.

For every positive rational d, the torsion of E_d is its rational two-torsion. A rational half of the largest root d would require both d and $2d$ square; the other roots have negative differences. The third-division polynomial

$$3x^4 - 6d^2x^2 - d^4$$

cannot vanish at a rational number, since it would give $x^2/d^2 = 1 \pm 2\sqrt{3}/3$. Mazur's torsion theorem [11], in the full-two-torsion case, leaves only $(\mathbf{Z}/2\mathbf{Z})^2$ after orders four and three have been excluded. This completes the point lists for E_1, E_2.

For A, the model has $a = -14, b = 48$. There are no real points with $x < 0$, so the first descent image is contained in $\{1, 2, 3, 6\}$; the four two-torsion points realize all four classes. The dual is $y^2 = x^3 + 28x^2 + 4x$, with candidates $1, -1, 2, -2$. For $q = -1$, equation (102) is nonsquare modulo three for every primitive pair. For $q = 2$ its right side is

$$2((U^2 - V^2)^2 + 16U^2V^2);$$

opposite parity gives valuation one at two, and both odd gives valuation five. For $q = -2$ the right side is

$$-2((U^2 - V^2)^2 - 12U^2V^2),$$

with valuations one and three, respectively. Thus the dual image is $\{1\}$, proving rank zero.

For E_0, the first descent image is $\{1, -1, 2, -2\}$, as witnessed by $O, (-1, 0), (2, 6), (-2, 2)$. Its dual is $y^2 = x(x-1)(x-9)$ and has only the candidates $1, 3$. The second class would give

$$W^2 = 3U^4 - 10U^2V^2 + 3V^4.$$

If both variables are units at three the right side is 2 modulo three; if just one is a unit its valuation is one. Thus the dual image is $\{1\}$ and the rank is zero.

To determine the last two torsion groups, the following table gives all cubic values at the good primes five and seven, in the order $x = 0, 1, \ldots, p-1$:

curve	p	$f(0), \ldots, f(p-1)$	$\#E(\mathbf{F}_p)$
A	5	$0, 0, 3, 0, 2$	4
A	7	$0, 0, 6, 3, 4, 1, 0$	8
E_0	5	$0, 0, 1, 4, 0$	8
E_0	7	$0, 3, 1, 0, 6, 4, 0$	8.

The nonzero squares are $1, 4$ modulo five and $1, 2, 4$ modulo seven. Each point count follows by adding one for infinity and counting the two square roots of each nonzero square. Prime-to-p torsion injects under good reduction. Combining the two primes bounds the torsion orders by four and eight, respectively: the five-primary part is excluded using seven, and the seven-primary part using five. The displayed rational points attain these bounds, completing both point lists. □

7.3 Torsion of the six factors

We shall need a convenient parametrization of a rational flex.

Lemma 7.3. *If $y^2 = x(x-A)(x-B)$ is nonsingular and has a rational point of order three, then there is $k \in \mathbf{Q} \setminus \{0, 1, -1, -\frac{1}{2}, -2\}$ such that*

$$\frac{B}{A} = \frac{1+2k}{(1-k)(1+k)^3}. \tag{104}$$

There is also a nonzero rational r with $A = r^2(1-k^2)$.

Proof. Write the tangent at the order-three point (x, y) as $h(X) = m(X-x)+y$. Since it meets the cubic with multiplicity three,

$$X(X-A)(X-B) - h(X)^2 = (X-x)^3.$$

For each root $e \in \{0, A, B\}$, the rational number $r_e = h(e)/(x-e)$ satisfies $r_e^2 = x - e$. These numbers are nonzero and distinct. The identity $h(e) = y - mr_e^2 = r_e^3$ shows that they are the three roots of $R^3 + mR^2 - y$. Hence their pairwise products sum to zero. Put $r = r_0$ and $k = r_A/r$. Then $r_B/r = -k/(1+k)$, and

$$A = r^2(1-k^2), \qquad B = r^2\frac{1+2k}{(1+k)^2}.$$

This gives (104). The values $0, \pm 1$ contradict nonzero roots or $A \neq 0$; $-1/2$ makes $B = 0$, and -2 makes $A = B$. □

Lemma 7.4. *For rational $t \notin \{0, 1, -1\}$, the curve G_t has no rational three-torsion. For $t > 1$ the following additional assertions hold:*

1. *if T is a rational square, neither H_+ nor F_- has rational three-torsion;*
2. *if T is twice a rational square, F_+ has no rational three-torsion.*

Proof. Set

$$D = (1-k)(1+k)^3, \quad R = \frac{1+2k}{D}, \quad N = 2 + 4k - 2k^3 - k^4.$$

Then

$$R - 1 = \frac{k^3(k+2)}{D}, \qquad R + 1 = \frac{N}{D}. \tag{105}$$

Suppose first that H_+ has rational three-torsion and T is a square. Take $A = 2T, B = T^2$ in Lemma 7.3. It gives $T/2 = R$ and

$$r^2 = \frac{4(1+2k)}{(1-k)^2(1+k)^4}.$$

Thus $1+2k=a^2$ and $1-k^2=2b^2$ for rational a,b, whence $8b^2=3+2a^2-a^4$. Write $a=U/V$ in lowest terms. The rational number $2bV^2$ is an integer, since its square multiplied by two is integral. Calling it W gives

$$2W^2 = 3V^4 + 2U^2V^2 - U^4.$$

Modulo sixteen the right side is 4 if U,V are both odd, is 7 or 15 if only U is odd, and is 3 or 11 if only V is odd. None is $0,2,8$, the possible residues of $2W^2$.

For F_- use $A=2d, B=2td$, so $t=R$. If T is square, (105) and $A=r^2(1-k^2)$ show that

$$\frac{r^2}{T} = \left(\frac{k}{(1-k)(1+k)^2}\right)^2 2k(k+2).$$

Hence $2k(k+2)$ is square. If $v_2(k)<0$ its valuation is $1+2v_2(k)$, and if $v_2(k)=0$ its valuation is one. Consequently $v_2(k)>0$. But then D is a dyadic unit and $N\equiv 2 \pmod 8$; thus $v_2(T)=v_2(N/D)=1$, contradicting that T is square.

For F_+ take $A=2d, B=-2td$, so $-t=R$. If T is twice square, (105) gives

$$-2k(k+2)(1-k^2) = 2T\left(\frac{(1-k)(1+k)^2}{k}\right)^2 \in \mathbf{Q}^{\times 2}.$$

On the quartic $w^2=-2k(k+2)(1-k^2)$, the transformation

$$X = 4 - 4/k, \qquad Y = -4w/k^2$$

gives the curve A of Lemma 7.2. The branch point $k=0$ maps to infinity. The quartic has no rational points at infinity, since its leading coefficient is 2. The complete point list on A therefore leaves just $k=0,-2,1,-1$, all excluded by Lemma 7.3.

Finally, rational three-torsion on G_t would give $A=d^2, B=-d$, and hence $R=-1/d$. Since $d+1=t^2$,

$$t^2 = \frac{R-1}{R} = \frac{k^3(k+2)}{1+2k}.$$

Consequently

$$(X,Y) = \left(2k, \frac{2t(1+2k)}{k}\right)$$

is a point on E_0. Its complete list in Lemma 7.2 leaves $k\in\{0,-\frac{1}{2},-2,1,-1\}$, again impossible. □

7.4 Positive rank on every factor

Theorem 7.5. *Let a strict rational-distance point have product parameter $t=r_0r_3+r_1r_2$, and let the curves be as in* (96)*. Each of $E_d, G_t, F_+, F_-, H_+, H_-$ has positive rational rank. In particular,*

$$\operatorname{rank}\operatorname{Jac}(D_t)(\mathbf{Q}) \geq 9. \tag{106}$$

Proof. Every ordinate in (99) is nonzero at the strict point. The point $(-s\ell, sPM\ell)$ on E_d therefore has infinite order, by the torsion argument in Lemma 7.2. If d were square, E_d would be rationally isomorphic to E_1; if $2d$ were square it would be isomorphic to E_2. Both are rank zero. Thus

$$d \notin \mathbf{Q}^{\times 2}, \qquad 2d \notin \mathbf{Q}^{\times 2}. \tag{107}$$

We next exclude rational order four on the remaining curves. A two-torsion point can have a rational half only at the largest root, because any other root has a negative root difference. For F_+ its half would require $2d$ square. For H_+ and H_- it would require d square, after removing square factors. For G_t it would require $d^2 + d = dt^2$ square. These contradict (107).

For F_- the largest root is $2td$. Its half requires both $2td$ and $2d(t-1)$ square. Their quotient gives $t/(t-1)$ square, so $b^2 = t(t-1)$ for a nonzero rational b; the second condition also gives $a^2 = 2(t+1)$. Then $(X, Y) = (a^2, 2ab)$ lies on

$$Y^2 = X(X-2)(X-4)$$

with nonzero ordinate. Translation $X \mapsto X - 2$ identifies this curve with E_2, contradicting its point list.

Mazur's theorem now implies that torsion on each of the five curves is either $(\mathbf{Z}/2\mathbf{Z})^2$ or $\mathbf{Z}/2\mathbf{Z} \times \mathbf{Z}/6\mathbf{Z}$. Lemma 7.4 excludes the second case for G_t, so its nonzero-ordinate image is of infinite order. For the other four curves, odd torsion is divisible by two. Hence its Kummer image is trivial, and the Kummer image of all torsion is just the image of the rational two-torsion.

The common roots yield the identities

$$\begin{aligned} X_{F_+} - 2d &= 2(TC/P)^2, & X_{F_-} - 2d &= 2(sB/M)^2, \\ X_{H_+} &= TP^2, & X_{H_-} - 2s &= -sC^2. \end{aligned} \tag{108}$$

On H_-, the possible torsion classes of the coordinate $X - 2s$ are 1 and $[-2s]$. The actual class $[-s]$ differs from the first by sign and from the second by [2]. This image has infinite order. On H_+, the torsion classes of X are $1, [2T]$. The actual class $[T]$ differs from both unless T is square. In that case Lemma 7.4 excludes order three, and the nonzero ordinate excludes two-torsion.

For F_-, the torsion classes of $X - 2d$ are

$$1, \quad [-2d], \quad [-s], \quad [2T].$$

The actual class [2] can meet this list only when T is square; the same lemma handles that case. For F_+ the list is

$$1, \quad [-2d], \quad [T], \quad [-2dT].$$

The actual class [2] meets it only when T is twice square, which is the remaining case of that lemma. Thus all six ranks are positive. Their multiplicities in (97) give $2+1+2+2+1+1 = 9$. □

7.5 Two independent images and the rank strata

Two quotient maps can furnish independent points on a single elliptic curve. The following argument uses the full Kummer rows of the points and of their sum. We write pairs of rational numbers for their classes in $(\mathbf{Q}^\times/\mathbf{Q}^{\times 2})^2$.

On E_d, take the two diagonal images

$$Q = (-s\ell, sPM\ell), \qquad R = (T\ell, TBC\ell).$$

For the coordinates $(X, X - d)$ their rows are

$$\delta Q = (-s\ell, -s), \qquad \delta R = (T\ell, T), \qquad \delta(Q+R) = (-d, -d). \tag{109}$$

For example $X(Q) - d = -sP^2$ and $X(R) - d = TC^2$. The four torsion rows are

$$(1,1), \quad (-1,-d), \quad (d,2), \quad (-d,-2d). \tag{110}$$

Since $d > 0$ is nonsquare, the sum row in (109) is absent from this list. Comparing the individual rows, with signs retained, shows that either one can meet a torsion row only if T is square or twice square. Thus

$$T \notin \mathbf{Q}^{\times 2} \cup 2\mathbf{Q}^{\times 2} \quad \Longrightarrow \quad \operatorname{rank} E_d(\mathbf{Q}) \geq 2. \tag{111}$$

Indeed the images of Q, R in $E_d(\mathbf{Q})/(2E_d(\mathbf{Q}) + E_d(\mathbf{Q})_{\mathrm{tors}})$ are then nonzero and distinct, hence linearly independent over $\mathbf{F}_2$.

For F_+ and F_-, take the image in (99) and its image after the involution (100). For both pairs the coordinates $(X, X - 2d)$ give

$$\delta Q_\pm = (tT\ell, 2), \qquad \delta R_\pm = (ts\ell, 2), \qquad \delta(Q_\pm + R_\pm) = (d, 1). \tag{112}$$

The first-coordinate products are, explicitly,

$$X(Q_+)X(R_+) = d\left(\frac{4tT}{PB}\right)^2, \qquad X(Q_-)X(R_-) = d\left(\frac{4ts}{MC}\right)^2.$$

The second coordinates follow from (108). The complete torsion rows are

$$\begin{array}{c|cccc} F_+ & (1,1) & (-t,-2d) & (2d,T) & (-2td,-2dT) \\ F_- & (1,1) & (t,-2d) & (2d,-s) & (2td,2T). \end{array} \tag{113}$$

For F_+, the sum row $(d, 1)$ never occurs: the positive rows would require d or 2 to be square. An individual row in (112) can occur only if T is twice square. It follows that

$$T \notin 2\mathbf{Q}^{\times 2} \quad \Longrightarrow \quad \operatorname{rank} F_+(\mathbf{Q}) \geq 2. \tag{114}$$

For F_-, an individual row can meet torsion only when T is square. The sum row can meet torsion only if both $2t$ and $2T$ are squares. Therefore

$$\begin{aligned} T \notin \mathbf{Q}^{\times 2}, \qquad & \text{at least one of } 2t, 2T \text{ is nonsquare} \\ \Longrightarrow \quad & \operatorname{rank} F_-(\mathbf{Q}) \geq 2. \end{aligned} \tag{115}$$

Theorem 7.6 (Rank strata)**.** *For the product parameter of any strict rational-distance point, the following lower bounds hold:*

condition	$\operatorname{rank} E_d$	$\operatorname{rank} F_+$	$\operatorname{rank} F_-$	$\operatorname{rank} \mathrm{Jac}(D_t)$
$T \notin \mathbf{Q}^{\times 2} \cup 2\mathbf{Q}^{\times 2}$	2	2	2	15
$T \in \mathbf{Q}^{\times 2}$	1	2	1	11
$T \in 2\mathbf{Q}^{\times 2},\ 2t \notin \mathbf{Q}^{\times 2}$	1	1	2	11
$T \in 2\mathbf{Q}^{\times 2},\ 2t \in \mathbf{Q}^{\times 2}$	1	1	1	9.

(116)

Every entry in the table is a lower bound.

Proof. Combine (111), (114), and (115) with Theorem 7.5. Formula (97) gives the rank identity

$$2\operatorname{rank} E_d + \operatorname{rank} G_t + 2\operatorname{rank} F_+ + 2\operatorname{rank} F_- + \operatorname{rank} H_+ + \operatorname{rank} H_-.$$

Substitution gives the final column. The four rows exhaust the positive rational values of T, since 2 is nonsquare. □

The theorem gives a useful exclusion criterion for a fixed parameter: a rank bound below the relevant entry in (116) excludes a strict point on that fibre. It also places every fibre arising from a strict point outside the classical Chabauty inequality $\operatorname{rank} \mathrm{Jac}(D_t)(\mathbf{Q}) < g(D_t)$.

8 Descent with a common source

The square conditions at different places constrain the same rational points. This section retains that dependence through finite quotients of the Mordell–Weil group. The first construction combines characters modulo 4 into an invariant of a complete coset modulo 2. A second application uses a quotient modulo 8 to exclude the parameter $n = 15569$.

8.1 The source coset and the formal parameter

Write

$$E_n: \quad y^2 = x(x-n)(x+n), \qquad T_0 = (0,0), \quad T_+ = (n,0), \quad T_- = (-n,0),$$

where $n > 1$ is odd and squarefree. The ordered Kummer map is

$$\delta : E_n(\mathbf{Q})/2E_n(\mathbf{Q}) \longrightarrow (\mathbf{Q}^*/\mathbf{Q}^{*2})^2, \qquad P \longmapsto ([x(P)], [x(P)-n]).$$

Its values at the two-torsion points are

P	O	T_0	T_+	T_-
$\delta(P)$	$(1,1)$	$(-1,-n)$	$(n,2)$	$(-n,-2n).$

(117)

Products and equalities in Kummer coordinates are understood modulo rational squares. To recall the kernel statement, let e_1, e_2, e_3 be the roots $0, n, -n$. Intersecting a line with the cubic shows that, for each i, the product of $x - e_i$ over the three intersection points is a square. This proves multiplicativity of each Kummer coordinate. If all $x(P) - e_i$ are squares, choose $r_i^2 = x(P) - e_i$ with $r_1 r_2 r_3 = -y(P)$, and put $s_1 = r_1 + r_2 + r_3$, $s_2 = r_1 r_2 + r_1 r_3 + r_2 r_3$. The point

$$(x(P) + s_2, -y(P) - s_1 s_2)$$

doubles to P, as substitution and its tangent of slope $-s_1$ show. The formulas extend at two-torsion by taking one $r_i = 0$. Thus the kernel is $2E_n(\mathbf{Q})$.

For $j \geq 2$, put

$$E_j(\mathbf{Q}_2) = \{O\} \cup \{P \in E_n(\mathbf{Q}_2) : v_2(x(P)) \leq -2j\}, \qquad H = E_n(\mathbf{Q}) \cap E_2(\mathbf{Q}_2).$$

Thus $E_j(\mathbf{Q}_2)$ is the indicated neighbourhood of the point at infinity. For a nonzero $P \in H$, its *depth* is $j(P) = -v_2(x(P))/2$. In the general criterion below we assume

$$E_n(\mathbf{Q}) = H \oplus E_n[2](\mathbf{Q}). \tag{118}$$

This specifies the subgroup and every torsion translation used in the argument.

Here are the formal-group estimates we shall need. With $z = -x/y$ and $-1/y = z^3 W$, the equation of E_n gives

$$W = 1 - n^2 z^4 W^2, \qquad W \in 1 + z^4 \mathbf{Z}_2[[z^4]], \qquad x = z^{-2} W^{-1}. \tag{119}$$

Moreover

$$\frac{dx}{2y} = \left(1 + \frac{zW'}{2W}\right) dz, \qquad \ell(z) = \int_0^z \frac{dx}{2y} \in z + 2z^5 \mathbf{Z}_2[[z^4]].$$

Indeed, the coefficients of $zW'/(2W)$ are even and its exponents are positive multiples of 4; integration divides by odd integers. The formal logarithm is additive and isometric on $4\mathbf{Z}_2$. Consequently H is torsion-free, its nonzero depths are integers, and

$$x(P), \quad x(P) - n, \quad x(P) + n \in \mathbf{Q}_2^{*2} \qquad (P \in H \setminus \{O\}). \tag{120}$$

The last assertion follows from (119), since all the unit corrections are 1 modulo 8.

Choose $\nu \geq 2$ with $H \subseteq E_\nu(\mathbf{Q}_2)$, and define

$$\phi_0(P) = 2^{-\nu}\ell(P) \pmod 4, \qquad \eta_0(P) = \phi_0(P) \pmod 2. \tag{121}$$

These are homomorphisms. Formula (119) also permits $\ell(P)$ to be replaced by $z(P)$ in (121).

We consider rational points $Q = (a, b)$ and $R = (c, e)$ satisfying

$$-n < a < 0 < c - n, \qquad \delta(Q)\delta(R) = (-n, -n), \qquad f(Q, R) = \frac{a + c}{2(c + n)}. \tag{122}$$

In particular $f(Q, R) > 0$. The correspondence of Theorem 5.2 sends a rational-distance configuration to such a pair with $f(Q, R)$ a nonzero rational square.

Suppose $n \equiv 1$ or $7 \pmod 8$, and choose $W \in H$ with

$$\delta(W) = \begin{cases} (n, 1), & n \equiv 1 \pmod 8, \\ (-n, -n), & n \equiv 7 \pmod 8. \end{cases} \tag{123}$$

Write $Q = U + \tau_Q$ and $R = V + \tau_R$ according to (118). The two-torsion rows are distinct over $\mathbf{Q}_2$. Equations (120) and (122) therefore give $\tau_Q + \tau_R = T_0$ when $n \equiv 1$, and $\tau_Q = \tau_R$ when $n \equiv 7$. Applying (117) over $\mathbf{Q}$ then gives

$$U + V \in W + 2H. \tag{124}$$

Thus the undetermined part of the source is one element of $2H$.

8.2 Local conditions on the two remainders

The following calculation extracts exactly the information needed from the place 2. Put $u = x(U)$ and $v = x(V)$. Translation by two-torsion gives

$$x(P + T_0) = -\frac{n^2}{x(P)}, \quad x(P + T_+) = n + \frac{2n^2}{x(P) - n}, \quad x(P + T_-) = -n + \frac{2n^2}{x(P) + n}. \tag{125}$$

For $n \equiv 1$, the orders (T_0, O) and (O, T_0) have respectively $v_2(f) = -1$ and $v_2(f) = v_2(u) - 1$. For $n \equiv 7$, the orders (T_+, T_+) and (T_-, T_-) have respectively $v_2(f) = -1$ and $v_2(f) = v_2(v) - 1$. These four valuations are odd. The other four orders give

$n \pmod 8$	(τ_Q, τ_R)	$f(Q, R)$
1	(T_-, T_+)	$n(u + v)/(2v(u + n))$
1	(T_+, T_-)	$(u + v)/(2(u - n))$
7	(O, O)	$(u + v)/(2(v + n))$
7	(T_0, T_0)	$-n(u + v)/(2u(v - n))$.

(126)

Each expression in the last column differs from $(u + v)/2$ by a $\mathbf{Q}_2$-square. If U, V have different depths, $u + v$ has even valuation and $(u + v)/2$ has odd valuation. If their depths agree, write $u = 2^{-2j}r^2$, $v = 2^{-2j}s^2$ with $r, s \in \mathbf{Z}_2^*$; then $v_2(r^2 + s^2) = 1$. We obtain

$$v_2(f) \text{ even} \quad \Longrightarrow \quad j(U) = j(V), \qquad \phi_0(V) = \pm\phi_0(U). \tag{127}$$

For the last assertion, at depth ν both character values are odd; at depth $\nu + 1$ both equal 2; at larger depths both vanish.

We next specify the odd primes used in the criterion. Let $q \equiv 5 \pmod 8$ be a prime with $q \nmid n$ and $n \in \mathbf{Q}_q^{*2}$. We allow either of the following reduction conditions on the entire subgroup H:

1. $\mathrm{red}_q(H) \subseteq \{O, T_0\}$. Write $\mathrm{red}_q(P) = \zeta_q(P)T_0$ and set $\phi_q = 2\zeta_q : H \longrightarrow \mathbf{Z}/4\mathbf{Z}$.

2. There is $A_q \in E_n(\mathbf{F}_q)$ with $2A_q = T_0$ such that $\mathrm{red}_q(H) \subseteq \langle A_q \rangle$. Define $\mathrm{red}_q(P) = \phi_q(P)A_q$ and put $i_q = x(A_q)/n$. Require that

$$\begin{cases} (1 - i_q)/2, & n \equiv 1 \pmod 8, \\ (1 + i_q)/2, & n \equiv 7 \pmod 8 \end{cases} \quad \text{is a nonsquare in } \mathbf{F}_q. \tag{128}$$

In the second case $A_q + T_0 = -A_q$, so (125) gives $i_q^2 = -1$. In either case put $\eta_q = \phi_q \bmod 2$.

Lemma 8.1. *For such a prime, $\eta_q(W) = 0$. If a pair (122) has $v_2(f)$ even and $f \in \mathbf{Q}_q^{*2}$, then*

$$\phi_q(U) = \phi_q(V) \in \{0, 2\}.$$

Proof. Only the cyclic case requires a proof that $\eta_q(W) = 0$. An odd multiple of A_q has first Kummer coordinate ni_q, a nonsquare since $q \equiv 5 \pmod 8$. The first coordinate of $\delta(W)$ is n or $-n$, both local squares. Thus W has even reduction coordinate. Equation (124) now implies $\eta_q(U) = \eta_q(V)$.

If both parities are odd, their abscissae reduce to ni_q. Translation by T_0 preserves this abscissa, whereas translation by T_+ or T_- changes it to $-ni_q$. The four orders in (126) give exactly the residues in (128). Their numerators and denominators are units, so this case is excluded.

Both remainders therefore reduce to O or T_0. Suppose their reductions differ. When $n \equiv 1 \pmod 8$, the corresponding Q, R both reduce to T_+ or both to T_-. In the first case f has residue $1/2$. In the second case use the exact identity

$$(2e)^2 f = 2(a + c)c(c - n).$$

The right side is a unit with residue $-8n^3$, of squareclass 2. The identity applies at every positive depth of approach to T_-, because $e \neq 0$.

When $n \equiv 7 \pmod 8$, Q, R reduce in opposite orders to O, T_0. Near O the abscissa has negative even valuation and square unit part; near T_0 it has positive even valuation and square unit part. These facts follow by writing the cubic as $y^2 = x^3(1-n^2/x^2)$ or as $y^2 = -n^2x(1-x^2/n^2)$, respectively. The abscissa near O dominates $a + c$, and substitution in f again gives squareclass 2. Since 2 is a nonsquare at q, the reductions cannot differ. Their equal labels O or T_0 give the asserted equal character values. □

8.3 A secondary invariant of the complete source

Take finitely many primes $q_1, \ldots, q_s$ satisfying the preceding conditions, and abbreviate their characters to ϕ_j, η_j. Set

$$V_H = H/2H, \qquad M : V_H \longrightarrow \mathbf{F}_2^{s+1}, \qquad M(\bar{Z}) = (\eta_0(Z), \ldots, \eta_s(Z)).$$

For $\kappa \in \ker M$, choose a representative $W \in H$ and define

$$b(W) = \left(\frac{\phi_0(W)}{2}, \ldots, \frac{\phi_s(W)}{2} \right), \qquad \beta(\kappa) = b(W) \pmod{\mathrm{im}\, M}. \tag{129}$$

Division by 2 denotes the isomorphism $2(\mathbf{Z}/4\mathbf{Z}) \simeq \mathbf{F}_2$. For any $Z \in H$,

$$b(W + 2Z) = b(W) + M(\bar{Z}). \tag{130}$$

Thus (129) defines a linear map $\beta : \ker M \longrightarrow \mathrm{coker}\, M$.

Theorem 8.2 (Common-source character criterion)**.** *Assume* (118)*, $n \equiv 1$ or 7 (mod 8), and the reduction hypotheses above. Let W satisfy* (123) *and $\eta_0(W) = 0$, and put $\kappa = \bar{W}$. If $\beta(\kappa) \neq 0$, no pair* (122) *has both $v_2(f)$ even and $f \in \mathbf{Q}_{q_j}^{*2}$ for every j.*

Equivalently, an exclusion certificate is a vector $\mathbf{c} = (c_0, \dots, c_s) \in \mathbf{F}_2^{s+1}$ such that

$$\sum_{j=0}^{s} c_j \eta_j = 0 \text{ on } H, \qquad \frac{1}{2}\sum_{j=0}^{s} c_j \phi_j(W) = 1 \quad \text{in } \mathbf{F}_2.$$

Proof. Suppose a pair exists. By (127), one of $K = U - V$ and $K = U + V$ satisfies $\phi_0(K) = 0$. By Lemma 8.1, every other row is equal and even on U, V; both their sum and their difference vanish modulo 4. Hence this single choice of K satisfies $\phi_j(K) = 0$ for all j. Both choices lie in $W + 2H$, so $K = W + 2Z$ for some $Z \in H$. Dividing $0 = \phi_j(W) + 2\phi_j(Z)$ by 2 gives $b(W) = M(\bar{Z})$, contrary to $\beta(\kappa) \neq 0$.

Finally, a vector lies outside $\operatorname{im} M$ precisely when a linear functional annihilating that image takes value 1 on it. Writing the functional as dot product with $\mathbf{c}$ gives the stated certificate. The parity relation makes $\frac{1}{2}\sum c_j \phi_j$ a homomorphism $H/2H \to \mathbf{F}_2$, so its value is independent of the representative W. □

For example, if $s = 1$ and $\eta_1 = \eta_0$, the secondary character is

$$\Delta = \frac{\phi_1 - \phi_0}{2} : H/2H \longrightarrow \mathbf{F}_2.$$

On $\ker \eta_0$, both individual half-characters change by $\eta_0(Z)$ under $W \mapsto W + 2Z$. Their difference is unchanged. This explains why the same correction must be retained at both places.

8.4 An exclusion using the common quotient modulo eight

Theorem 8.3. *Let $n = 15569$. There are no finite rational points $Q = (a, b)$, $R = (c, e)$ satisfying* (122) *for which $f(Q, R)$ is a square in each of $\mathbf{Q}_2, \mathbf{Q}_3, \mathbf{Q}_{113}$.*

Proof. We first determine the exact quotient of the rational group that is needed. The integer n is prime and is 1 modulo 8. One short primality certificate is

$$n - 1 = 112 \cdot 139, \quad 2^{112} \equiv 6064, \quad 2^{n-1} \equiv 1 \pmod{n}, \quad 803n - 2062 \cdot 6063 = 1.$$

Since 139 is prime, every prime divisor p of n has $139 \mid p - 1$; as $139 > \sqrt{n}$, this proves primality. For a rational point, both Kummer coordinates are supported on $-1, 2, n$: at an odd prime not dividing n, either all three cubic factors have the same negative valuation or at most one has positive valuation, and the equation forces the relevant valuations to be even.

Over $\mathbf{Q}_2$, choose $v^2 = n$ and use $X = x/n$, $Y = y/(nv)$. The curve becomes $Y^2 = X(X - 1)(X + 1)$, with the same local Kummer coordinates. Its non-two-torsion points have the following rows; the two-torsion centres have the rows in (117).

condition	δ in $(\mathbf{Q}_2^*/\mathbf{Q}_2^{*2})^2$
$v_2(X) \leq -4$	$(1, 1)$
$v_2(X) = -2$	$(1, 5)$
$v_2(X) = 2$	$(-1, -5)$
$v_2(X) \geq 4$	$(-1, -1)$
$X \equiv 1 \pmod 8$, $v_2(X - 1) = 3$	$(1, 10)$
$X \equiv 1 \pmod 8$, $v_2(X - 1) \geq 5$	$(1, 2)$
$X \equiv -1 \pmod 8$, $v_2(X + 1) = 3$	$(-1, -10)$
$X \equiv -1 \pmod 8$, $v_2(X + 1) \geq 5$	$(-1, -2)$.

(131)

In the first four rows the nonzero valuation is even; in the last four rows the displayed positive valuation is odd. The table follows from even valuation and unit part 1 modulo 8 in the cubic. For example, $X = u/4$ forces $u \equiv 1 \pmod 8$, $X = 4u$ forces $u \equiv 7 \pmod 8$, and near ± 1 the other two factors have valuations $0, 1$. At depth 3 the additional unit class is 5; at greater odd depth it is a square. These cases exhaust the parity of the valuations and the odd unit residues.

Since n is a $\mathbf{Q}_2$-square, a globally supported Kummer coordinate cannot have the extra local unit class 5. The four remaining rows are precisely the distinct rows of $E_n[2]$. Every rational point has a unique two-torsion translate in H, proving (118). Furthermore

$$\delta(H) \subseteq \{1, n\}^2, \qquad \ker(\delta|_H) = 2H.$$

Thus $\dim_{\mathbf{F}_2} H/2H \le 2$.

Two points $P, G \in H$ are specified by the following factorizations. The three entries in each row are $x, x - n, x + n$:

	d	d^2x	$d^2(x-n)$	$d^2(x+n)$
P	7760	$n\,7809^2$	108983^2	$n\,11009^2$
G	13382920	$n\,13522049^2$	$n\,1934751^2$	2373852137^2.

(132)

Choose the negative ordinate in each row:

$$y(P) = -\frac{n(7809)(108983)(11009)}{7760^3}, \qquad y(G) = -\frac{n(13522049)(1934751)(2373852137)}{13382920^3}.$$

The displayed differences are n, and their products are these ordinates squared. Hence the two rows specify rational points with

$$\delta(P) = (n, 1), \qquad \delta(G) = (n, n).$$

They form a basis of $H/2H$. The depths are $j(P) = 4$, $j(G) = 3$. The homomorphism $\ell/4 \pmod 2$ vanishes on these representatives and therefore on H, proving $H \subseteq E_3(\mathbf{Q}_2)$. Every element of H has a representation

$$h = kG + mP + 8h_1, \qquad 0 \le k, m < 8. \tag{133}$$

Indeed, successive substitution of $H = \{0, G, P, G + P\} + 2H$ three times gives this assertion. Direct reduction of the fractions in (132) gives

$$z(G)/8 \equiv 7, \qquad z(P)/8 \equiv 6 \pmod 8. \tag{134}$$

The formal logarithm shows that $z(h)/8 \pmod 8$ is a homomorphism on $H/8H$.

Every nonzero H-point has positive first Kummer coordinate and hence lies on the real identity component. Equations (122) and (125), followed by the dyadic valuation test, leave only

$$Q = U + T_-, \qquad R = V + T_+, \qquad U + V \in P + 2H. \tag{135}$$

Write

$$U = kG + mP + 8h_1, \qquad V = k'G + m'P + 8h_2.$$

The source relation is

$$k + k' \equiv 0, \qquad m + m' \equiv 1 \pmod 2. \tag{136}$$

At 3, $n \equiv 2$, and $E_n(\mathbf{F}_3)$ consists of its four two-torsion points. Their local Kummer rows are

$$O : (1, 1), \quad T_0 : (2, 1), \quad T_+ : (2, 2), \quad T_- : (1, 2).$$

The entries $1, 2$ denote unit squareclasses in $\mathbf{Q}_3$. Every root neighbourhood has an even positive offset valuation and the indicated constant unit class; near O the negative valuation is even. Since $-n$ is a local square, Q, R have equal Kummer rows and therefore the same reduction. A common reduction T_+ gives $f \equiv 1/2$, a nonsquare. For common reduction T_- the identity $(2e)^2 f = 2(a+c)c(c-n)$ has nonsquare unit residue, including every positive pole depth. Thus a square return requires their common reduction to be O or T_0.

The rational coordinates give $\operatorname{red}_3 G = T_+$ and $\operatorname{red}_3 P = T_0$. In view of (135), both k and k' must be odd. It follows from (134) that U, V have depth exactly 3. Put

$$\alpha = z(U)/8 \equiv 7k + 6m, \qquad \gamma = z(V)/8 \equiv 7k' + 6m' \pmod 8.$$

Both are odd. Writing $u = 2^{-6} r^2$ and $v = 2^{-6} s^2$ in the formal chart gives $r \equiv \alpha^{-1}$ and $s \equiv \gamma^{-1}$ (mod 8). The square condition at 2 is

$$\frac{r^2 + s^2}{2} \equiv 1 \pmod 8 \quad \Longleftrightarrow \quad \alpha \equiv \pm\gamma \pmod 8. \tag{137}$$

Here odd squares modulo 16 are $1, 9$, so their average is 1 modulo 8 exactly when the two squares agree modulo 16.

It remains to use the same coefficients at 113. At this prime $n \equiv 88 \equiv 38^2$, and the curve is $y^2 = x^3 + 53x$. Its complete group structure follows from the short doubling certificate

A	$2A$	$4A$	$8A$
$(23, 39)$	$(81, 112)$	$(77, 79)$	T_0
B	$2B$	$4B$	$8B$
$(2, 1)$	$(7, 6)$	T_+	O.

(138)

Thus A has order 16 and B order 8. Their cyclic subgroups have different points of order 2, so intersect trivially. They generate a subgroup of order 128. Hasse's bound places the total order between 93 and 135, whence

$$E_n(\mathbf{F}_{113}) \simeq \mathbf{Z}/16\mathbf{Z} \oplus \mathbf{Z}/8\mathbf{Z}, \qquad \exp(2E_n(\mathbf{F}_{113})) = 8.$$

Both coordinates of $\delta(H)$ are squares in $\mathbf{Q}_{113}$, so every element of H is twice a local point. Consequently $\operatorname{red}_{113}(8H) = O$.

Set

$$g = \operatorname{red}_{113} G = (81, 112), \qquad w = \operatorname{red}_{113} P = (36, 58).$$

The addition law gives

$$4g = T_0, \quad w = 2g + T_+, \quad x(g) = 81,\ x(3g) = 16,\ x(g + T_+) = 87,\ x(3g + T_+) = 11.$$

Put $\lambda = k + 2m$ and $\mu = k' + 2m'$ modulo 8. Then $\alpha \equiv -\lambda$, $\gamma \equiv -\mu$, so λ, μ are odd and (137) gives $\mu = \pm\lambda$. Equations (136) and (135) now imply

$$\operatorname{red}_{113} Q = (\lambda + 4)g + (m + 1)T_+, \qquad \operatorname{red}_{113} R = \pm\lambda g + mT_+.$$

The possible abscissae and return residues are therefore

$\bar{a}$	$\bar{c}$	$\bar{f}$	$\bar{f}^{56}$
11	81	21	-1
16	87	71	-1
81	11	29	-1
87	16	101	-1.

(139)

All numerators and denominators are nonzero. For a short check of the last column, $21^4 = 8$ and $21^{28} = 98$ in $\mathbf{F}_{113}$, while $98^2 = -1$. The other entries satisfy $71 = 26^2 \cdot 21$, $29 = 30^2 \cdot 21$, and $101 = 8^2 \cdot 21$. Thus each return is a nonsquare. This contradicts $f \in \mathbf{Q}_{113}^{*2}$ and completes the proof. □

Corollary 8.4. *For a strict interior rational-distance configuration, put $t = r_{00}r_{11} + r_{10}r_{01}$. Then $\mathrm{sf}(t^2 - 1) \neq 15569$.*

Proof. Theorem 5.2 gives the source (122) with $n = \mathrm{sf}(t^2-1)$ and $f(Q, R) = 1/(r_{00}+r_{11})^2$. Apply Theorem 8.3. □

9 A reciprocity obstruction on the complete source curve

We now retain the fourth square in the ordered distance construction. For $p \in \mathbf{Q}$, $p > 1$, put $a = p - 1$ and $b = p + 1$. The complete source is the intersection of three quadrics

$$C_p : \quad pV^2 = 2a(U^2 - pZ^2), \qquad pW^2 = 2b(U^2 + pZ^2), \qquad pT^2 = aU^2 + pbZ^2 \quad \subset \mathbf{P}^4_{[Z:U:V:W:T]}. \tag{140}$$

On $Z \neq 0$, the parameter $\rho = U^2/(pZ^2)$ gives the four ordered squares

$$p\rho, \quad 2(p-1)(\rho-1), \quad 2(p+1)(\rho+1), \quad (p-1)\rho + p + 1. \tag{141}$$

The smooth projective model has genus five: its degree-sixteen map to $\mathbf{P}^1_\rho$ has five branch values $0, 1, -1, -b/a, \infty$, with inertia of order two at each. Thus $2g - 2 = -32 + 5 \cdot 8 = 8$. Write $\mathrm{Loc}(p)$ for the assertion that the four expressions in (141) are simultaneous nonzero squares at a finite $\rho \notin \{0, 1, -1\}$ over each completion of $\mathbf{Q}$. The real solutions can be chosen in the strict distance region: take $\rho > 1$ sufficiently close to 1. Indeed, if $F = a\rho + b$, $u^2 = p\rho$, $v^2 = 2a(\rho - 1)$ and $w^2 = 2b(\rho + 1)$, the strict margin is $2F - 4|u| - |vw|$, whose limit is $4p - 4\sqrt{p} > 0$.

The obstruction below uses every source point on (140). Its formula depends only on the valuations of $p, p-1, p+1$. We use the additive Hilbert symbol notation $(c, d)_v \in \mathrm{Br}(\mathbf{Q}_v)[2]$, and write its bit as $2\,\mathrm{inv}_v(c, d)_v \in \mathbf{F}_2$. For an odd prime q, and $c = q^r c_0, d = q^s d_0$ with units c_0, d_0,

$$(-1)^{2\,\mathrm{inv}_q(c,d)_q} = (-1)^{rs(q-1)/2} \left(\frac{c_0}{q}\right)^s \left(\frac{d_0}{q}\right)^r. \tag{142}$$

For odd $u, v \in \mathbf{Z}_2^\times$, the dyadic rule is

$$2\,\mathrm{inv}_2(2^r u, 2^s v) = \frac{(u-1)(v-1)}{4} + r\frac{v^2-1}{8} + s\frac{u^2-1}{8} \pmod 2. \tag{143}$$

9.1 The obstruction and its arithmetic domain

Assume

$$p(p^2 - 1) = nk^2, \qquad n > 1 \text{ odd and squarefree}, \qquad k \in \mathbf{Q}^\times. \tag{144}$$

For parameters satisfying $\mathrm{Loc}(p)$ define

$$A(p) = \sum_{\substack{q \text{ odd},\ (2/q)=-1 \\ v_q(a)>0}} \frac{v_q(a)}{2}, \qquad B(p) = \sum_{\substack{q \text{ odd},\ (2/q)=-1 \\ v_q(b)>0}} \frac{v_q(b)}{2} \quad \text{in } \mathbf{F}_2. \tag{145}$$

If $v_q(p) < 0$ is odd, choose $e \in \mathbf{F}_q$ with $e^2 = 2$ and define $\sigma_q \in \mathbf{F}_2$ by

$$(-1)^{\sigma_q} = \left(\frac{2+e}{q}\right), \qquad P(p) = \sum_{\substack{q \text{ odd} \\ v_q(p)<0 \text{ odd}}} \sigma_q.$$

This definition is independent of the choice of e, since $(2+e)(2-e) = 2$ is a nonzero square. The existence of e and the integrality of the half-valuations follow from the next lemma. Put

$$\mathcal{T}(p) = \begin{cases} B(p) + P(p), & v_2(p) \neq 0, \\ [v_2(a) = 5] + B(p) + P(p), & p \equiv 1 \pmod{16}, \\ [v_2(b) = 5] + A(p) + P(p), & p \equiv 15 \pmod{16}. \end{cases} \tag{146}$$

Here brackets denote an indicator, and unit congruences are interpreted in $\mathbf{Z}_2^\times$.

Theorem 9.1. *Suppose* (144) *holds. If* Loc(p) *holds and* $\mathcal{T}(p) = 1$*, then* $C_p(\mathbf{Q}) = \varnothing$.

We prove the theorem by constructing a quaternion algebra on a coordinate-omission curve and calculating its invariant at every place. The sum is $\mathcal{T}(p)/2$. A rational point would make this sum zero by Hilbert reciprocity.

Omitting U and W, respectively, gives the genus-one curves

$$D_U: \quad v^2 = 2t^2 - 4pz^2, \qquad aw^2 = 2b(t^2 - 2z^2), \tag{147}$$

$$D_W: \quad V^2 = 2T^2 - 4pZ^2, \qquad aU^2 = pT^2 - pbZ^2. \tag{148}$$

These are projective equations in the four displayed coordinates. Both omission maps are everywhere defined: if all retained coordinates vanish, (140) forces the omitted coordinate to vanish too.

Lemma 9.2. *Under* Loc(p)*, the numbers* p, a, b *are norms from* $\mathbf{Q}(\sqrt{2})$*. At an odd prime where* 2 *is nonsquare their valuations are even. Under* (144)*, the possible dyadic branches are*

$$v_2(p) \geq 4 \text{ even}, \quad v_2(p) \leq -4 \text{ even}, \quad p \equiv 1 \pmod{16}, \quad p \equiv 15 \pmod{16}.$$

In the last two branches, respectively, $[a]_2 = [2]_2$ *and* $[b]_2 = [2]_2$*, and the positive valuation is odd and at least five.*

Proof. At a completion where 2 is nonsquare, a point of (147) has $z, w \neq 0$. Its equations give the exact norm witnesses

$$\mathrm{N}\left(\frac{v + t\sqrt{2}}{2z(1+\sqrt{2})}\right) = p, \qquad \mathrm{N}\left((2+\sqrt{2})\frac{b}{w}(t + z\sqrt{2})\right) = ab.$$

At an odd inert prime the valuation of a norm is even. At a pole of p, the valuations of p, a, b agree. Away from poles, a, b are integral and cannot both have positive valuation. The even valuation of ab therefore makes each positive valuation even. Thus $(2, a)_q = (2, b)_q = 0$ at every odd place; their real invariants also vanish. Reciprocity gives vanishing at 2. The quadratic norm criterion gives the asserted global norms.

The norm classes at 2 are $1, -1, 2, -2$. Since $v_2(pab)$ is even, a nonzero $v_2(p)$ is even. Depth 2 is excluded by the unit part of a: if p has depth 2 it is 3 (mod 8), and if $p = u/4$ with $u \equiv 1, 7$ (mod 8), it is $5, 3$ (mod 8). For a unit p, its norm condition gives $p \equiv 1, 7$ (mod 8); the norm

conditions on b, a sharpen these to $1, 15 \pmod{16}$. We spell out the four-square restriction at 2. Write $A = v_2(p)$ and $R = v_2(\rho)$. Since $p\rho$ is square, $A + R$ is even. If A, R are both nonzero and have the same sign, the valuation of $2(p-1)(\rho-1)$ is odd. If their signs are opposite and both are even, the same conclusion holds. If their signs are opposite and both are odd, the fourth radicand has a unique term of negative odd valuation: it is $(p-1)\rho$ when $A > 0$, and $p(\rho+1)$ when $A < 0$. Finally $A = R = 0$ is impossible: the first square gives $p \equiv \rho \pmod 8$, so one of the middle two radicands has valuation three. Exactly one of A, R is therefore zero and the other is even.

For a unit p, the middle two squares force both $v_2(p-1)$ and $v_2(p+1)$ odd, so $p \equiv 1$ or $7 \pmod 8$. If $R > 0$, the fourth square excludes $p \equiv 1 \pmod 8$, because its valuation would be one. Thus $[p]_2 = -1$; the product of the middle two squares gives $[p^2-1]_2 = [\rho^2-1]_2 = -1$. If $R < 0$, the fourth square excludes $p \equiv 7 \pmod 8$, because $(p-1)\rho$ would have odd valuation. Now $[p]_2 = 1$ and $[p^2-1]_2 = [\rho^2-1]_2 = 1$. In both cases $[pab]_2 = 1$. For $p \equiv 1 \pmod{16}$, $[p]_2 = 1$ and $[b]_2 = 2$, whence $[a]_2 = 2$; for $p \equiv 15 \pmod{16}$, $[p]_2 = -1$ and $[a]_2 = -2$, whence $[b]_2 = 2$. The corresponding valuation is odd and at least five. □

9.2 A rational anchor and the quaternion class

Temporarily suppose an actual rational point $Q_0 = [z : v : w : t] \in D_U(\mathbf{Q})$ is given. Its z, w, t are nonzero. For instance, $z = 0$ gives $v^2 = 2t^2$, and $w = 0$ gives $t^2 = 2z^2$; $t = 0$ is excluded over $\mathbf{R}$. Also $v \neq 0$: otherwise $2p = (t/z)^2$, contrary to the even valuation of pab at 2. Define

$$z_1 = aw - 2bz, \qquad u_1 = pb(w + 2t + 2z), \qquad t_1 = -b(aw + 2pt + 2az). \tag{149}$$

This is a rational point on $aU^2 - pT^2 + pbZ^2 = 0$, since

$$au_1^2 - pt_1^2 + pbz_1^2 = 2p^2b(aw^2 - 2bt^2 + 4bz^2) = 0.$$

The linear transformation $[z : t : w] \mapsto [z_1 : u_1 : t_1]$ has determinant $-8p^2b^2$ and hence never produces the zero tuple. Choose the sign of w so that $z_1t - zt_1 \neq 0$. Such a choice exists: the sum for the two signs is $4abz(t+z)$, and $t = -z$ is impossible in (147). Fix this choice. On D_W put

$$L_U = vV - 2tT + 4pzZ, \qquad L_V = au_1U - pt_1T + pbz_1Z, \qquad F = \frac{L_UL_V}{V^2}, \qquad \alpha = (-a, F). \tag{150}$$

Here is the divisor calculation underlying this construction. The projections from D_W to the two conics in (148) have degree two. Let E_U, E_V be the degree-two fibres over their respective rational anchors $[z : v : t]$ and $[z_1 : u_1 : t_1]$, and let H_V be the degree-four coordinate section $V = 0$. A tangent line on a smooth conic meets it twice at its point of tangency. Consequently

$$\operatorname{div}(F) = 2(E_U + E_V - H_V). \tag{151}$$

All residues of $(-a, F)$ vanish, so it defines a class on the smooth proper curve D_W.

For completeness, the associated elliptic curve and ordered classes are

$$E:\ y^2 = x(x-L)(x-R), \qquad L = \frac{8p}{a^2}, \quad R = -\frac{8pb}{a^3}, \qquad \chi = (2, pa, 2pa), \quad \beta = (-a, -a, 1). \tag{152}$$

The row χ describes the actual curve D_W. Indeed, with $\xi_0 = 2aU, \xi_L = 2V, \xi_R = 2T, \zeta = a^2Z$ its equations become

$$2\xi_0^2 - pa\xi_L^2 = L\zeta^2, \qquad 2\xi_0^2 - 2pa\xi_R^2 = R\zeta^2.$$

In particular its degree-four covering map is $x = 8U^2/(a^2Z^2)$, $y = 16pUVT/(a^4Z^3)$. Write T_0, T_L, T_R for the named two-torsion points, and σ_i for the sign change of a coordinate i. The diagonal degree-four covering map is the intrinsic map $P \mapsto [P - \sigma_Z(P)]$ to the Jacobian: its ordered root functions are $2U/(aZ)$, $2V/(a^2Z)$ and $2T/(a^2Z)$, with the constants in χ. It sends the boundary sections $U = 0, V = 0, T = 0, Z = 0$ to $T_0, T_L, T_R, 0$, respectively. Choose a geometric point O_C above $Z = 0$ to compute Abel sums. Then σ_Z acts by negation and the covering map acts by multiplication by two. Each other reflection acts as $P \mapsto t_i - P$, with $2t_i = 0$ because it commutes with σ_Z. At a fixed point of σ_i, the covering image is $2P = t_i$; the boundary images therefore give $t_U = T_0, t_V = T_L, t_T = T_R$. Every fibre of the corresponding conic projection has Abel sum $P + \sigma_i(P) = t_i$, including ramified fibres counted twice. Thus the Abel sums of E_U, E_V are T_0, T_L. Since $H_V \sim 2E_V$, we obtain $[E_U + E_V - H_V] = T_0 - T_L = T_R$.

For the cyclic algebra in (150), the divisor identity (151) identifies the Picard cocycle as $\gamma \mapsto \epsilon_{-a}(\gamma)T_R$: over $\mathbf{Q}(\sqrt{-a})$ the norm of $E_U + E_V - H_V$ is its double, which is $\mathrm{div}(F)$. The Weil pairings of T_R with T_0, T_L, T_R are $-1, -1, 1$, respectively, so the ordered two-descent row is $\beta = (-a, -a, 1)$.

We shall use only the constant-evaluation consequence of the Picard cocycle. Let $k = \mathbf{Q}_v$ and choose any point of $D_W(k)$, which exists by $\mathrm{Loc}(p)$. This point splits the degree map on $\mathrm{Pic}(\overline{D_W})$, giving the Galois module decomposition $\mathrm{Pic}(\overline{D_W}) = E(\overline{k}) \oplus \mathbf{Z}$. The coefficient $-a$ becomes a square over $\overline{k}$, so α lies in the algebraic Brauer group $\mathrm{Br}_1(D_W) = \ker(\mathrm{Br}(D_W) \to \mathrm{Br}(\overline{D_W}))$. The Hochschild–Serre exact sequence [9, §2, (2.1)] contains

$$\mathrm{Br}(k) \longrightarrow \mathrm{Br}_1(D_W) \longrightarrow H^1(k, \mathrm{Pic}(\overline{D_W})).$$

A Kummer class $\beta \in E(k)/2E(k) \subset H^1(k, E[2])$ maps to zero in $H^1(k, E)$ by the multiplication-by-two exact sequence. Since $H^1(k, \mathbf{Z}) = 0$, an algebraic Brauer class with this Picard cocycle comes from $\mathrm{Br}(k)$ and has constant evaluation. For a torsor the cocycle is computed modulo its torsor class; that class becomes zero as soon as a local point is chosen. Thus it suffices to prove local solubility of the full cover representing β.

9.3 The partner cover at every place

Put $c = -a$. The complete partner cover is

$$cA^2 - cB^2 = L\zeta^2, \qquad cA^2 - C^2 = R\zeta^2. \tag{153}$$

Two useful projective points are

$$[1 : 1 : \sqrt{c} : 0] \qquad \text{if } c \text{ is square,} \tag{154}$$

$$\left[\frac{4p}{a^2}\sqrt{\frac{b}{2p}} : \frac{4p}{a^2} : 0 : 1\right] \qquad \text{if } b/(2p) \text{ is square.} \tag{155}$$

These follow from $(R - L)/c = (4p/a^2)^2$.

At an odd prime with $v_q(p) > 0$, c is square; with $v_q(a) > 0$, $b/(2p)$ is square. At a pole with 2 square, the latter criterion again applies. At a pole with 2 nonsquare, write $p = \pi^{-2h}p_0$, $a = \pi^{-2h}a_0$, $b = \pi^{-2h}b_0$. Replacing (A, B, C) by $(\pi^{2h}A', \pi^{2h}B', \pi^hC')$ gives a unit model whose three elliptic roots reduce to $0, 8/p_0, -8/p_0$. It is a smooth genus-one curve over $\mathbf{F}_q$, so the Hasse bound and Hensel's lemma supply a point. The same argument applies whenever p, a, b are units.

It remains to consider $v_q(b) > 0$. If 2 is square, c is square. Otherwise write $v_q(b) = 2h$ and let B_0 be the unit squareclass of $\pi^{-2h}b$. The Kummer row at T_0 is $(2B_0, 2, B_0)$. If $B_0 = 1$ this

is β. If $B_0 = 2$, choose a residue X with X nonsquare and $X+2$ a nonzero square. There are $(q + \left(\frac{-2}{q}\right))/4 > 0$ such residues: expanding the two character indicators reduces the count to $\sum_X \left(\frac{X(X+2)}{q}\right) = -1$, which follows from $(X+1-Y)(X+1+Y) = 1$. Lifting $x = X$ gives an actual point of $E(\mathbf{Q}_q)$ with row $(2,1,2)$, since $L \equiv -2, R \equiv 0$. Adding T_0, whose row is now $(1,2,2)$, gives β. The simultaneous square roots in the Kummer row give one point of (153).

At 2, the branches of Lemma 9.2 suffice. If $v_2(p) \geq 4$, c is square. If $v_2(p) = -2h \leq -4$, normalize $z = 1$ in the actual rational anchor. Its norm condition leaves $[p]_2 = 1$ or -1. In the first case the equation $v^2 = 2t^2 - 4p$ forces $v_2(t) = 1-h$: a smaller valuation has odd leading depth and a larger one gives squareclass -1. Then $(2b/a)t^2(1 - 2/t^2)$ has squareclass 2, contradicting the w-equation. Thus $[p]_2 = -1$ and c is square. If $p \equiv 1 \pmod{16}$, $b/(2p) \in 1 + 8\mathbf{Z}_2$, so (155) applies.

Finally suppose $p \equiv 15 \pmod{16}$. Normalizing $z = 1$, the first anchor equation forces $v_2(t) \geq 2$; the second then implies that $2b$ is square. Choose local r, s, h_d, h_Z with

$$r^2 = 2b, \quad s^2 = 3 - p, \quad h_d^2 = -a/2, \quad h_Z^2 = \frac{p^2 - 4p - 1}{2pa}.$$

Here $3 - p = 4(1 - b/4)$ and $1 - b/4 \in 1 + 8\mathbf{Z}_2$, so the indicated s exists. The last two unit radicands are squares because $-a/2 \equiv 1 \pmod 8$ and $h_Z^2 - 1 = -b^2/(2pa) \in 2^8\mathbf{Z}_2$. The actual point

$$[Z : U : V : T] = [h_Z/2 : r/4 : 1 : rs/(4h_d)] \in D_W(\mathbf{Q}_2)$$

has row $\chi = (2,2,1) = \beta$ over $\mathbf{Q}_2$ and hence supplies a point of (153). Over $\mathbf{R}$ one may use $[0 : \sqrt{8p/a^3} : \sqrt{8pb/a^3} : 1]$. The partner is therefore everywhere locally soluble, and the evaluation of α is constant at every place.

9.4 Evaluation of the quaternion algebra

We give the local calculations leading to (146). At a tangent fibre the extension in (151), rather than the value of a vanishing rational function, is used. When the missing coordinate is locally a square, the actual points

$$P_U = [z : u_0 : v : t], \quad u_0^2 = \frac{p}{a}(t^2 - bz^2), \qquad P_V = [z_1 : u_1 : v_1 : t_1], \quad v_1^2 = 2t_1^2 - 4pz_1^2$$

satisfy

$$\alpha(P_U) = (-a, -pL_V(P_U)), \qquad \alpha(P_V) = (-a, -2bL_U(P_V)). \tag{156}$$

To see this, the polar identities are

$$(zV - vZ)^2 - 2(zT - tZ)^2 = -2zZL_U, \qquad a(z_1U - u_1Z)^2 - p(z_1T - t_1Z)^2 = -2z_1ZL_V.$$

Dividing by the square of $tZ - zT$ in the first case gives the limiting coefficient $-4p/v^2$ for L_U. Dividing by the corresponding square in the second gives $-p^2b/(2au_1^2)$ for L_V; removing squares and norms from $\mathbf{Q}_v(\sqrt{-a})$ gives the factor $-2b$ in (156). If $u_1 = 0$, use U as parameter and $L_V/U^2 \to -a/2$. If $z_1 = 0$, use the chart $T = t_1$. The choice $z_1t - zt_1 \neq 0$ ensures the other polar is nonzero.

For an odd prime the evaluation table is

parameter branch	$2\,\mathrm{inv}_q\,\alpha$
$v_q(p), v_q(a), v_q(b) = 0$	0
$v_q(p) > 0$	0
$v_q(a) > 0$	$2\,\mathrm{inv}_q(-a, -2)$
$v_q(b) > 0,\ (2/q) = 1$	0
$v_q(b) > 0,\ (2/q) = -1$	$v_q(b)/2$
$v_q(p) < 0$ even	0
$v_q(p) < 0$ odd	bit of $\left(\frac{-(2+e)}{q}\right), \quad e^2 = 2.$

(157)

Here and below equality of entries is in $\mathbf{F}_2$. We justify the entries, including arbitrary valuation depths. At a good unit prime, the source and its two conics have smooth proper models. Normalize the anchor primitively. The anchor transformation has unit determinant, so both tangent sections are primitive. Their horizontal divisors are even and neither has a vertical divisor. The quaternion extends over the model and specializes to $\mathrm{Br}(\mathbf{F}_q) = 0$. At a positive-$p$ prime, $-a$ is square.

At a positive-a prime where 2 is square, use the actual point $[Z : U : V : T] = [1 : \sqrt{p} : 0 : \pm\sqrt{2p}]$. Normalize $\min(v_q(t), v_q(z)) = 0$ and choose the sign so that L_U is a unit. Direct substitution of (149) gives $L_V/L_U \equiv -p^2 b \pmod q$; all terms involving aw have positive valuation because $v_q(a) + 2v_q(w) \geq 0$. Removing the square pole V^2 leaves the class $-b = -2$. If 2 is nonsquare, $v_q(a)$ is even. In the same normalization $z_1 \equiv -4z$, $t_1 \equiv -4t$, and the missing square is $2t_1^2 - 4pz_1^2 \equiv 16v^2$. Choose $v_1 \equiv 4v$; then $L_U(P_V) \equiv 8v^2$ is a unit. Both entries of the symbol in (156) have unit squareclass, so its invariant is zero, as is $(-a, -2)_q$.

At a positive-b inert prime put $v_q(b) = 2h$ and normalize $\min(v_q(t), v_q(z)) = 0$. The unramified norm $t^2 - 2z^2$ gives $v_q(w) = h$ and $v_q(z) = 0$. The anchor transformation has $v_q(z_1) = h$, $v_q(t_1) \geq 2h$. Hence the missing square at P_V has class $-p = 1$. Choose its sign so that $L_U(P_V)/z_1 \equiv \pm 2v - 4z$ is nonzero. Thus $v_q(L_U(P_V)) = h$. The first entry $-a$ has class 2, and (142) gives the bit $2h + h = h$. If 2 is square, $-a$ is square instead.

At a pole $v_q(p) = -2h$ with 2 nonsquare, normalize $z = 1$. The w-equation forbids $v_q(t) < 0$, and the v-equation then forces $-p$, and hence $-a$, to be square. If 2 is square, set $Z = \pi^h Z'$. Writing $p = \pi^{-2h} p_0$, $a = \pi^{-2h} a_0$, $b = \pi^{-2h} b_0$ gives the good model

$$V^2 - 2T^2 + 4p_0 Z'^2 = 0, \qquad a_0 U^2 - p_0 T^2 + p_0 b_0 Z'^2 = 0.$$

Normalize $[v : t : z/\pi^h]$ primitively. If t is nonunit, $-p_0$ is square. Otherwise the transformed second anchor, multiplied by π^{4h}, reduces to $[0 : p_0^2(w + 2t) : -p_0^2(w + 2t)]$, a primitive point. The original second polar differs from its primitive-model polar by π^{-6h}, a square factor. The good-model argument therefore again gives zero.

At an odd-depth pole $v_q(p) = -(2h + 1)$, 2 is square. Normalize $t = 1$. The first anchor equation forces $v_q(z) \geq h + 1$ and $v, w \equiv \pm\sqrt{2}$. On D_W choose $[Z : U : V : T] = [0 : u : e : 1]$, where $u^2 = p/a$, $u \equiv 1$ and $e^2 = 2$ has sign opposite to v. Then $L_U \equiv -4$; L_V has valuation $-3(2h + 1)$ and leading unit $2p_0^3(w + 2)$. Thus $[F] = [(-a)(w + 2)]$ and $\alpha = (-a, -(w + 2))$. Formula (142) gives the last line of (157).

We next calculate the dyadic entries. In either nonunit branch $-a$ is square, as proved above, and the value is zero. For $p \equiv 1 \pmod{16}$ put $h = v_2(a)$, $d = (h - 1)/2$ and normalize $z = 1$. The anchor has $t = 2t_0$, $v = 2v_0$, with t_0, v_0 odd and $v_0^2 = 2t_0^2 - p$; moreover $v_2(w) = (3 - h)/2$. Set

$$C = \frac{aw}{2b}, \quad g = \frac{t_1}{2z_1}, \quad z_1 = -2b(1 - C), \quad v_2(C) = d.$$

Then g is odd and

$$g - t_0 = \frac{a(t_0 + 1/2) + C(t_0 + b/2)}{1 - C}, \qquad v_2(g - t_0) \geq d + 1.$$

Choose $e^2 = 2g^2 - p$ close to $-v_0$, and set $K = v_0 e - 2t_0 g + p$. Factoring $e^2 - v_0^2 = 2(g - t_0)(g + t_0)$ gives $v_2(e + v_0) \geq d + 2$, whence $[K] = [-2]_2$. The actual point P_V with $v_1 = 2z_1 e$ has $L_U = 4z_1 K$. Formula (156) becomes $\alpha_2 = (-2, (1 - C)K)$. By (143) the K term has bit one, and the $1 - C$ term has bit one exactly when $d = 2$. Therefore

$$2 \operatorname{inv}_2 \alpha = 1 + [v_2(a) = 5]. \tag{158}$$

For $p \equiv 15 \pmod{16}$ put $h = v_2(b)$, $d = (h - 1)/2$ and again $z = 1$. Now $t = 2t_0$ with t_0 even, $v = 2v_0$, $v_0^2 = -p + 2t_0^2$, and $v_2(w) = (h + 1)/2$. The transformed anchor has $v_2(z_1) = (h + 3)/2$, $v_2(t_1) = h + 2$. Thus $g = t_1/(2z_1)$ has depth d, and $e^2 = 2g^2 - p$ is a square. Choose its sign so that $v_0 e \equiv 3 \pmod 4$. Since $p \equiv -1 \pmod{32}$, the number $K = v_0 e - 2t_0 g + p$ is 14 or 2 (mod 16); its valuation is one and its odd part is 7 or 1 (mod 8). Hence $(2, K)_2 = 0$ and (156) gives

$$2 \operatorname{inv}_2 \alpha = 2 \operatorname{inv}_2(2, w/z) + [v_2(b) = 5]. \tag{159}$$

Indeed $z_1 = aw(1 - 2b/(aw))$, the correction has depth d, and p, a, b are norms from $\mathbf{Q}_2(\sqrt{2})$.

The invariant at $\mathbf{R}$ is $1/2$. To verify the sign in the chosen rational gauge, note that

$$(2pt + 2az)^2 - a^2 w^2 = 2\big((p^2 + 1)t^2 + 4patz + 4paz^2\big) > 0.$$

The quadratic is positive definite, and $2pt + 2az$ has the sign of t. Thus t_1 has the opposite sign to t. Cauchy–Schwarz on the two positive conics shows that L_U and L_V have respective signs $-\operatorname{sgn}(tT)$ and $-\operatorname{sgn}(t_1 T)$ away from their zeros. Consequently $F < 0$. Since $-a < 0$, the real invariant is $1/2$. The evaluation at a zero or pole is obtained by continuity of the proper class, so this value holds on every real component.

9.5 Reciprocity and the exclusion theorem

The dyadic ratio in (159) has an exact global support formula. At an odd prime where 2 is nonsquare,

$$v_q(w/z) = \begin{cases} -v_q(a)/2, & v_q(a) > 0, \\ v_q(b)/2, & v_q(b) > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{160}$$

For a unit or positive-p branch, normalize $\min(v_q(t), v_q(z)) = 0$. The unramified norm $t^2 - 2z^2$ forces w to be a unit, and z cannot be nonunit, since then v^2 has leading squareclass 2. For $v_q(a) = 2h > 0$ normalize $w = 1$: the same norm has depth $2h$, and the v-equation forces $v_q(z) = h$. For $v_q(b) = 2h > 0$ the normalization above gives $v_q(w) = h$, $v_q(z) = 0$. At a pole, normalize $z = 1$. A negative valuation of t would make w^2 have class 2; with t integral, $t^2 - 2$ is a unit, so w is a unit. These arguments prove (160) at all depths. Hilbert reciprocity for the literal rational number w/z now gives

$$2 \operatorname{inv}_2(2, w/z) = A(p) + B(p). \tag{161}$$

Apply reciprocity also to the constant symbol $(-a, -2)$. Its real bit is one; its odd contributions are the positive-a entries of (157) and, at odd-depth poles, the character $\left(\frac{-1}{q}\right)$. At 2 its bit is one

in the $p \equiv 1 \pmod{16}$ branch and zero in the other branches. Adding this zero global sum to the sum for α cancels the real entry and every positive-a entry. At odd poles the remaining character is $\left(\frac{2+e}{q}\right)$. Hence

$$2\sum_v \operatorname{inv}_v \alpha = 2\operatorname{inv}_2 \alpha + 2\operatorname{inv}_2(-a,-2) + B(p) + P(p) = \mathcal{T}(p),$$

by (158), (159) and (161). This proves the claimed invariant formula.

To finish Theorem 9.1, suppose a rational point of C_p exists. Its U-omission supplies the single rational anchor used throughout the construction. A rational point on C_p has $Z, V, W, T \neq 0$ by the arguments following (147), and $U = 0$ would make $pV^2 = -2apZ^2 < 0$. It therefore gives the nondegenerate four squares in (141). The construction applies to this anchor and the image point on D_W. Evaluation there is a quaternion algebra over $\mathbf{Q}$, so the sum of its local invariants is zero. This contradicts $\mathcal{T}(p) = 1$. The theorem follows.

9.6 A conic family with nonzero obstruction

Let

$$r^2 + 2s^2 = 8, \qquad p = 3 - s^2 > 1. \tag{162}$$

Then $r^2 = 2(p+1)$ and $[z : v : w : t] = [s : 2(p-1) : 2r : r]$ is an actual rational point of D_U. Every odd inert prime dividing $p-1$ would make $s^2 = 2 - (p-1)$ a nonsquare unit; thus $A(p) = 0$. Every pole of p has even depth, so $P(p) = 0$. For rational points of this conic, $v_2(s) = 1$ and $v_2(r) \geq 3$. Indeed write $r = R/D$, $s = S/D$ with $\gcd(R, S, D) = 1$. If D were even, $R^2 + 2S^2 = 8D^2$ modulo 4 would force R, S even. Thus D is odd. Reduction modulo 8 gives R, S even; writing $R = 2R_1$, $S = 2S_1$ gives $R_1^2 + 2S_1^2 = 2D^2$. Here R_1 is even and S_1 odd. With $R_1 = 2R_2$, the identity $2R_2^2 + S_1^2 = D^2$ modulo 8 forces R_2 even. Consequently $p \equiv -1 \pmod{32}$ and $v_2(p+1) = 2v_2(r) - 1$. Formula (146) reduces to

$$\mathcal{T}(p) = [v_2(r) = 3]. \tag{163}$$

Thus every locally soluble complete source in this conic family with $v_2(r) = 3$ is a genus-five failure of the Hasse principle.

Theorem 9.3. *For every integer $j \geq 0$, set*

$$N = 3 + 154j, \quad M = N + 2, \quad D = N^2 + 2M^2, \qquad r = \frac{8MN}{D}, \quad s = \frac{2(N^2 - 2M^2)}{D}, \quad p = 3 - s^2.$$

The curves C_p form an infinite family of smooth genus-five curves with points at every completion of $\mathbf{Q}$, including points in the strict real distance region, and with no rational point.

Proof. The displayed parametrization satisfies (162). The integers M, N, D are odd, $\gcd(M, N) = 1$ and $v_2(r) = 3$. The ratio $m = M/N$ decreases strictly from $5/3$ towards 1. On $1 < m \leq 5/3$, the function $s = 2(1 - 2m^2)/(1 + 2m^2)$ is negative with derivative $-16m/(1 + 2m^2)^2 < 0$; thus $p = 3 - s^2$ decreases as a function of m. It follows that the distinct parameters p increase from $3719/3481 > 1$. Their squarefree kernels in $p(p^2 - 1)$ are odd, since $[p(p^2-1)]_2 = 1$. Such a kernel cannot equal one: if $p(p^2 - 1) = y^2$, the legs $(p^2 - 1)/y, 2p/y$ and hypotenuse $(p^2 + 1)/y$ give a rational right triangle of area one, contradicting Lemma 2.2. By (163), it remains to prove $\mathrm{Loc}(p)$. We give the local proof directly.

At 2 use $\rho = b/(3p+1)$. Its four radicands are

$$\frac{r^2}{4}\frac{2p}{3p+1}, \qquad 4\frac{-p(p-1)}{3p+1}, \qquad r^2\frac{2(2p+1)}{3p+1}, \qquad 4p\rho. \tag{164}$$

The three unit factors displayed after their square factors lie in $1+8\mathbf{Z}_2$, because $p \equiv -1 \pmod{32}$. At an odd prime with $v_q(p) > 0$ or $v_q(p-1) > 0$, the identities $r^2 = 2(p+1)$ or $s^2 = 3-p$, respectively, make 2 square locally. Take $\rho = p$. The first three radicands are then explicit squares, and the fourth, p^2+1, has square residue 1 or 2. If $v_q(p+1) > 0$, use (164) again: the first and third radicands divided by r^2 have residues $1/4$ and 1; the second divided by 4 has residue 1. At a pole of p, the identity $s^2 = 3-p$ makes $-p$ a local square. Put $h = 1/p$ and $\rho = -(p+1)/(p+3)$. The radicands become

$$(-p)\frac{1+h}{1+3h}, \quad (-4p)\frac{(1-h)(1+2h)}{1+3h}, \quad 4\frac{1+h}{1+3h}, \quad 4\frac{1+h}{1+3h}.$$

Every remaining factor is a principal unit and hence square. These witnesses treat all nonordinary odd primes and every depth.

At an ordinary prime q, the model has five distinct branch points and smooth genus-five reduction. If $q \geq 101$, Hasse–Weil gives

$$\#C_p(\mathbf{F}_q) \geq q + 1 - 10\sqrt{q} > 0.$$

Any such point gives a nondegenerate $\mathbf{Q}_q$-point. At a finite branch point ρ_0, write its one vanishing radicand as $u(\rho - \rho_0)$ with u a unit, and take $\rho = \rho_0 + q^2/u$. The other three radicands remain square units. At infinity the three leading ratios to $p\rho$ are squares; taking $\rho = 1/(pq^2)$ gives four nonzero roots. Thus normalized branch points are sufficient for lifting.

Here is a short finite-field proof for the remaining primes. It also records the explicit arithmetic used for the small-prime part. Let χ be the quadratic character of $\mathbf{F}_q$, extended by $\chi(0) = 0$, and put

$$H = \chi(-1), \quad T = \chi(2), \quad A = \chi(p), \quad B = \chi(p-1), \quad c = -\frac{p+1}{p-1}.$$

The conic identity gives $\chi(p+1) = T$. The five elliptic character quotients are

$$\begin{aligned}
E_0 &: y^2 = p(p^2-1)x(x-1)(x+1),\\
E_1 &: y^2 = 2p\,x(x-1)(x-c),\\
E_2 &: y^2 = 2p(p^2-1)x(x+1)(x-c),\\
E_3 &: y^2 = (p+1)(x-1)(x+1)(x-c),\\
E_4 &: y^2 = p(p+1)c(x-1)(x+1)(x-c^{-1}).
\end{aligned}$$

If $t_i = q + 1 - \#E_i(\mathbf{F}_q)$, expansion by the characters of the four square roots gives

$$\#C_p(\mathbf{F}_q) = q + 1 - \sum_{i=0}^{4} t_i. \tag{165}$$

Indeed the products of three radicands give E_0, E_1, E_2, E_3; the product of all four, after $x = 1/\rho$, gives E_4. The remaining characters have genus zero and their finite sums cancel with their contributions at infinity.

Each E_i has rational two-torsion. On $y^2 = d\prod(x - e_i)$, the point $(e_i, 0)$ is twice a rational point if and only if $d(e_i - e_j)$ and $d(e_i - e_k)$ are both squares. This follows by writing the tangent at a putative half-point through $(e_i, 0)$ and factoring the resulting quadratic; conversely those two square roots give the two tangent slopes and a half-point. Consequently the group order is 4 (mod 8) when there is no rational point of order four, is divisible by eight when such a point exists, and is divisible by sixteen when all four-torsion is rational.

Let U_q, V_q, W_q be the largest integers at most $2\sqrt{q}$ congruent to $q+1 \pmod 8$, $q-3 \pmod 8$, and $q+1 \pmod{16}$, respectively. Set $\tau_q = -\sum_{x\in\mathbf{F}_q} \chi(x^3 - x)$. Then $t_0 = TAB\tau_q$; for $q \equiv 3 \pmod 4$, $\tau_q = 0$ by pairing x and $-x$. Applying the preceding halving criterion to the displayed roots gives the following upper bounds on the sum in (165). Only the sign patterns without an immediate rational branch point need be listed:

q mod 8	sign pattern	upper bound for $\sum t_i$
3	all A, B	$3U_q + V_q$
7	$A = -1, B = 1$	$2U_q + 2V_q$
1	$A = B = -1$	$\tau_q + 2U_q + 2V_q$
5	$A = -1, B = 1$	$\tau_q + 4U_q$
5	$A = B$	$-\tau_q + 2U_q + V_q + W_q$

For example, in the first row exactly three of $E_1, \ldots, E_4$ have order divisible by eight, and the remaining order is 4 (mod 8). In the second and third rows two have each type; in the fourth all four have order divisible by eight; in the last, one has full rational four-torsion, two have an order divisible by eight, and one has order 4 (mod 8). These statements follow by inserting A, B, H, T into $d(e_i - e_j)$ and $d(e_i - e_k)$. The omitted branch patterns are also explicit: when $T = 1$, $\rho = 1$ works for $A = 1$, and $\rho = 0$ works for $HB = 1$; when $H = 1, T = -1$, $\rho = c$ works for $A = 1, B = -1$.

The trace values needed below are

q	5	13	17	29	37	41	53	61	73	89	97
τ_q	-2	6	2	-10	-2	10	14	-10	-6	10	18.

They can be evaluated from the displayed finite sum; equivalently, for $q = u^2 + v^2$, u odd, v even and $u + v \equiv 1 \pmod 4$, the usual Gaussian evaluation of this sum is $\tau_q = 2u$. Substitution in the bounds proves positivity except for the following cases, where the two conic conditions $\chi(2(p+1)) = 1$ and $\chi(3 - p) \in \{0, 1\}$ are retained. The congruence $N \equiv 3$, $M \equiv 5 \pmod{7 \cdot 11}$ gives $p \equiv 1 \pmod 7$ and $p \equiv 9 \pmod{11}$. At 17 the remaining residues are $7, 12$, and at 41 they are $7, 35$; taking $\rho = p$ works because $p^2 + 1$ has respective square residues $16, 9$ and $9, 37$. At 13 the sign case $A = -1, B = 1$ would require $p = 5$, but $3 - p = 11$ is nonsquare. At 53 the remaining case is $A = -1, B = 1$, and at 61 the case $A = B = -1$ is also settled by $\rho = -2$, whose radicands are $-2p, -6(p-1), -r^2, s^2$; use $(3/53) = -1$ and $(3/61) = 1$. The residual cases have the following explicit nonzero square roots:

q	$p \bmod q$	$\rho \bmod q$	roots in the order (141)
11	9	4	$5, 2, 1, 3$
19	2	3	$5, 2, 9, 5$
19	11	5	$6, 2, 7, 9$
19	13	10	$4, 8, 2, 1$
19	17	4	$7, 1, 3, 5$
23	17	5	$4, 6, 3, 11$
29	7	22	$3, 7, 7, 13$
29	25	4	$10, 12, 12, 8$
43	2	3	$7, 2, 14, 7$
43	6	10	$19, 2, 5, 10$
43	11	9	$20, 17, 5, 4$
43	21	14	$6, 2, 12, 1$
43	25	35	$12, 16, 18, 7$
43	29	5	$4, 3, 4, 16$
43	31	10	$3, 14, 4, 17$
43	32	3	$15, 9, 7, 13$
43	33	12	$3, 4, 14, 17$
43	36	23	$21, 11, 20, 5$
61	16	3	$29, 11, 21, 1$
61	42	12	$4, 29, 9, 13$
61	49	12	$10, 18, 18, 4$

Every row is checked by squaring its four entries in (141); all roots are nonzero and therefore lift by Hensel's lemma. This completes the proof of finite-place solubility. At the real place the strict open interval described after (141) supplies the required points. Now (163) and Theorem 9.1 give $C_p(\mathbf{Q}) = \varnothing$ for every parameter in the family.

There are also infinitely many geometric isomorphism classes among these curves. A fixed smooth curve of genus five has finite automorphism group: its automorphism scheme is of finite type, for example through the tricanonical embedding, and its tangent space is $H^0(C, T_C) = 0$ because $\deg T_C = -8$. It therefore has finitely many subgroups isomorphic to $(\mathbf{Z}/2)^4$. Each such subgroup gives one quotient with its unordered branch configuration. Here that configuration is $\{0, 1, -1, c, \infty\}$, with $c = -(p+1)/(p-1)$. For a fixed configuration, there are finitely many ways to choose and normalize three ordered branch points, and hence finitely many values of c. Since $p = (c-1)/(c+1)$, a fixed geometric curve occurs for only finitely many of our distinct parameters. □

10 Theta characteristics and a complete double cover

The fixed opposite-distance difference $1/5$ gives a particularly explicit connection between the distance equations, the geometry of a plane quartic, and descent on its Jacobian. In this section we determine all the odd theta characteristics of that quartic and identify the double cover which restores the four distances. A rational point at infinity then gives a local solubility theorem for every finite multiplication cover, including the pullbacks which retain this double cover.

10.1 The quartic quotient and its metric interpretation

Let $C \subset \mathbf{P}^4_{[P:Q:R:z:T]}$ be the curve

$$\begin{aligned} P^2 &= z^2 + 50T^2, \\ Q^2 &= z^2 + 14zT + 51T^2, \\ R^2 &= z^2 - 14zT + 51T^2. \end{aligned} \tag{166}$$

We use C as a complete source: all three equations are imposed at the same value of z/T. On $T = 1$, put

$$\begin{aligned} x = \frac{50 - P - 7z}{100}, \qquad y = \frac{50 - P + 7z}{100}, \\ (r_0, r_1, r_2, r_3) = \left(\frac{P-1}{10}, \frac{Q}{10}, \frac{R}{10}, \frac{P+1}{10} \right). \end{aligned} \tag{167}$$

The strict positive locus $\mathcal{U} \subset C(\mathbf{R})$ is defined by

$$T = 1, \qquad P > 1, \quad Q > 0, \quad R > 0, \qquad P + 7|z| < 50. \tag{168}$$

It corresponds exactly to the strict distance configurations with $r_3 - r_0 = 1/5$. For completeness, the first distance identity follows from

$$5000(r_0^2 - x^2 - y^2) = 49(P^2 - z^2 - 50) = 0.$$

The other three follow from

$$r_1^2 - r_0^2 = 1 - 2x, \qquad r_2^2 - r_0^2 = 1 - 2y, \qquad r_3^2 - r_0^2 = 2 - 2x - 2y.$$

Conversely, the inverse formulas are

$$P = 5(r_3 + r_0), \quad Q = 10r_1, \quad R = 10r_2, \quad z = \frac{50(y - x)}{7}.$$

The inequalities in (168) are precisely the positivity of the chosen roots and $0 < x, y < 1$.

Proposition 10.1. *The curve C is smooth of genus five and has eight rational points at $T = 0$. The fixed-point-free involution*

$$\sigma[P : Q : R : z : T] = [-P : -Q : -R : z : T]$$

has quotient the smooth plane quartic $D \subset \mathbf{P}^2_{[X:Y:Z]}$ defined by

$$F = 9801(X^4 + Y^4) + 19598X^2Y^2 - 39592(X^2 + Y^2)Z^2 + 39984Z^4 = 0. \tag{169}$$

The quotient map is

$$\pi : C \longrightarrow D, \qquad [P : Q : R : z : T] \longmapsto [Q : R : P].$$

In particular, π is a connected étale *double cover. The points*

$$O_C = [1 : 1 : 1 : 1 : 0], \qquad O_D = [1 : 1 : 1]$$

are rational and satisfy $\pi(O_C) = O_D$.

Proof. On the chart $T = 1$, the radicands in (166) have discriminants -200, -8 and -8, and pairwise resultants 9801, 9801 and 39984. Their six roots are therefore distinct. A simple zero of each polynomial shows that its square class is independent of those of the other two over $\overline{\mathbf{Q}}(z)$. The map to the $[z : T]$-line has degree eight, with six branch values and inertia order two at each. Riemann–Hurwitz gives

$$2g(C) - 2 = 8(-2) + 6 \cdot 4 = 8.$$

The simple branch points are smooth. At $T = 0$ we may set $z = 1$, and the points are exactly

$$[\epsilon_1 : \epsilon_2 : \epsilon_3 : 1 : 0], \qquad \epsilon_i \in \{1, -1\}.$$

The three derivatives with respect to P, Q, R are nonzero there, so these points are smooth as well.

There is no point of C with $P = Q = R = 0$: subtracting the last two equations gives $zT = 0$, and the equations then force $z = T = 0$. Hence π is a morphism. A projective fixed point of σ would have either $P = Q = R = 0$ or $z = T = 0$, both impossible.

On the chart $P \neq 0$, set

$$\xi = Q/P, \qquad \eta = R/P, \qquad v = T/P, \qquad w = z/P, \qquad S = \xi^2 + \eta^2.$$

The three quadrics give

$$v^2 = \frac{S-2}{2}, \qquad wv = \frac{\xi^2 - \eta^2}{28}, \qquad w^2 = 51 - 25S. \tag{170}$$

Eliminating v, w gives

$$(\xi^2 - \eta^2)^2 + 9800(S-2)^2 - 392(S-2) = 0, \tag{171}$$

which is (169) on $Z = 1$.

Here is also a direct smoothness proof for the plane equation. Write

$$a = 9801, \quad b = 19598, \quad c = -39592, \quad d = 39984, \qquad M = \begin{pmatrix} 2a & b & c \\ b & 2a & c \\ c & c & 2d \end{pmatrix}.$$

At a singular point, the rows indexed by the nonzero coordinates annihilate the corresponding vector of coordinate squares. The nonempty principal minors of M have values

$$19602, \quad 79968, \quad 156800, \quad 6272, \quad -1229312,$$

with repetitions. Every such submatrix is invertible. This proves smoothness; a smooth plane quartic is geometrically integral and has genus three. The map to it has degree two, since generically v determines w in (170), and the two choices are interchanged by σ. The fixed-point calculation proves that the map is *étale*. □

Proposition 10.2. *A rational point $[\xi : \eta : 1] \in D(\mathbf{Q})$ is the image of a point of $\mathcal{U} \cap C(\mathbf{Q})$ if and only if*

$$\xi > 0, \qquad \eta > 0, \qquad v = \sqrt{\frac{\xi^2 + \eta^2 - 2}{2}} \in \mathbf{Q}_{>0}, \qquad \left| \frac{\xi^2 - \eta^2}{28v} \right| < \frac{7}{11}. \tag{172}$$

For this choice of v, the unique point on the positive branch is

$$P = \frac{1}{v}, \qquad Q = \frac{\xi}{v}, \qquad R = \frac{\eta}{v}, \qquad z = \frac{w}{v}, \qquad w = \frac{\xi^2 - \eta^2}{28v}. \tag{173}$$

Proof. Equation (171) gives $w^2 + 50v^2 = 1$, and substitution in (173) gives all three equations of C. Moreover $P \geq \sqrt{50} > 1$. The remaining strict inequality is

$$P + 7|z| < 50 \quad \Longleftrightarrow \quad 1 + 7|w| < 50v.$$

Both sides on the right are positive. Squaring and using $50v^2 = 1 - w^2$ reduces it to

$$99|w|^2 + 14|w| - 49 < 0,$$

whose positive root is $7/11$. The inverse implication follows from the same calculation. □

Every real point of D lies on $Z \neq 0$. Indeed,

$$F(X, Y, 0) = 9801(X^2 + Y^2)^2 - 4X^2Y^2 > 0$$

for a nonzero real pair (X, Y). The four points where $v^2 = 0$ are

$$E_{\epsilon,\eta} = [\epsilon : \eta : 1], \qquad \epsilon, \eta \in \{1, -1\}; \tag{174}$$

each has two rational preimages at $T = 0$ on C.

10.2 The complete bitangent configuration

Put $r = \sqrt{2}$, $s = \sqrt{51}$, and $K = \mathbf{Q}(r, s)$. Independent signs in the following statement range over $\{1, -1\}$.

Theorem 10.3. *The twenty-eight bitangents of D are precisely the following lines:*

1. *$Z = uX$ and $Z = uY$, where $u = \epsilon r/2$ or $u = 5\epsilon r$;*
2. *$Z = (\epsilon hX + \eta hY)/14$, where $h = 9$ or $h = 11$;*
3. *$Z = s(\epsilon hX + \eta jY)/102$, where $(h, j) = (9, 11)$ or $(11, 9)$;*
4. *$X = tY$, where $t = \epsilon r(s + 7\eta)/2$.*

Each has two distinct contact points. The associated odd-theta étale *algebra is*

$$L_{\mathrm{odd}} = \mathbf{Q}^8 \times \mathbf{Q}(\sqrt{2})^4 \times \mathbf{Q}(\sqrt{51})^4 \times \mathbf{Q}(\sqrt{2}, \sqrt{51}). \tag{175}$$

The full Galois action is induced by the independent sign changes of r and s in the displayed lines.

Proof. We first give the contact identities. These also specify the contact divisors, since each is the zero divisor of the displayed quadratic on its line. For type 1, on $Z = uX$ one has

$$F|_{Z=uX} = 9801(Y^2 + \alpha X^2)^2, \quad \alpha = \begin{cases} -1/99, & u^2 = 1/2, \\ -9899/99, & u^2 = 50. \end{cases} \tag{176}$$

Exchange X, Y for the other four lines. For type 2,

h	k_h	q_h
9	$-818/81$	$13122/49$
11	$1222/121$	$29282/49$

$$F|_{Z=h(\epsilon X+\eta Y)/14} = q_h(X^2 + \epsilon\eta k_h XY + Y^2)^2. \tag{177}$$

For type 3 the identity is

$$F|_{Z=s(\epsilon hX+\eta jY)/102} = \frac{6534}{17}(X^2+\epsilon\eta k_{h,j}XY-Y^2)^2, \quad k_{9,11}=-\frac{980}{99}, \quad k_{11,9}=\frac{980}{99}. \tag{178}$$

Finally, $t^2 = 50+7\eta s$ for type 4, and

$$F|_{X=tY} = 39984\left(Z^2-\frac{39592(t^2+1)}{79968}Y^2\right)^2. \tag{179}$$

All these homogeneous quadratics have nonzero discriminant: in (176) the coefficient α is nonzero; in (177) one has $k_h \neq \pm 2$; in (178) one has $k_{h,j}^2+4>0$; and in (179) one has $t^2+1\neq 0$.

We prove completeness by comparing coefficients of binary quartics. A line with one vanishing coefficient is of the form $X=a_0Z$, $Y=a_0Z$, or $X=a_0Y$. For the first two forms the discriminant condition for the restricted even quartic to be a square is

$$-3136(a_0^2-2)(50a_0^2-1)=0.$$

For the third form it is

$$-6272(a_0^4-100a_0^2+1)=0.$$

The resulting lines are exactly types 1 and 4. The three coordinate axes give nonzero discriminants and hence contribute no line.

Now write a line with all coefficients nonzero as $Z=\alpha X+\beta Y$, and set $U=\alpha^2$, $V=\beta^2$. With a,b,c,d as in the proof of Proposition 10.1, its restricted quartic has coefficients

$$\begin{aligned} A_4 &= a+cU+dU^2, & A_3 &= 2\alpha\beta(c+2dU),\\ A_2 &= b+c(U+V)+6dUV, & A_1 &= 2\alpha\beta(c+2dV),\\ A_0 &= a+cV+dV^2. \end{aligned}$$

For a square quartic, $A_4=0$ would imply $A_3=0$. The latter forces $U=101/204$, for which $A_4=2/51$. Thus A_4, A_0 are nonzero. If one of A_3, A_1 vanishes, both vanish, giving $U=V=101/204$; at that pair

$$A_2^2-4A_4A_0=\frac{78368326400}{51}\neq 0.$$

In the remaining case, comparison with the square of a quadratic gives

$$\begin{aligned} &(c+2dU)^2A_0-(c+2dV)^2A_4 \\ &\qquad = 2458624(U-V)(102U+102V-101)=0, \end{aligned} \tag{180}$$

$$E := A_0(c+2dV)A_2-UV(c+2dV)^3-2A_0^2(c+2dU)=0. \tag{181}$$

More explicitly, when these four coefficients are nonzero, the restricted quartic is a square precisely when

$$A_4=A_0(A_3/A_1)^2, \qquad A_2=\frac{A_1^2}{4A_0}+\frac{2A_0A_3}{A_1}.$$

The square is then

$$A_0\left(\frac{A_3}{A_1}X^2+\frac{A_1}{2A_0}XY+Y^2\right)^2.$$

Substitution gives exactly (180)–(181). On the two lines singled out by (180), the second expression factors as

$$E|_{V=U} = -1568(196U - 121)(196U - 81)(204U - 101),$$
$$E|_{V=101/102-U} = \frac{76832}{17}(68U - 27)(204U - 121)(204U - 101).$$

After discarding $U = 101/204$, already treated, these yield

$$(U, V) = \left(\frac{81}{196}, \frac{81}{196}\right), \quad \left(\frac{121}{196}, \frac{121}{196}\right), \quad \left(\frac{27}{68}, \frac{121}{204}\right), \quad \left(\frac{121}{204}, \frac{27}{68}\right).$$

These are precisely types 2 and 3, and their sufficiency was established by (177)–(178).

Since D is a smooth plane quartic, its canonical bundle is $\mathcal{O}_D(1)$. A bitangent ℓ therefore cuts out $2B_\ell$, where B_ℓ is an effective theta characteristic of degree two. It has $h^0(B_\ell) = 1$: a degree-two pencil would make D hyperelliptic, whereas its canonical map is a plane embedding. Conversely, an odd theta characteristic has a unique effective divisor of degree two and gives a bitangent by taking twice that divisor. The preceding classification is thus the full odd-theta set.

The eight type 2 lines are rational. Type 1 consists of four quadratic orbits over $\mathbf{Q}(r)$, and type 3 consists of four quadratic orbits over $\mathbf{Q}(s)$. The four type 4 lines form one orbit over K: the value of t^2 recovers s up to sign, and t then recovers r up to sign. This proves (175) and the Galois assertion. □

10.3 Theta fractions and the literal lifting class

Choose the rational bitangent

$$\ell_0 = 14Z - 9X - 9Y.$$

None of the twenty-eight bitangents contains O_D: this follows directly from their equations (for type 3 the coefficient of s is rational and nonzero, and for type 4 $t^2 = 50 \pm 7s \neq 1$). Thus, for every line in Theorem 10.3, the function

$$f_\ell = \frac{\ell}{\ell_0}\frac{\ell_0(O_D)}{\ell(O_D)} \tag{182}$$

is defined at O_D and satisfies

$$f_\ell(O_D) = 1, \qquad \operatorname{div}(f_\ell) = 2(B_\ell - B_{\ell_0}).$$

The line equations and their Galois action therefore give explicitly labelled theta fractions in each factor of (175). Alternatively, relative to the canonical divisor $H = (Z = 0)|_D$, the functions ℓ/Z have divisor $2B_\ell - H$.

Proposition 10.4. *Let*

$$C_0 = X^2 + Y^2 - 2Z^2, \qquad N = 97X^2 + 99Y^2 - 196Z^2, \qquad h = \frac{C_0}{2Z^2}, \qquad \ell_\pm = Z \pm \frac{X}{\sqrt{2}}.$$

The cover $\pi : C \to D$ is the normalization of $v^2 = h$. If $E_0 = \sum_{\epsilon,\eta} E_{\epsilon,\eta}$ is the sum of the four points in (174)*, its geometric covering class is*

$$\delta = [E_0 - H] \in \operatorname{Jac}(D)(\mathbf{Q})[2] \setminus \{0\}.$$

Over $\mathbf{Q}(\sqrt{2})$ *it is identified with* $[B_{\ell_+} - B_{\ell_-}]$ *by the exact function-field identity*

$$h = \frac{\ell_+}{\ell_-}\left(\frac{N}{28\sqrt{2}\,Z\ell_+}\right)^2. \tag{183}$$

Proof. The polynomial identity

$$N^2 = F + 784C_0\left(Z^2 - \frac{X^2}{2}\right) \tag{184}$$

gives (183), including its scalar, upon restriction to D. The function-field description in (170) identifies this cover with C.

On the conic $C_0 = 0$, equation (171) becomes $(X^2 - Y^2)^2 = 0$. The conics $C_0 = 0$ and $X^2 - Y^2 = 0$ meet transversely at the four points $E_{\epsilon,\eta}$. They have no common point on $Z = 0$. It follows that

$$(C_0 = 0)|_D = 2E_0, \qquad \operatorname{div}(h) = 2E_0 - 2H.$$

Taking divisors in (183) now gives the asserted identity of two-torsion classes. The latter class is nonzero because the two distinct contact divisors B_{ℓ_+}, B_{ℓ_-} each have $h^0 = 1$; if they were linearly equivalent, they would be equal. □

The zeros of h give useful rational charts on the normalized cover. Near O_D, take $X = 1 + u$, $Z = 1$, and solve the quartic equation for Y. Its first terms are

$$Y = 1 - u - \frac{48}{49}u^2 + O(u^3), \qquad h = \frac{u^2}{49} + O(u^3). \tag{185}$$

Indeed substitution of $Y = 1 - u + au^2 + O(u^3)$ gives the coefficient $-784a - 768$ of u^2 in F. In the normalized chart $v = u\omega$, the two points above O_D have $\omega = \pm 1/7$. The same conclusion at the other three points follows by the sign changes of X, Y. This also exhibits directly the eight rational boundary points of the cover.

10.4 Local survival at every finite descent level

The rational boundary and the real topology have a consequence independent of the explicit bitangent calculation. We formulate it for an arbitrary abelian variety, so it applies both to $\mathrm{Jac}(C)$ and to the quartic quotient.

Theorem 10.5. *Let* $X/\mathbf{Q}$ *be a smooth projective geometrically integral curve, let* $B \subset X$ *be a finite closed subset, and suppose* $O \in B(\mathbf{Q})$. *Let* $A/\mathbf{Q}$ *be an abelian variety and* $a : X \to A$ *a morphism with* $a(O) = 0$. *Let* $\mathcal{V} \subset (X \setminus B)(\mathbf{R})$ *contain a point joined to* O *by a continuous path in* $X(\mathbf{R})$. *For every integer* $m \geq 1$, *form*

$$X_m = X \times_{a,A,[m]} A.$$

The geometrically connected component X_m^0 *containing* $(O, 0)$ *is defined over* $\mathbf{Q}$. *It has a point above* $X \setminus B$ *over every* $\mathbf{Q}_p$, *and a real point above* $\mathcal{V}$.

Proof. In characteristic zero, multiplication by m on A is finite *étale*. Consequently $X_m \to X$ is finite *étale*, and X_m is smooth and projective. Its geometric connected components are disjoint.

The unique one containing the rational point $(O, 0)$ is Galois invariant and therefore descends to $\mathbf{Q}$.

Fix a finite prime p. The p-adic inverse function theorem, applied to the projection at $(O, 0)$, identifies a neighborhood of this point with a neighborhood of O in $X(\mathbf{Q}_p)$. A local parameter on the smooth curve takes infinitely many values in every sufficiently small p-adic neighborhood. We may therefore choose a nearby point outside the finite set B and lift it to $X_m^0(\mathbf{Q}_p)$. This argument includes primes dividing m: the differential of $[m]$ is multiplication by the nonzero element m of the field $\mathbf{Q}_p$.

For the real assertion, let γ be a path from O to a point $P \in \mathcal{V}$. The path $a \circ \gamma$ starts at zero and lies in $A(\mathbf{R})^0$. This identity component is a compact connected real Lie group; its multiplication map $[m]$ is a surjective covering map. Lift $a \circ \gamma$ starting at zero. Together with γ this gives a path on $X_m(\mathbf{R})$ beginning at $(O, 0)$, and hence lying in X_m^0. Its endpoint is the required point above P. □

Corollary 10.6. *Put $J_C = \mathrm{Jac}(C)$, $J_D = \mathrm{Jac}(D)$, and*

$$i_C(P) = [P - O_C], \qquad i_D(S) = [S - O_D].$$

For every $m \geq 1$, the distinguished components of both covers

$$C \times_{i_C, J_C, [m]} J_C, \qquad C \times_{i_D \circ \pi, J_D, [m]} J_D \tag{186}$$

have nonboundary points at every finite place and a real point whose image on C belongs to $\mathcal{U}$ and gives $x \neq y$.

Proof. Take $B = C \cap (T = 0)$ and choose the positive square roots in (166) as z decreases from $+\infty$ to $1/6$. All three radicands are positive, since they are $z^2 + 50$, $(z+7)^2 + 2$, and $(z-7)^2 + 2$. In projective coordinates this path starts at O_C. At $z = 1/6$ its coordinates are

$$P = \frac{\sqrt{1801}}{6}, \qquad Q = \frac{\sqrt{1921}}{6}, \qquad R = \frac{\sqrt{1753}}{6}.$$

Here $1 < P < 8$ and $P + 7z < 8 + 7/6 < 50$, so it belongs to $\mathcal{U}$. Formula (167) gives $y - x = 7/300 \neq 0$. Apply Theorem 10.5 with $a = i_C$ and $a = i_D \circ \pi$, respectively. □

There is also an elementary uniform description of a finite-prime square sieve on this source.

Proposition 10.7. *For a fixed $z \in \mathbf{Q}_2$, the three equations* (166) *with $T = 1$ have simultaneous roots in $\mathbf{Q}_2$ if and only if $v_2(z) < 0$. The analogous assertion over $\mathbf{Q}_3$ holds with $v_3(z) < 0$. For every finite set S of primes, there are infinitely many rational values of z for which all three radicands are squares in every $\mathbf{Q}_p$, $p \in S$, and whose positive real roots belong to $\mathcal{U}$.*

Proof. For integral z at two, $z^2 + 50$ is $2, 3$, or 6 modulo eight, so cannot be a square. Write $z = 2^{-k}u$ with $k \geq 1$ and u a unit. For $k \geq 2$, division by z^2 makes all three radicands units congruent to one modulo eight. For $k = 1$, multiplication by four gives

$$u^2 + 200, \qquad u^2 + 28u + 204, \qquad u^2 - 28u + 204,$$

which are again one modulo eight. The complete square criterion in $\mathbf{Q}_2$ proves sufficiency.

At three, the residue classes $z = 0, 1, -1$ respectively make the first, third, and second radicand a nonsquare modulo three. If $v_3(z) < 0$, division by z^2 gives units congruent to one modulo three.

More generally, at any odd prime p, a negative valuation of z gives this same unit congruence modulo p, so Hensel's lemma supplies all roots.

Set $M = 6\prod_{p\in S} p$ and take $z = M^{-k}$ for $k \geq 1$. All the required valuations are negative, while $0 < z \leq 1/6$. At the real place, the positive roots satisfy $1 < P < 8$ and $P + 7z < 8 + 7/6 < 50$. These distinct rational parameters have all the asserted properties. □

Proposition 10.7 establishes survival of the simultaneous square conditions at any specified finite set of places. Corollary 10.6 establishes the stronger statement after imposing a fixed multiplication level on either Jacobian: a single geometrically connected cover retains nonboundary points at every finite place and a strict real distance configuration. For the second cover in (186), this conclusion includes the literal quadratic lift identified in Proposition 10.4. Thus these finite descent levels have nonempty local-solubility sets with the complete source and the strict real inequalities imposed.

A The distance surface and its Brauer–Manin sets

The four-distance equations also define a fixed surface, independent of the square-free parameter in Section 5. The surface approach is developed in [1], and its product-quotient description belongs to the general construction in [6]. We give the equations and their geometry explicitly, and then determine the projection of the proper Brauer–Manin set onto the positive real region.

A.1 A canonical complete intersection

Let $S \subset \mathbf{P}^4_{[A:B:C:D:T]}$ be defined by

$$\begin{aligned} Q &= A^2 + D^2 - B^2 - C^2 = 0, \\ F &= (T^2 + A^2 - B^2)^2 + (T^2 + A^2 - C^2)^2 - 4T^2A^2 = 0. \end{aligned} \tag{187}$$

On $T = 1$, the identification with the distance equations is

$$(r_0, r_1, r_2, r_3) = (A, B, C, D), \qquad x = \frac{1 + A^2 - B^2}{2}, \qquad y = \frac{1 + A^2 - C^2}{2}. \tag{188}$$

Indeed, $F = 0$ gives $x^2 + y^2 = A^2$, and subtraction gives the other three identities. Let $X \to S$ be the minimal resolution. Write U for the open subset $T = 1$, $ABCD \neq 0$, viewed as an open subset of X, and put

$$\mathcal{C} = \{(x, y, r_0, r_1, r_2, r_3) \in U(\mathbf{R}) : 0 < x, y < 1,\ r_j > 0 \text{ for all } j\}.$$

Proposition A.1. *The surface S is geometrically integral and has precisely forty ordinary double points. Its resolution X is a simply connected minimal surface of general type, with*

$$K_X^2 = 8, \qquad p_g(X) = 5, \qquad q(X) = 0, \qquad c_2(X) = 64, \qquad b_2(X) = 62.$$

The geometric Picard group of X is a finitely generated free abelian group.

Proof. Work first over an algebraically closed field of characteristic zero. Put $z = x+iy$, $w = x-iy$, and

$$(\alpha_0, \alpha_1, \alpha_2, \alpha_3) = (0, 1, i, 1 + i).$$

The affine distance equations become

$$r_j^2 = (z - \alpha_j)(w - \overline{\alpha}_j), \qquad 0 \le j \le 3. \tag{189}$$

Their four square classes are independent in $\overline{\mathbf{Q}}(z,w)^*/\overline{\mathbf{Q}}(z,w)^{*2}$: the divisor $z = \alpha_j$ occurs with odd multiplicity only in the jth radicand. Thus the affine surface is geometrically integral. The complete intersection (187) is Cohen–Macaulay and purely two-dimensional. Its intersection with $T = 0$ has dimension one, so every irreducible component meets the affine chart. There is consequently only one component, and it is generically reduced. The Cohen–Macaulay property excludes embedded components, hence S is reduced and geometrically integral.

The branch grid in (189) determines all affine singularities. Away from the grid the covering is étale. Along a single branch line, its square root is a local parameter. At an intersection $z = \alpha_j$, $w = \overline{\alpha}_k$ with $j \ne k$, the two corresponding square roots are independent local parameters. At an intersection with $j = k$, the local equation, after extracting square roots of units, is $r_j^2 = uv$. This is an ordinary double point. There are eight choices for the other three roots at each of the four vertices, giving thirty-two affine nodes. In distance coordinates the three nonzero roots at a vertex are $\pm 1, \pm 1, \pm\sqrt{2}$.

At infinity, write $H_\pm = A^2 - B^2 \pm i(A^2 - C^2)$. The boundary is the union of the two intersections $Q = H_\pm = 0$ in $\mathbf{P}^3$. Each is smooth: the ratios of the four diagonal coefficients of $H_\pm$ to those of Q are

$$1 \pm i, \quad 1, \quad \pm i, \quad 0,$$

which are pairwise distinct. Proportional gradients would therefore leave at most one coordinate nonzero, contrary to $Q = 0$. Their intersections are exactly

$$[1 : \epsilon_1 : \epsilon_2 : \epsilon_3 : 0], \qquad \epsilon_1, \epsilon_2, \epsilon_3 \in \{1, -1\}. \tag{190}$$

At any such point set $A = 1$, $B = \epsilon_1 + b$, $C = \epsilon_2 + c$, $T = t$, and eliminate D using Q. The quadratic term of F is $4(b^2 + c^2 - t^2)$, which is nondegenerate. These are eight more nodes. The preceding smoothness calculation also exhausts the points at infinity. In particular S is normal.

Adjunction for a complete intersection gives $\omega_S = \mathcal{O}_S(1)$. Ordinary double points are rational and their resolutions are crepant, so

$$K_X = \pi^* \mathcal{O}_S(1), \qquad K_X^2 = \deg S = 8.$$

The Koszul resolution

$$0 \longrightarrow \mathcal{O}_{\mathbf{P}^4}(-6) \longrightarrow \mathcal{O}_{\mathbf{P}^4}(-2) \oplus \mathcal{O}_{\mathbf{P}^4}(-4) \longrightarrow \mathcal{O}_{\mathbf{P}^4} \longrightarrow \mathcal{O}_S \longrightarrow 0$$

gives $H^1(S, \mathcal{O}_S) = 0$ and $h^0(S, \mathcal{O}_S(1)) = 5$. Rationality of the singularities then gives $q = 0$ and $p_g = 5$ on X. The canonical divisor is nef and big; adjunction excludes a (-1)-curve. Noether's formula gives $c_2 = 12(1 - q + p_g) - K_X^2 = 64$, and $b_1 = 2q = 0$ gives $b_2 = 62$.

For simple connectedness, deform the two defining equations slightly to a smooth complete intersection of type $(2, 4)$, which is simply connected by the Lefschetz hyperplane theorem. Remove disjoint small neighbourhoods of the forty nodes. The smoothing and the resolution have the same differentiable exterior. Each boundary is $\mathbf{RP}^3$. Both the corresponding Milnor fibre and the resolution neighbourhood are simply connected: the former retracts onto a two-sphere, and the latter onto the exceptional projective line. Van Kampen's theorem therefore presents both fundamental groups as the fundamental group of the common exterior modulo the normal closures of the boundary-group images. They are the same quotient. Thus $X(\mathbf{C})$ is simply connected.

Consequently $H_1(X(\mathbf{C}),\mathbf{Z}) = 0$, and the universal coefficient theorem makes $H^2(X(\mathbf{C}),\mathbf{Z})$ torsion-free. The exponential sequence and $q = 0$ inject $\mathrm{Pic}(X_{\mathbf{C}})$ into this group. Finally $\mathrm{Pic}^0(X) = 0$ and the Néron–Severi theorem give finite generation of the geometric Picard group. □

A.2 A fixed product of genus-five curves

Let $K = \mathbf{Q}(i)$. Define C_0/K by its function field

$$K(C_0) = K(z)(u_0, u_1, u_2, u_3), \qquad u_j^2 = z - \alpha_j, \tag{191}$$

and let $\overline{C}_0$ denote its coefficient conjugate. Both curves are understood to be smooth and projective.

Proposition A.2. *The curve C_0 has genus five. The diagonal sign action of $G = (\mathbf{Z}/2\mathbf{Z})^4$ gives a birational identification*

$$X_K \sim (C_0 \times \overline{C}_0)/G.$$

Its invariant functions include the four labelled products $r_j = u_j v_j$, where $v_j^2 = w - \overline{\alpha}_j$. The Jacobian of C_0 is geometrically isogenous to a product of five elliptic curves: four have $j = 128$ and one has $j = 1728$.

Proof. The four radicands in (191) are independent square classes by their valuations at their distinct zeros. The covering $C_0 \to \mathbf{P}^1$ has degree sixteen. Its inertia elements are the four individual sign changes at the finite branch points and their sum at infinity. Each inertia group has order two, so

$$2g(C_0) - 2 = 16(-2) + 5 \cdot 8 = 8.$$

On the product, the diagonal action fixes $z, w, u_j v_j$. The field they generate has degree sixteen over $K(z,w)$ by the square-class argument for (189). This is the degree of the invariant field, since the product field has degree 256 over $K(z,w)$. The invariant field is therefore precisely the distance function field.

For a nonempty subset $J \subset \{0,1,2,3\}$, the corresponding character quotient is

$$y_J^2 = \prod_{j \in J} (z - \alpha_j).$$

Only the four subsets of size three and the subset of size four have positive genus; all five quotients have genus one. The pullbacks of their differentials give the five distinct character lines in $H^0(C_0, \Omega^1)$, hence their product induces an isogeny on Jacobians. The branch cross-ratios can be chosen as i for the three-element quotients and 2 for the four-element quotient. Substitution in

$$j(\lambda) = 256 \frac{(1 - \lambda + \lambda^2)^3}{\lambda^2 (1 - \lambda)^2}$$

gives the asserted values. □

A.3 The positive real component

We write Br for the cohomological Brauer group and Br_0 for the image of the Brauer group of the ground field. The rational point used below lies on the boundary of U.

Lemma A.3. *There is a point $P_* \in X(\mathbf{Q})$ in the same real connected component as every point of $\mathcal{C}$.*

Proof. At the node $[1:1:1:1:0]$, use the local coordinates b, c, t from the proof of Proposition A.1. The exceptional conic is $b^2 + c^2 = t^2$. Take the rational point $P_* = [-1:0:1]$ on this conic.

Start with a point of $\mathcal{C}$, fix its $y \in (0,1)$, and let x tend to $+\infty$, choosing all four positive distance roots. This is a path in the smooth affine locus because none of the distances vanishes. In the chart $A = 1$ its coordinates are $B = r_1/r_0$, $C = r_2/r_0$, $T = 1/r_0$. Rationalization gives

$$\frac{b}{t} = r_1 - r_0 = \frac{1-2x}{r_1 + r_0} \longrightarrow -1, \qquad \frac{c}{t} = r_2 - r_0 = \frac{1-2y}{r_2 + r_0} \longrightarrow 0.$$

The lifted path extends to P_* on the resolution. □

Proposition A.4. *The group $\mathrm{Br}(X)/\mathrm{Br}_0(X)$ is finite.*

Proof. Resolve the rational map $C_0 \times \overline{C}_0 \dashrightarrow X_K$ from Proposition A.2. This gives a smooth projective surface Y, birational to the product, and a generically finite morphism $f : Y \to X_K$ of degree sixteen. The product theorem of Skorobogatov–Zarhin [14, Theorem A], with its corrected proof [15], states that, over a field finitely generated over $\mathbf{Q}$, the map

$$\mathrm{Br}(\overline{V})^{\Gamma} \oplus \mathrm{Br}(\overline{W})^{\Gamma} \longrightarrow \mathrm{Br}(\overline{V} \times \overline{W})^{\Gamma}$$

has finite cokernel for smooth projective geometrically integral varieties V, W. The geometric Brauer group of a smooth projective curve is zero. Applied to our two curves, this theorem and birational invariance therefore show that $\mathrm{Br}(\overline{Y})^{\Gamma_K}$ is finite.

Restriction–corestriction on function fields shows that the kernel of $f^* : \mathrm{Br}(\overline{X}) \to \mathrm{Br}(\overline{Y})$ is killed by sixteen. This kernel is finite, because the Kummer sequence places $\mathrm{Br}(\overline{X})[16]$ in a quotient of the finite group $H^2_{\text{ét}}(\overline{X}, \mu_{16})$. Taking invariant subgroups thus proves that $\mathrm{Br}(\overline{X})^{\Gamma_K}$, and hence $\mathrm{Br}(\overline{X})^{\Gamma_{\mathfrak{Q}}}$, is finite.

By Proposition A.1, $\mathrm{Pic}(\overline{X})$ is a finitely generated free abelian group. Its Galois action factors through a finite group. Inflation–restriction and the vanishing of continuous homomorphisms from a profinite group to a discrete free abelian group identify its first Galois cohomology with that of a finite group acting on a lattice; this group is finite. The Hochschild–Serre sequence now embeds $\mathrm{Br}_1(X)/\mathrm{Br}_0(X)$ in $H^1(\mathbf{Q}, \mathrm{Pic}(\overline{X}))$, while $\mathrm{Br}(X)/\mathrm{Br}_1(X)$ embeds in $\mathrm{Br}(\overline{X})^{\Gamma_{\mathfrak{Q}}}$. Both are finite. □

A.4 Adelic points with a prescribed strict real component

For a subgroup $B \subset \mathrm{Br}(U)$, let $U(\mathbf{A}_{\mathbf{Q}})^B$ denote the points in the usual restricted-product adelic space orthogonal to B under the Brauer–Manin pairing. We identify $\mathrm{Br}(X)$ with its restriction to U.

Theorem A.5. *Every $w \in \mathcal{C}$ is the real component of an adelic point in $U(\mathbf{A}_{\mathbf{Q}})^{\mathrm{Br}(X)}$. Moreover, for every finite subgroup $B \subset \mathrm{Br}(U)$ and every $w \in \mathcal{C}$, there is a point $(P_v) \in U(\mathbf{A}_{\mathbf{Q}})^B$ with $P_\infty = w$.*

Proof. First let B_0 be a finite subgroup of $\mathrm{Br}(X)$. Choose a finite set Σ of primes such that X, U and P_* extend to a smooth proper model $\mathscr{X}$, a smooth geometrically integral open model $\mathscr{U}$, and a section over $\mathbf{Z}[1/\Sigma]$, and all the classes in B_0 extend to $\mathrm{Br}(\mathscr{X})$. The section is denoted

by the same symbol P_* here. Lang–Weil and Hensel's lemma, after enlarging Σ, give a point of $\mathscr{U}(\mathbf{Z}_p)$ for every $p \notin \Sigma$. Every class in B_0 evaluates to zero at this point and at the section, since $\mathrm{Br}(\mathbf{Z}_p) = 0$.

At each $p \in \Sigma$, the evaluation of a Brauer class on $X(\mathbf{Q}_p)$ is locally constant. Choose a sufficiently small neighbourhood of the rational point P_* on which all the evaluations in B_0 agree with their values at P_*. This neighbourhood contains a point of $U(\mathbf{Q}_p)$: in smooth analytic coordinates a proper algebraic closed subset cannot contain an open ball. At the real place use the prescribed w. Evaluation is constant on real connected components, so Lemma A.3 gives agreement with P_* there as well.

These choices form an adelic point of U, integral in the single model $\mathscr{U}$ outside Σ. For each $\beta \in B_0$ its evaluations coincide place by place with those of $\beta(P_*) \in \mathrm{Br}(\mathbf{Q})$. Global reciprocity therefore gives orthogonality to B_0.

Proposition A.4 supplies finitely many representatives for $\mathrm{Br}(X)/\mathrm{Br}_0(X)$. Because the Brauer group is torsion, they generate a finite subgroup B_0. Applying the construction to this group gives orthogonality to all of $\mathrm{Br}(X)$, as constants have zero total invariant.

For the last assertion, use Harari's formal lemma in the form recalled in [7]: if V is smooth and geometrically integral over a number field, $U \subset V$ is a nonempty open subset, and $B \subset \mathrm{Br}(U)$ is finite, then an adelic point of U orthogonal to $B \cap \mathrm{Br}(V)$ can be replaced by one orthogonal to B, preserving its components at any prescribed finite set of places. Apply the lemma with $V = X$, starting with the point just constructed, and preserve the real component. □

Thus the positive real region lies in the projection of the adelic set for the entire proper Brauer group, and also in the projection for each finite subgroup of the affine Brauer group. In the second assertion the finite subgroup is fixed before the adelic point is chosen; this is the quantifier order used in the theorem.

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